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Joseph Henry Briggs

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Dedication

Declaration

Acknowledgements

Abstract

Chapter 1

Introduction

Towards the end of the 19th century, it was becoming clear that Newton's theory of gravity and Maxwell's theory of electromagnetism were not in agreement; the force law arising from Maxwell's equations that governs the motion of charged particles in magnetic fields did not have an absolute frame of reference and the mechanics of the situation follow Lorentz transformations; in contrast, Newton's laws of motion dictate that objects obey Galilean transformations.

In 1915, Einstein resolved this contradiction by formulating special relativity. This naturally lead to him establishing a general theory of relativity. This theory explained the long-standing mystery surrounding the precession of the perihelion of Mercury's orbit. It predicted the the deflection of light in a gravitational field and the dilation caused gravitational fields. Measurements verifying these predictions were made during the 20th century.

The final prediction of general relativity to be measured was gravitational waves. These signals are very weak even in the most extreme of environments, such as the

coalescing of black holes. The existence of these could be inferred from the discovery of a pulsar signal by Taylor and Hulse, for which they won the 1995 Nobel prize. It was not until 2015 that LIGO directly measured a signal generated by the merging of two black holes that the existence gravitational waves was finally confirmed, and all the predictions of Einstein's theory were measured.

However, the story of gravitational wave detectors is far from over. Current observatories are reaching their design sensitivities, and soon gravitational wave detections will become routine. These will give astronomers large datasets with which they can develop our understanding of how our universe is structured.

In the coming decades, new observatories capable of probing to larger red shifts will be constructed. With these, our understanding of physics may be revolutionised. The theory of general relativity does not translate into the quantum world, there is still mystery surrounding dark matter and dark energy and the earliest movements of our universe after the big bang cannot be studied with current tools.

With every new method of observing the universe, new phenomena have been discovered. Telescopes allowed astronomers to study planets and stars, and radio telescopes revolutionised our understanding of cosmology. Gravitational waves have given us a new way of probing the universe and detecting the existence of objects that are not predicted by our current model of physics.

Chapter 2

Gravitational Wave Interferometers

2.1 Introduction

On the 14th of September 2015, the detection of a gravitational wave signal emanating from two coalescing black holes signalled the beginning of a new era for astronomy. The information carried by a gravitational wave is undisturbed by the matter between the observer and the astrophysical body that is emitting them, thus it gives astronomers valuable information which cannot be obtained from the electromagnetic radiation emitted by that body. Gravitational wave astronomy also allows for direct observations of objects that do not emit electromagnetic radiation e.g. black holes. It is of great importance to astronomy that large numbers of gravitational wave signals are measured, so it is imperative that gravitational wave detectors are made more sensitive.

However, gravitational waves are incredibly weak and the detection of them is a science in itself. This chapter will outline how a gravitational wave interferome-

ter, with a particular emphasis on LIGO, can be used to detect these signals and the fundamental noise sources these detectors face. While there are other important sources of noise, the suppression of quantum noise is crucial to achieving the sensitivity required to detect gravitational waves and so this is the focus of this chapter.

2.2 Astrophysical Origins of Gravitational Waves

Gravitational waves emanating BBH (e.g. [1,2]), BNS [3], and a source which is either a BBH or a binary consisting of a black hole and a neutron star (NSBH) [4] have been detected; this is summarised in [5]. The waveform of gravitational radiation emitted from these systems is known as a chirp signal. These signals increase in frequency as the two objects get closer to merging. Due to the geometry of these systems, the frequency of the gravitational waves are in the audioband with the cut-off frequency, determined by the radius of the objects that will merge, between 100 Hz-10 kHz. A typical gravitational wave strain amplitude of these types of systems, all of which have a total mass of $10M_{\odot} - 100M_{\odot}$ and are of the order 100 Mpc to 10 Gpc away from Earth, is $h \sim 10^{-21} - 10^{-23}$.

Einstein's general theory of relativity accurately describes the gravitational waves measured with the LIGO and Virgo gravitational wave detectors [6]. Gravitational waves are emitted from mass quadrupole systems (see e.g. [7]). As the coupling between mass and the curvature of space-time is weak, gravitational waves are only observable in systems that have large masses involved.

Other astrophysical phenomena will be sources of gravitational waves. Asymmetries in pulsars [8,9] and magnetars [10] will cause them to emit a gravitational wave with a constant frequency. When the core of a star collapses as it goes supernova, a burst of gravitational radiation may be emitted [11,12]. Burst signals may be generated immediately after a BNS merges [13]. Weaker sources that the future gravitational wave interferometers may detect include: the stochastic background of gravitational waves [14], similar to the cosmic microwave background radiation, as well as a background level of gravitational waves due to the merging of compact binaries [15]; dark matter [16,17] and cosmic strings [18].

To improve statistical models of the systems that emit gravitational waves, the number detections needs to be increased. To detect continuous wave signals, the signal to noise ratio improves with the total observing time of the detector. With longer observation times, it is more likely that an exotic event will be measured. For these reasons, the gravitational wave detectors are operated for year long time-scales with minimal interruption. These are known as observation runs. Between observation runs, the detectors undergo hardware upgrades that increase the detector sensitivity to gravitational waves. The current observation runs are summed up in Table 2.1.

2.3 Measuring Gravitational Waves with an Interferometer

Gravitational waves are propagating longitudinal perturbations in the curvature of space-time, and can be split into a basis set containing two polarisations, plus and

Observation Run	Dates	Detections	Detectors
O1	12/09/2015 - 19/01/2016	3	aLIGO
O2	30/11/2016 - 25/08/2017	8	aLIGO, Virgo
O3a	01/04/2019 - 01/10/2019	39	aLIGO, Virgo
O3b	01/11/2019 - 04/30/2020	23	aLIGO, Virgo
O4	2021	-	aLIGO, Virgo, KAGRA
O5	date?	-	LIGO A+, Advanced Virgo, KAGRA, LIGO India

Table 2.1: table of observation runs

cross (e.g. [7]). The effect these waves have, in the local Lorentz frame, on test masses on Earth is to move them by a small amount ¹.

To detect plus polarised gravitational waves, the end mirrors of a Michelson interferometer can be used as the test masses. These are referred to as *end test masses* (ETMs). A sketch of a gravitational wave interacting with a Michelson interferometer is shown in Figure 2.1. For test masses L away from the origin of the local Lorentz frame, the distance they move ΔL due to an incoming plus polarised gravitational wave is

$$\Delta L = Lh. \quad (2.1)$$

¹In the local Lorentz frame of the beam splitter, for any feasible Earth based detector, the effect of a gravitational wave on the wavelength of the light is negligible compared to its effect on the positions of the test masses due to the strong equivalence principle: all physical laws that are expressible as a tensor equation in special relativity have the exact same form in a local Lorentz frame of a curved space time. In the local Lorentz frame, the metric has the form $g_{\alpha\beta} = \eta_{\alpha\beta} + \mathcal{O}(h)$, where $\eta_{\alpha\beta}$ is the Minkowski space-time metric and $\mathcal{O}(h)$ is a perturbation of the order $\frac{4\pi L}{\lambda_{\text{EM}}}^2 h$. The Minkowski space-time metric is flat, so in the local Lorentz frame the metric is, essentially, flat and the gravitational waves cause the test masses to move by $\Delta L = hL$. Maxwell's equations are modified by a term that is of the order $\frac{\delta\lambda_{\text{EM}}}{\lambda_{\text{EM}}} \sim \frac{4\pi L}{\lambda_{\text{EM}}}^2 h$, and so if $\frac{4\pi L}{\lambda_{\text{EM}}}$ is sufficiently small, as it would be for an Earth based interferometer, then the effect of the gravitational wave on the light is much smaller than the effect of the gravitational wave on the test masses.

The interferometer is sensitive to changes in the phase between the two arms corresponding of the order of one wavelength,

$$\Delta\phi = \frac{4\pi}{\lambda_{\text{EM}}}Lh, \quad (2.2)$$

allowing for small distances when laser light is used. The choice of $\lambda_{\text{EM}} = 1064 \text{ nm}$ as the wavelength which the detector operates with is determined by the low loss and low thermal noise coatings [19] and test masses that can be fabricated for 1064 nm light.

The gravitational wave signal is encoded in intensity fluctuations of the light at the *antisymmetric* port (see Figure 2.1). Gravitational wave detectors are operated on the *dark fringe* meaning that there is a low amount of power at the antisymmetric port for the signal light to beat with and the variation in the intensity of the light depends linearly on the gravitational wave strain. The interferometer is set to operate on the dark fringe by introducing a small static differential offset, ϕ_0 , is the differential arm length. At the beam splitter, the electric field, Ψ_{AS} , emerging from the antisymmetric port due to the interference of the beams, with differential phase $\phi = \phi_0 + \Delta\phi$, returning from the end test masses, Ψ_y and Ψ_x , is

$$\Psi_{\text{AS}} = \frac{\Psi_y}{\sqrt{2}} - \frac{\Psi_x}{\sqrt{2}}, \quad (2.3)$$

$$= \sqrt{\frac{I_0}{2}} \left(\frac{e^{i(\phi_0 + \Delta\phi)}}{\sqrt{2}} - \frac{e^{-i(\phi_0 + \Delta\phi)}}{\sqrt{2}} \right). \quad (2.4)$$

The intensity of the beam at the anti-symmetric port is

$$I_{AS} = \Psi_{AS}^* \Psi_{AS}, \quad (2.5)$$

$$\simeq I_0 \phi_0^2 + 2I_0 \phi_0 \Delta\phi, \quad (2.6)$$

$$= I_0 \phi_0^2 + 2I_0 \phi_0 \frac{4\pi}{\lambda_{EM}} Lh, \quad (2.7)$$

In the absence of a gravitational wave signal, the DC light level at the antisymmetric port is related to the static offset by $\phi_0 = \sqrt{\frac{I_{AS}}{I_0}}$. As we are interested in a fluctuating signal, the DC offset from equation 2.7 can be removed, allowing for the signal at the antisymmetric port to be written as

$$I_{AS} = 2\sqrt{I_0 I_{AS}} \frac{4\pi}{\lambda_{EM}} Lh \quad (2.8)$$

and so by placing a photo detector in this beam, it is possible to measure the differential motion caused by the strain generated by the gravitational wave.

2.3.1 Using a Fabry Perot Cavity in the Arms of a Michelson Interferometer to enhance its Sensitivity

The optimal arm length of the interferometer is determined by 1/4 of the wavelength of a gravitational wave as this gives the maximal $\Delta\phi$. For BBH and BNS, the gravitational wave signal will be at 100 Hz, and so the wavelength of the gravitational wave will be of the order 3000 km. Its not possible to construct such an interferometer on Earth.

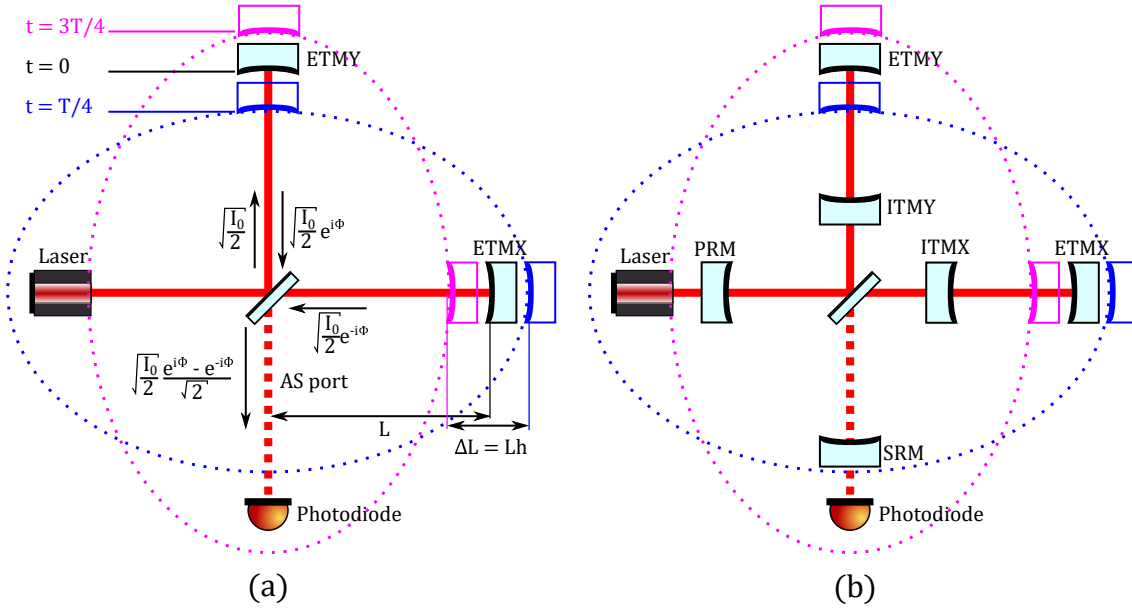


Figure 2.1: (a) A sketch of the effect a gravitational wave has on a Michelson interferometer. The positions of the test masses, ETMX and ETMY, at different points during the gravitational wave's period is shown by the blue and pink lines. The difference in phase shift between the fields in the two arms affects the intensity of the light at the AS port. By detecting this light with a photodiode, the gravitational wave strain h can be determined. (b) To reach the sensitivity needed to detect gravitational wave, several mirrors need to be added to the basic Michelson interferometer: ITMX and ITMY boost the signal generated in the arms, the PRM increases the power at the beam splitter, and the SRM can be used to shape the quantum noise of the interferometer.

The sites that were available to construct the current generation of gravitational wave detectors allow for 4 km arms. To reach the optimal arm length on a 4 km site, the number of times that the light bounces off the test mass, B , needs to be increased. Numerically, B should be ~ 250 . This means that equation 2.2 becomes

$$\Delta\phi = \frac{4\pi}{\lambda_{\text{EM}}} BLh. \quad (2.9)$$

Fabry-Perot cavities are used² to increase the number of bounces of the light from the test masses. The cavity is formed by placing an *initial test mass* (ITM) near the beam splitter. When a photon enters a cavity, it remains in the cavity for an average number of bounces that is determined by the reflectivity of the mirrors. When a Fabry-Perot cavity goes through a resonance, the phase of the reflected light rapidly changes in response to a change in arm length. This is illustrated in Figure 2.2. The interferometer measures $\Delta\phi_{\text{EM}} = \frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}} \Delta\phi_{\text{ETM}}$, so when a Fabry Perot cavity is near or on resonance, as the slope $\frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}}$ is much greater for a Fabry Perot than it is for a mirror, the signal is enhanced. The exact way this happens is dependent of the frequency of the gravitational wave and the reflectivity of the mirrors, with the arm cavity cavity as a low pass filter (e.g. see [20, Chapter 3] or [21, Appendix D]).

²Herriott delay lines were considered as a means of achieving more bounces from the test masses, however due to difficulty in manufacturing suitable mirrors which do not scatter light, thus generating noise, delay line interferometers fell out of favour. In principle, both delay lines and Fabry Perot cavities would perform equally well due to the Mizuno limit [20].

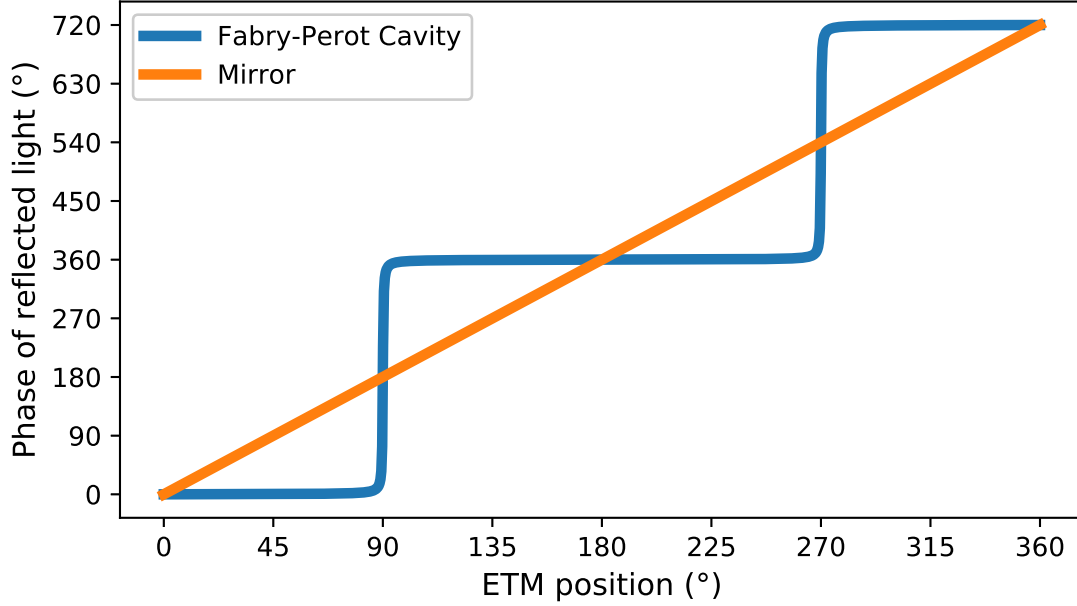


Figure 2.2: Comparison between the phase of the light reflected from a mirror versus the phase of light reflected from a cavity. The blue line shows the phase of the light that is reflected by a Fabry Perot cavity. Initially, the cavity is on anti-resonance so perfect destructive interference occurs as the round-trip phase is 180° . The end mirror (ETM) is moved 90° , resulting in an extra 180° round-trip phase, and so the cavity resonates. The phase of the light reflected from the cavity rapidly changes as it moves through the resonance. The ETM is moved further, the cavity stops being resonant and the phase of the light is not changed as its reflected by the cavity. The orange line shows the phase of the light when a mirror is used instead of a cavity. The interferometers sensitivity is proportional to the slop $\frac{d\phi_{EM}}{d\phi_{ETM}}$, and so by inserting another mirror in the arm near the beamsplitter, a Fabry-Perot cavity is formed and the sensitivity of the interferometer is increased.

2.4 Sources of noise in Gravitational wave Interferometers.

As the gravitational wave strain causes a minuscule change in the phase accumulated by light in the arm cavities, the challenge of detecting gravitational waves lies in suppressing the noise below this signal. From equation 2.9, for a gravitational wave of strain $h_{\text{rms}} = 10^{-23}$ interacting with LIGO the change in phase shift that the light experiences is $\sim 10^{-11}$ rad.

2.4.1 Laser Power and Shot Noise

The shot noise associated with detecting a coherent state, such as that created by the laser, can limit the sensitivity of the detector. To reduce the level of shot noise, the energy, E , of the laser needs to be high. The required energy to measure the phase shift that the light experiences due to the gravitational wave strain is determined by the time-energy uncertainty principle $\Delta\phi = \omega_{EM}\Delta T$ [22]. The measurement time, T , required corresponds to half the period of the gravitational wave $T = \frac{1}{2f_{\text{GW}}}$. The power of the laser is $I_0 = \hbar\omega_{EM}N$, where N is the number of photons detected in the time interval T , and so the energy used to measure the time is $E = I_0T = I_0\frac{\hbar\omega_{EM}N}{2f}$. As the laser is in a coherent state, the uncertainty in the number of photons detected in time T is $\Delta N = \sqrt{N}$. The uncertainty principle may now be written as

$$N \geq \frac{1}{4(\Delta\phi)^2}. \quad (2.10)$$

The amplitude spectrum of the intensity measured by the photodiode over a time corresponding to a bandwidth of Δf at the antisymmetric port will depend the

amount of light at the dark port,

$$I_{\text{ASrms}} = \sqrt{2\bar{I}_{\text{AS}}\hbar\omega_{EM}\Delta f}. \quad (2.11)$$

The conversion from light power at the antisymmetric port to gravitational wave strain can be inferred from Equation 2.8

$$\frac{h}{I_{\text{AS}}} = \frac{1}{2\sqrt{I_0\bar{I}_{\text{AS}}}} \frac{\lambda_{\text{EM}}}{4\pi LB} \quad (2.12)$$

and so the amplitude spectrum of the strain corresponding to the the shot noise of the light at the AS port is

$$I_{\text{ASrms}} = \sqrt{\frac{\hbar c^2}{8I_0\omega_{EM}L^2B^2}\Delta f}. \quad (2.13)$$

Equation 2.13 assumed the frequency response to a gravitational wave is flat, however the arm cavities have a low pass filtering effect on the signal, and so the shot noise will rise as f above the characteristic frequency of the arm cavity. This can be seen in the spectral density of the LIGO detector's noise (Figure 2.3). Equation 2.13 is more rigorously found in [23].

To beat the photon shot noise by a factor of 10, the required number of photons to measure a gravitational wave with a strain amplitude of $h_{\text{rms}} = 10^{-23}$ is $\sim 10^{22}$. For a gravitational wave with a frequency of 100 Hz, the laser power required entering the arm cavity would need to be $\sim 1\text{MW}$. No such laser exists.

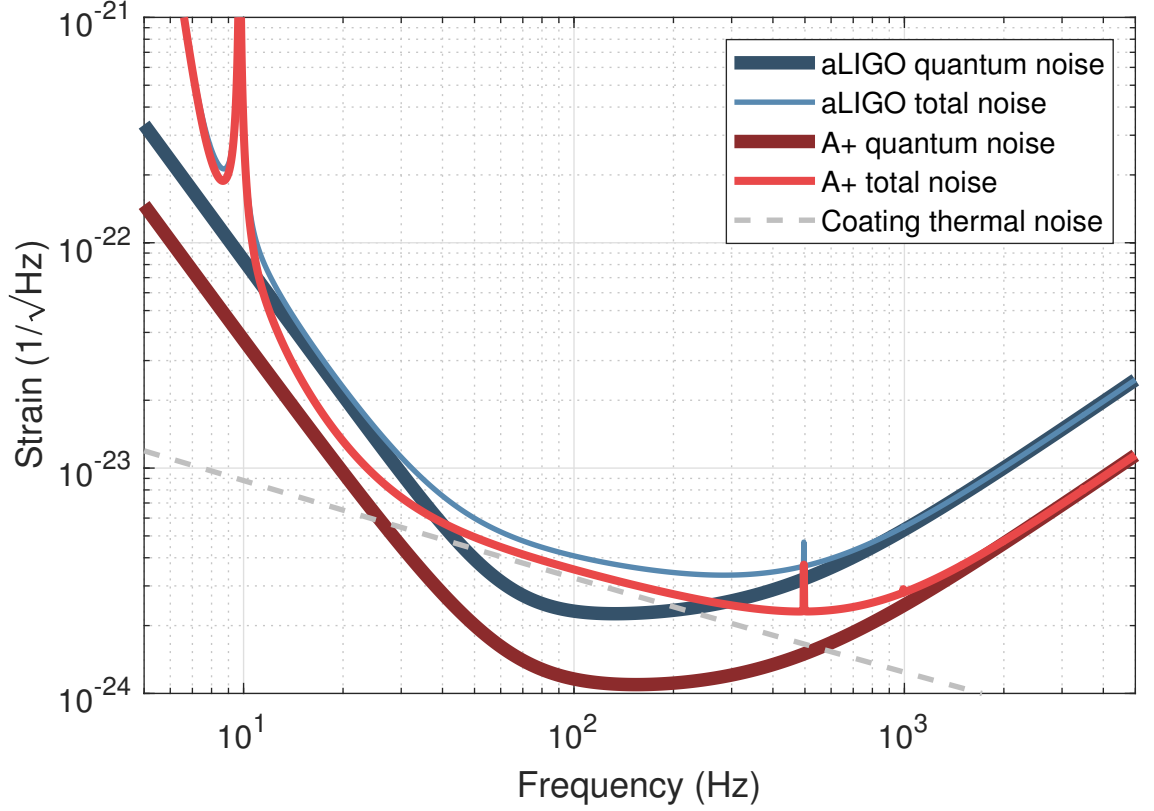


Figure 2.3: The strain Sensitivity of the aLIGO and A+ detectors. The noise due to radiation pressure and photon shot noise is shown in dark-blue (aLIGO) and dark-red (A+). Above 10 Hz, quantum effects are one of the limiting sources of noise; with the other being the noise due to Brownian motion of the mirror's coating. Below 10 Hz, seismic noise begins to dominate the noise of the detector. Above 100 Hz, shot noise dominates the noise of the detector. This rises with f as the shot noise is filtered by the arm cavities. Below 100 Hz, the quantum noise goes as $1/f$ due to radiation pressure.

2.4.2 Power Recycling to reduce Shot Noise

To increase the laser power that is injected into the beam splitter, a power recycling cavity can be used [?]. Most of the light from the laser that is incident on the beam splitter gets reflected, effectively making the interferometer a mirror. Drever [?] proposed that a mirror, called the *power recycling mirror* (PRM), could be placed in between the laser and the beam splitter and this would form a cavity. By tuning the position of the PRM such that the light reflected from the interferometer and the light from the laser add in phase, the power would be ‘recycled’ and the total power impinging on the beam splitter would be increased. Current laser sources that can be operated for year-long time scales are capable of producing of the order 100W, see Chapter 3, and so the power recycling factor needs to be $\sim 10^4$. This technique has been successfully implemented at LIGO and by many other large scale interferometers. [?]

2.4.3 Using Signal recycling to shape Quantum noise

The bandwidth of the detector depends on how long the signal carrying light is in the arms for. This detection bandwidth can be altered by introducing another mirror: the signal recycling mirror (SRM). This mirror is placed between the photodetector and the beamsplitter, and will form a coupled cavity with the arm cavities. The finesse of the arm cavities as seen by the light emerging at the antisymmetric port depends on the tuning of the SRM. There are two common modes of operation: resonant sideband extraction (RSE) [24, 25] and signal recycling .

In RSE mode, the signal recycling mirror is tuned so that the SRC is anti-resonant, thus broadening the sensitivity bucket at the expense of losing peak sensitivity. This was motivated by wanting to reduce the thermal load on the beam splitter by increasing the finesse of the arm cavities (see Equation 2.13) . However, by increasing the finesse of the arm cavity, the bandwidth of the interferometer is reduced. To increase the recover this lost bandwidth, the SRM is tuned so that the finesse of the arm cavity experienced by the signal is lowered.

In signal recycling mode, the SRM is tuned so that SRC is resonant, deepening the sensitivity bucket at the expense of bandwidth. This comes from the conservation of energy. From an energy argument, the sensitivity of a detector to gravitational waves is determined by the detectors bandwidth and the energy within this bandwidth. This is known as the Mizuno limit [20].

2.4.4 Using a Mode cleaning Cavity to Remove Junk Light and Reduce Shot noise.

All light that is incident on the output photodiode contributes to the total noise of the measurement; therefore, only signal carrying light should be incident on the photodiode to enhance the signal to noise ratio of the measurement. In an ideal interferometer, the light emerging at the AS port will be a beam that is defined by mode resonant in the arm cavity. Due to manufacturing tolerances of mirrors and to minimise diffraction losses, the light in the arm cavity will be in the *fundamental Gaussian mode*. However, due to optical imperfections and RF side-bands used to control the interferometer, the light emerging from the AS port contains light in higher order modes. This undesired light is referred to as *junk light*.

To remove junk light, mode cleaners are used. A mode cleaner is a high finesse cavity that strips away higher order mode light and its control sidebands. Higher order modes acquire a different amount of phase as they propagate compared to the fundamental Gaussian mode, an optical cavity can be designed such that the fundamental mode resonates while the higher order modes do not. These higher order modes will be suppressed by a factor of the cavity's finesse. The control sidebands are attenuated if they are outwith the cavities linewidth. The beam transmitted by the mode cleaner will be predominantly in the fundamental mode, and so the signal to noise ratio of the light measured by the photodiode will increase.

2.4.5 Radiation Pressure Noise and the Standard Quantum Limit

Radiation pressure noise is a quantum effect that arises due to the uncertainty in the momentum that light imparts to the test mass when it is reflected from it. These quantum fluctuations will give rise to a differential force on the ETMs, meaning that this noise couples as $1/f^2$ to the measurement of the ETMs' positions. As the force exerted by a beam is $F = \hbar\omega_{EM}N/c = P/c$, the amplitude spectral density of the position of a free mass is

$$\Delta x_{\text{RPN Free mass}} = \frac{\sqrt{\hbar\omega_{EM}P}}{2\sqrt{2}mc\pi^2 f^2}. \quad (2.14)$$

To reduce the effect of radiation pressure noise, the LIGO test masses are 40 kg. The effect of radiation pressure noise on the LIGO detectors' sensitivity curves can be seen in Figure 2.3.

Radiation pressure noise scales with the quantum fluctuations entering the anti-symmetric port whereas shot noise is inversely proportional to these fluctuations. For a laser which produces a coherent state, i.e. a classical laser, the standard quantum limit (SQL) [26–28] sets an optimal the laser power for a given ETM mass which minimises the total quantum noise. The SQL results in a $1/f$ locus that is determined by the mass of the ETM and the duration of the measurement. The sensitivity curve a Michelson interferometer can only reach the SQL at only one frequency.

The sensitivity curve of an interferometer with a signal recycling mirror that is tuned to either signal recycling mode or RSE mode will ‘touch’ the SQL at one frequency; however, for an arbitrary detuning of the SRM, the sensitivity curve of the interferometer will drop below the SQL at some frequencies due to non-linear optical effects [29].

2.4.6 Coating Thermal Noise and other Noise Sources

There are many classical sources of noise to contend with in an gravitational wave interferometer. The most notable of these is the noise associated with the Brownian motion of atoms in amorphous mirror coating layers of the test masses. Current detectors are designed around the expected noise level that can be achieved with these coatings. This will be one of the limiting sources of noise in the A+ detector. Other noise sources include seismic noise from the motion of the Earth, Newtonian noise from pressure waves in earth crust, residual gas pressure noise that alters the refractive index of the arms, noise associated with the interferometer’s electronics and other technical noises.

2.5 Reducing Shot noise with Non-classical states of Light

The two-photon formalism developed by Caves and Schumaker [30,31] can be used to see how non-classical states of light can improve the sensitivity of an interferometer. In this framework, light is analogous to a quantum simple harmonic oscillator (SHO). A SHO can be described by two orthogonal components such as the momentum and position quadrature; for light a convenient basis to use is the *amplitude* and *phase* quadrature. The amplitude and phase quadratures, $\hat{\mathbf{X}}$ and $\hat{\mathbf{Y}}$ respectively, defined in terms of creation and annihilation operators of the electromagnetic field are (N.B repetition with BHD chapter)

$$\hat{\mathbf{X}}(t) = \frac{1}{2} (\hat{a}(t) + \hat{a}^\dagger(t)) , \quad (2.15)$$

$$\hat{\mathbf{Y}}(t) = \frac{1}{2i} (\hat{a}(t) - \hat{a}^\dagger(t)) , \quad (2.16)$$

and from this formalism, an uncertainty relation between phase and amplitude can be derived:

$$\Delta\hat{\mathbf{X}}\Delta\hat{\mathbf{Y}} \geq \frac{1}{2} \quad (2.17)$$

If the uncertainty in one quadrature is reduced, the uncertainty in the other is increased. Such a light field is *squeezed*. The phase between the squeezed field and the detected quadrature is defined as the squeezing angle.

As the gravitational wave signal we are trying to measure is in the phase quadrature of the light, it would be advantageous for, by the uncertainty principle, move noise to the undetected amplitude quadrature from the detected phase quadrature;

this can be achieved by using squeezed light. Squeezed light is created with an optical parametric amplifier; this is a device which generates correlated photons by converting higher energy one photon into two lower energy photons. This squeezed vacuum light is injected into the interferometer from the AS port so that the observed noise within the interferometer is squeezed. This technique has been implemented at LLO [32] and GEO600 (citations).

However, fluctuations in the amplitude quadrature of the light perturb the position of the test masses; thus, they couple to the signal at the antisymmetric port. During LIGO’s third observation run, the squeezing angle had to be tuned to prevent radiation pressure noise limiting the BNS range of the detector [32].

To infer the quantum state of a light field, a technique known as *balanced homodyne detection* (BHD) can be used to observe one quadrature of the light field. By combining the signal light field that is to be measured with a much stronger *local oscillator* light field on a beam splitter, the quantum state of the signal light field is encoded in the the intensity of the combined beam. The relative phase between the signal and local oscillator determines the quadrature of the quantum state that will be observed. Chapter 5.1 explores this in more detail.

2.6 Interferometry Techniques to beat Quantum Limits

Squeezed light and the natural correlations, called ponderomotive squeezing [27], created in the interferometer can be used to surpass the SQL at some frequencies.

This has been demonstrated the LIGO Livingston detector [33]. However, this technique results in noise at other frequencies being enhanced.

To avoid the increase in radiation pressure noise that is experienced when squeezed light is used, a frequency-dependent squeezer can be used. The light fields squeezing angle can be rotated as a function of frequency. This is done by using a high finesse detuned filter cavity. At frequencies below the linewidth of the cavity, the light reflected from the filter cavity will experience a phase shift and so the squeezing angle will be rotated; in contrast, at frequencies above the linewidth, the light incident upon the cavity is reflected without experiencing any phase shift and so the squeezing angle is not rotated. At low frequencies, the amplitude quadrature can be squeezed and the radiation pressure noise can be decreased. At high frequency, the phase quadrature can be squeezed and the shot noise can be decreased. This will be part of the A+ detector.

Variational readout is a technique where quantum noise due to radiation pressure is removed from the measurement by using a frequency dependent homodyne angle [27]. This is realised in a similar way to frequency dependent squeezing; the AS beam is coupled to a filter cavity and thus the detected quadrature of the beam emerging from this cavity is frequency dependent. This beam is then measured on photodiodes using a conventional BHD scheme. Variational readout schemes are very sensitive to losses within the filter cavity, and so this technology is not mature enough to be implemented into current gravitational wave detectors.

To beat the bandwidth limit, energy has to be injected into the system. White light cavities as a means of increasing the detector bandwidth have been studied

(cite Haixing), although these solutions have yet to be demonstrated in the lab. Alternatively, PT-symmetry schemes may be used to beat the bandwidth limit, again these have yet to be demonstrated in the lab.

There are several sub-SQL schemes which avoid the quantum noise problems related to making a position measurement of the test masses. If a momentum measurement of the test masses could be made, then a random impartation of ‘position’ would not affect future measurements of the momentum. Such measurements are known as quantum non-demolition. As a speed measurement is being made, the displacement noise would rise as $1/f$ instead of $1/f^2$.

2.7 The Effect Losses have on Measuring Quantum States of Light

To preserve the benefits gained by using quantum states of light, it is essential to have minimal losses in the read-out of a gravitational wave detector. Real optical components have physical attributes that result in incident light power being wasted. These mechanisms are known as losses, and examples of these include absorption and scatter. A lossy optical component can be modelled as an ideal one downstream of a beam splitter whose splitting ratio represents the loss. The incoming enters side of the beam splitter, some light is directed to the ideal component and some gets lost. In the remaining input to the beam splitter, vacuum fluctuations can enter. Some of these fluctuations will make it to the ideal component, and so the state of the light that is detected is contaminated with extra vacuum fluctuations compared to if there was no loss [34].

Therefore, it is critical that the photodiode used to measure the AS beam has low loss. Photodiodes convert photons into electrons, ideally with a one-to-one ratio. The measure of how well this is done is the photodiode's *quantum efficiency*, η . The factor by which the shot noise increases, ϵ , due to a simple loss mechanism, such as a non-ideal photodiode, is

$$\epsilon = \sqrt{\frac{1}{\eta} - 1}. \quad (2.18)$$

It is acceptable in a gravitational wave interferometer to have a quantum efficiency of at least 99%. This can be achieved by photodiodes fabricated from InGaAs.

Mode matching losses increase have a greater contribution to the total shot noise compared to losses of the same size generated by a simpler mechanism (cite lee's work). This source of loss can be reduced through careful design of the optical layout. Mirrors with a tunable radius of curvature can be used to ensure that there is minimal mode matching loss.

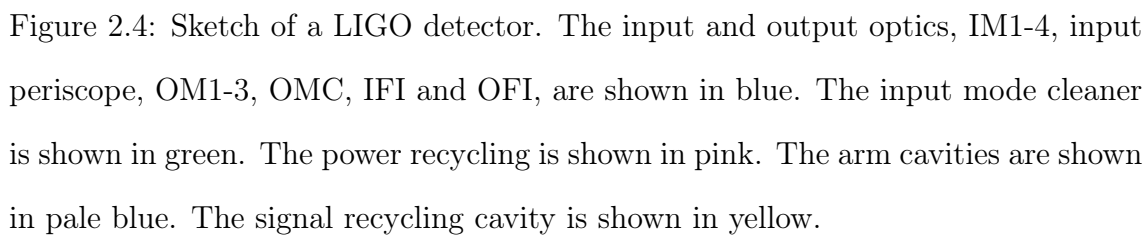
2.8 Overview of the LIGO detector

The LIGO detector is a dual-recycled-Fabry-Perot Michelson interferometer with 4 km arms. The layout of the detector is shown in Figure 2.4 and the noise spectral density is shown in Figure 2.3. The term dual-recycled refers to the fact that the interferometer uses both power and, the mathematical dual of power recycling, signal recycling. The interferometer is housed within a vacuum system comprised of several horizontal access modules (HAM) and beam splitter chambers (BSC). Figure 2.4 includes all the optics from the pre-stabilised-laser (PSL) to the readout photodiodes that the main beam interacts with.

The first cavity after the PSL is the input mode cleaner; this provides a beam with high spatial purity to the interferometer and is used in the initial locking sequence as a stable reference frequency. The power recycling cavity features two extra mirrors in addition to the PRM known as PR2 and PR3. The role of these mirrors is to make round trip gouy phase sufficient so that it is stable; the signal recycling cavity also has three mirrors for the same reason. The beam passes through the output Faraday isolator (OFI), and is steered into the output mode cleaner. The gravitational signal is encoded on the beam transmitted by the OMC, so photodiode monitor the light transmitted by the OMC.

2.9 The future of Ground based Gravitational Wave Detectors

The next generation of ground based gravitational wave detectors include the Einstein telescope and LIGO cosmic explorer. These interferometers will be ten times larger than the current generation of interferometers and will be able to test the no-hair theorem, test alternative theories of gravity, test models of string theory, measure objects with large redshifts to learn about the formation of cosmological structures, measure closer to the big bang than is possible with electromagnetic telescopes, detect cosmic strings, probe the physics of neutron stars and measure what happens during a supernova. As well as this, the third generation of gravitational wave detectors may measure signals from exotic signals which have not yet been theorised.



As the limiting sources of of current gravitational wave detectors is the coating thermal noise, research into cryogenic mirror coatings is underway. Fused silica is opaque at these frequencies and does the noise characteristics do not improve when cooled, so silicon test masses are being developed.

It is expected that low noise, low loss coatings will not be compatible with 1 μm light, so these detectors will operate with either a 1.5 μm or 2 μm wavelength laser. Laser technology for 2 μm light is still in its infancy, and so there is lots of uncertainty in the practicality of using 2 μm as the operating wavelength of a gravitational wave detector; for instance, low noise, high quantum efficiency photodiodes that can detect 2 μm light have never been fabricated.

2.10 Conclusion

Further detections of gravitational waves will lead to a better understanding of the universe. To reach the design sensitivity of the LIGO detectors, among other upgrades, the input laser power needs increasing. To further lower the LIGO detectors noise floor, the A+ upgrade will utilise frequency dependent squeezed light. In order for this to be successfully implemented, the quantum state of the light field must be accurately detected with a balanced homodyne readout and losses in the read-out must be minimal.

As there is potential to reduce the noise floor that will limit the A+ detector by using cryogenic mirror coatings that are compatible with 2 μm light, future detectors may use a 2 μm laser; this leads to the question: is laser technology at this wavelength sufficiently mature enough to enable quantum limited measurements? Balanced

homodyne detection and photodiodes with high quantum efficiency and low dark noise will be required for any future interferometer.

Chapter 3

A Prototype High Power Laser for LIGO's Fourth Observation Run

3.1 Introduction

The LIGO detectors require high input power, nominally 125 W at 1064 nm [35], to decrease the shot noise which limits its sensitivity at frequencies above 100 Hz. The current laser cannot reach the required power without generating excess noise. In this chapter, the new laser layout is discussed. The PMC throughput power, beam quality, intensity noise, pointing noise and frequency noise of the prototype laser were measured. It was found that this layout provided a suitably low noise laser. This laser was able to produce 100 W of intensity stabilised light. While the 125 W nominal power level was not obtained, this layout provides more power than the current laser. With additional work in optimising the layout, the maximum output power of this laser may be increased.

If a laser is in a coherent state, the shot noise associated with detecting the laser is $\sigma = \sqrt{\frac{hc}{\lambda}P}$, so the signal to noise ratio at high frequencies scales with $\sqrt{P}$. Alternatively, to decrease the shot noise limit the wavelength could be decreased; however, the LIGO detectors operate at 1064 nm due to the availability of mirror coatings with low absorption and low thermal noise. Future interferometers may operate at longer wavelengths due to the potential reduction of mirror coating thermal noise that can be achieved at longer wavelengths [36].

In the field of gravitational wave detection, high power refers to lasers with powers of the order 100 W. High power is a loosely defined term as lasers ranging from 100 W to 100 kW are defined as high power in the literature [37–47]. As the power is increased, other properties of the laser tend to be degraded. It needs to operate uninterrupted on year-long timescales; this poses a major engineering challenge.

The laser for LIGO must be continuous wave, linearly polarised, have high spatial purity, low intensity noise and low frequency noise. These requirements are detailed in [35]. To overcome the technical challenges that come with the high power and low noise requirements for the LIGO laser, the master-oscillator-power-amplifier (MOPA) topology was selected. This type of laser is based on a high performance seed laser which is then amplified. As amplifiers are linear and coherent, the low noise properties of the seed laser characterises the noise of the high power beam.

LIGO’s original laser used a high power oscillator (HPO) as the means for obtaining the target power. The HPO was a ring cavity containing laser crystals that was seeded by a 35 W low noise laser. This design went through extensive testing and refinement at the AEI [48] and was installed as part of LIGO’s pre-stabilised-laser

(PSL). However, due to jitter noise from the water cooling of the crystals within the HPO coupling to the differential arm motion signal (DARM), as well as one HPO being destroyed by accident, the HPO was bypassed and the LIGO detectors operated using the 35 W laser [49].

To reach higher powers during LIGO’s third observation run, instead of the HPO, a neoVAN-4S single-pass amplifier was used. This stepped up the laser power from 35 W to 70 W [50, 51]. The next stage of development was a complete redesign of the LIGO laser based on using two upgraded versions of the neoVAN-4S. The neoVAN-4S-HP amplifier showed promising results [46, 47], so a prototype laser was constructed and tested at LIGO Livingston. This design will be installed as the new PSL for the start of LIGO’s fourth observation run [49].

3.2 Concept for LIGO’s New Laser

LIGO will use a MOPA laser that is seeded by a non-planar ring oscillator (NPRO). The NPRO beam will be amplified by two single-pass amplifiers in series and be followed by a mode cleaner cavity to produce a low-noise high-power laser in the fundamental mode. This concept is shown in Figure 3.1. As the NPRO has good free running noise characteristics the amplified beam will as well. Results from a setup using a single neoVAN-4S-HP showed these amplifiers can be continuously operated for many tens of days [47]. This led to the proposal to construct the PSL out of two of these in series. A similar set up using a 35 W seed laser and two neoVAN-4S-HPs was able to generate 195 W of light [46] for nearly 80 days with no significant drops in power.

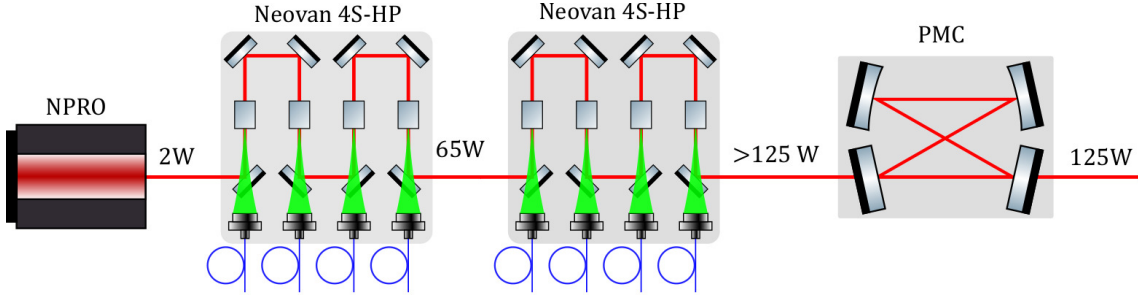


Figure 3.1: Conceptually, the PSL will be made up of four components: the NRPO, two neoVAN-4S-HP amplifiers and the PMC cavity. The NPRO provides seed light with good noise characteristics. These noise characteristics are preserved by the amplifiers, so the resulting beam will have high power and low noise. The PMC filters junk light and RF intensity noise so a spatially pure mode can be sent to the interferometer.

3.2.1 Non-Planar Ring Oscillators

The first element of the PSL is a non-planar ring oscillator [52–54]. The NPRO provides a low noise laser making it a suitable seed for a MOPA system. This is a single frequency laser constructed from one Nd:YAG crystal which acts as both the gain medium and the cavity, thus it produces a laser with a high amount of frequency stability. The NPRO used was a Mephisto, built by Coherent, and it can produce to 2W of 1064 nm light.

3.2.2 The neoVAN-4S-HP Solid-State Amplifier

To amplify the NPRO beam, two neoVAN-4S-HP amplifiers will be used. The neoVAN-4S-HP is a single-pass, solid-state amplifier manufactured by Neolase. It is constructed from four Nd : YVO₄ crystals which are pumped by fibre coupled,

low noise laser diodes. It should be seeded by s-polarised light at 1064 nm. These gain mediums are in contact with a metal plate which is cooled. This keeps jitter noise low as flowing water is not directly in contact with the gain medium. As these amplifiers have low noise pump-diodes and will be seeded by an NPRO, the noise of the light produced by them should be determined by the NPRO.

3.2.3 The Pre Mode Cleaner Cavity

The pre mode cleaner (PMC), see Figure 3.2, is used as the hand-off point between the PSL and the rest of the interferometer. Light transmitted by this cavity is used as the input beam for the interferometer. The PMC is a bow tie cavity with an FSR of 9 MHz, thus it passively filters RF intensity noise. Control sidebands are provided after the PMC but before the IMC, meaning that the IMC cannot be used to provide this stabilisation. The PMC is locked to the fundamental mode so the beam that is delivered to the rest of the interferometer will have a low amount of higher order mode content. This is important as the amplifiers produce light with spatial defects. The PMC has been redesigned [55] to prevent contamination due to glue on its mirrors and to make maintaining the PMC easier. The PMC design no longer includes a vacuum enclosure.

3.3 Layout of the Prototype Laser System

An optical layout based on the concept described in Section 3.2 was constructed. A sketch of the layout is shown in Figure 3.2. In addition to the main beam path, there are low power diagnostic beams after each amplifier that can be used to characterise the laser. The prototype layout did not include the tabletop frequency stabilisation

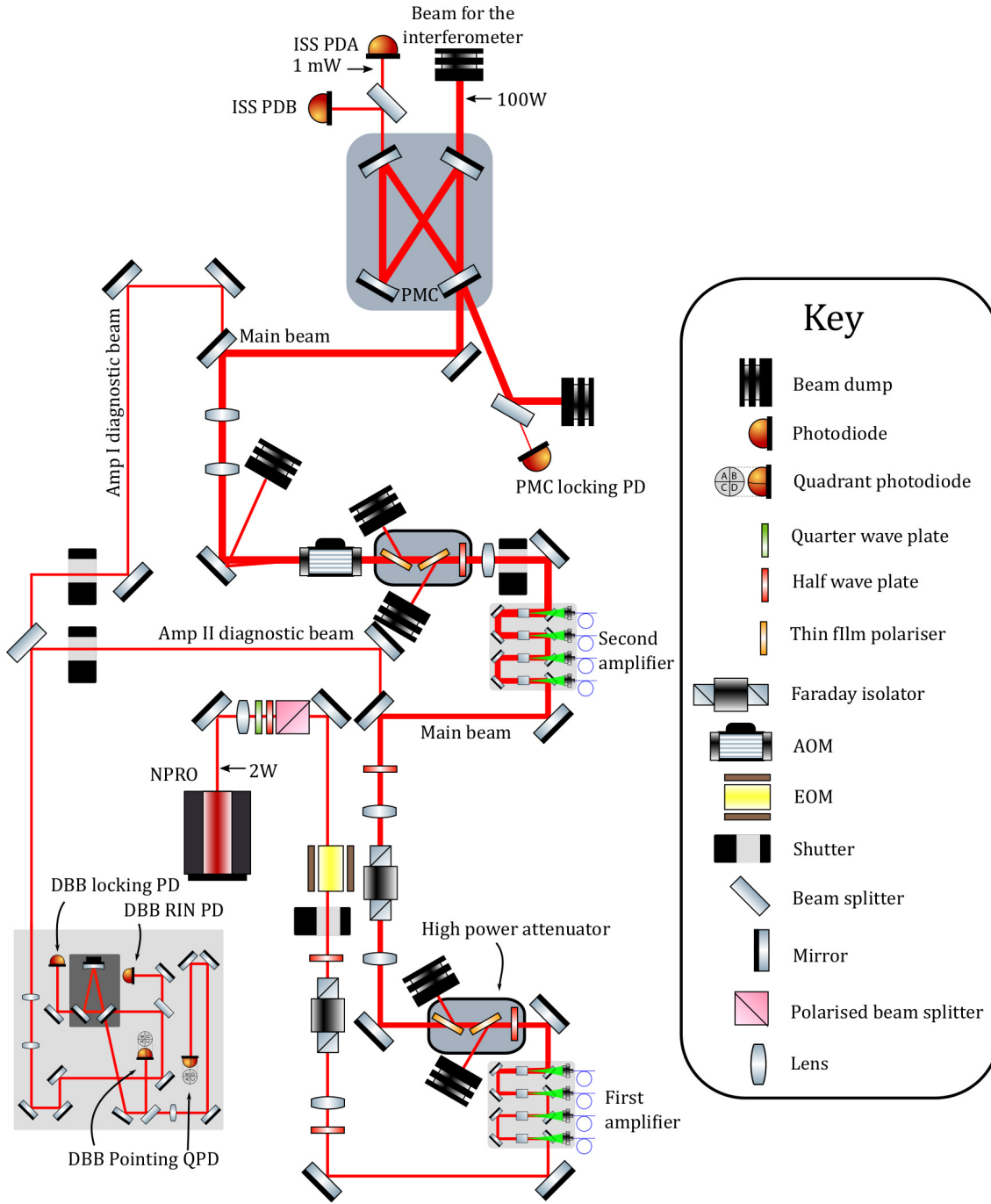


Figure 3.2: Sketch of the prototype optical layout. The main and diagnostic beam paths are shown. This sketch shows all the optics in the main beam path as well as the pick-off points for the diagnostic beams.

servo (TTFSS) so this has not been included in the layout description. However, there is space for the TTFSS on the optical table.

The NPRO beam passes through some optics to clean up the polarisation. Transmission through a steering mirror is used to monitor the power of the NPRO. An electro-optic modulator (EOM) is used to produce 35.5 MHz sidebands for the Pound-Drever-Hall (PDH) scheme used to lock the PMC. A Faraday isolator ensures light reflected back from the amplifiers does not damage the NPRO. The beam passes through a lens to mode match the beam to the first amplifier. Two mirrors are used to align the NPRO beam to the amplifier. A half-wave plate ensures the polarisation into the first amplifier is set correctly. These optics are shown in Figure 3.3.

After the first amplifier, the beam passes through a high power attenuation stage, which consists of a half wave-plate and two thin film polarisers (TFP), to allow for safe set up and low power operation. When setting up this layout, the amplifiers must be run at maximum power since they exhibit significant thermal lensing. The high power attenuators are not envisioned being used during observation runs. The laser is then directed through a Faraday isolator and wave-plate before reaching the second amplifier. The mode matching lenses for the second amplifier are on either side of the Faraday isolator. Two mirrors immediately before the second amplifier are used for its alignment. The transmission through one of the steering mirrors before the second amplifier is used for monitoring the beam produced by the first amplifier. This section of the layout is shown in Figure 3.3.

After the second amplifier, the beam goes through a high power attenuation stage before passing through an acousto-optic modulator (AOM). This AOM is used for

controlling the power of the laser. Downstream of the AOM, a pick-off beam is used to measure the properties of the laser produced by the second amplifier. Before entering the PMC, the beam is transmitted through a TFP so that the beam is set to be p-polarised. This part of the layout can be seen in Figure 3.4.

The beam produced by the second amplifier is then coupled to the PMC; this requires two mode matching lenses and two steering mirrors. There are three beams transmitted by the PMC: a high power one, which would be sent to the interferometer, and two low power ones for stabilising the frequency and intensity of the laser. The low power beams contain of the order 100 mW of light. The intensity stabilisation servo (ISS) inner-loop photodiodes are in a light-tight enclosure. One of the photodiodes is used as the in-loop sensor while the other is used as the out-of-loop sensor. The PMC is shown in Figure 3.5. The reflected light is used to lock the PMC; this path can be seen in Figure 3.4.

The diagnostic bread board (DBB) [56,57] is integrated into the layout via several pick-off points and can be used to monitor the status of both amplifiers. The DBB enables measurements of intensity noise, frequency noise, beam pointing noise and higher order mode content. The DBB is shown in Figure 3.6. The DBB consists of a cavity, motorised lenses, an automatic alignment system and several photodiodes. Despite having motorised lens mounts in the DBB, static mode matching lenses are required in the diagnostic beam paths. To characterise the NPRO beam, mirrors on magnetic mounts can be inserted into the layout.

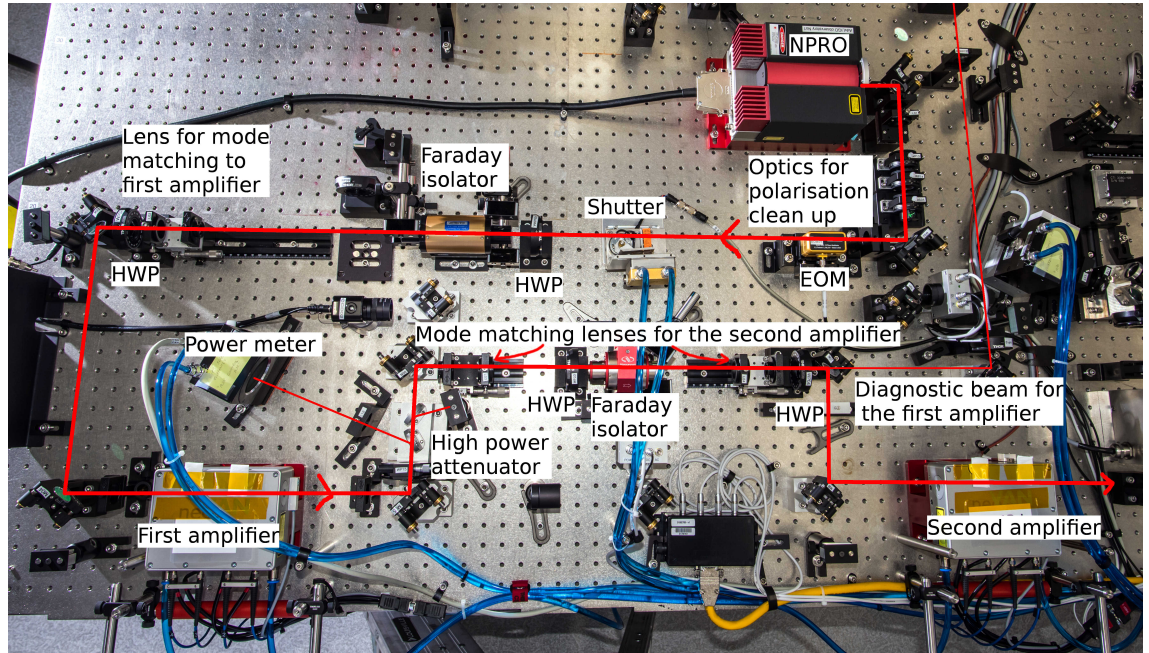


Figure 3.3: Photograph of the PSL prototype showing the beam path (red lines) from the NPRO (top right) to the second amplifier (bottom right). The beam path up to the first amplifier includes optics for setting the polarisation of the beam, an EOM, a Faraday isolator and lenses for mode matching. After the first amplifier, the beam passes through a high power attenuator, mode matching lenses and a Faraday isolator. The first amplifier (bottom left) is monitored using a low power pick-off beam (centre right).

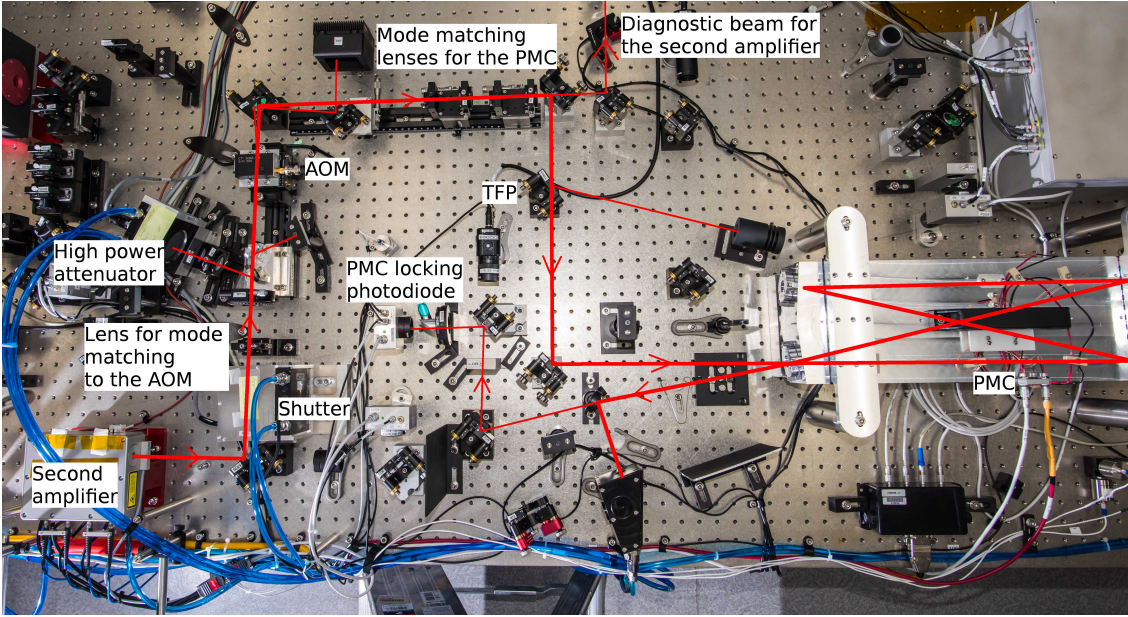


Figure 3.4: Photograph of the layout showing the beam path (red lines) from the second amplifier (bottom left) to the input of the PMC (centre right). This beam path includes a high power attenuator, an AOM, mode matching lenses for the PMC and a TFP. The PMC can be seen more clearly in Figure 3.5. The path taken by light reflected from the PMC is shown. This light is used to lock the PMC. A low power beam for monitoring the second amplifier is obtained using the transmission through a steering mirror (centre top) for the alignment to the PMC.

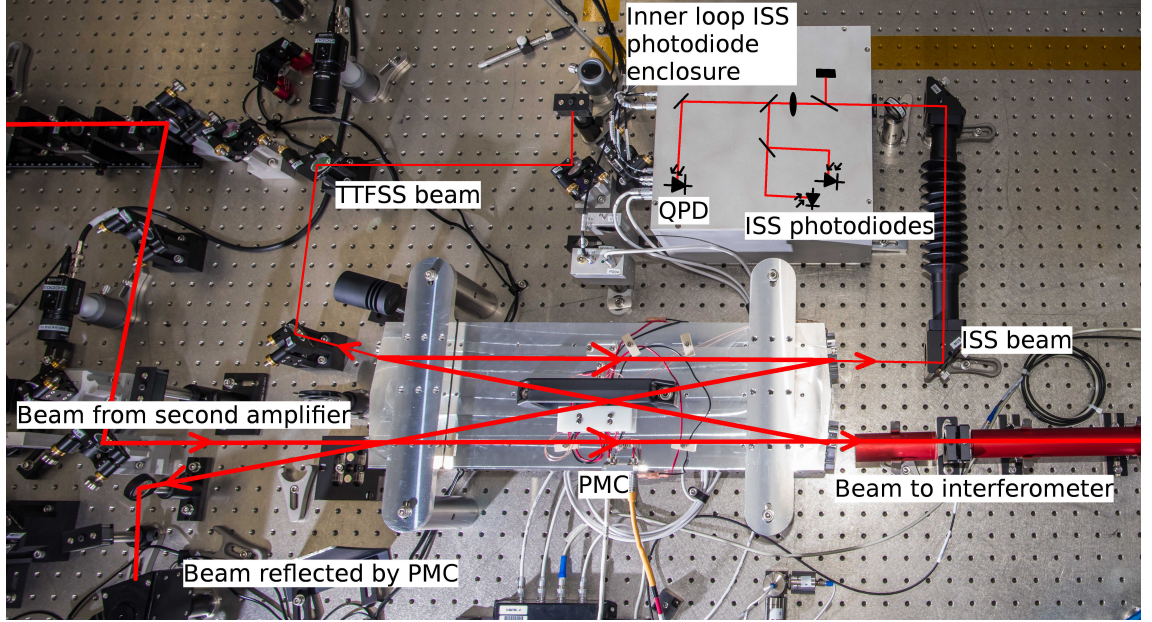


Figure 3.5: Photograph of the PMC (centre) and the inner ISS photodiode box (top right). The beam (shown in red) from the second amplifier enters the PMC. The majority of this light is sent to the interferometer. A small fraction of the light in transmission of the PMC is used to stabilise the intensity and frequency of the laser. These beams are labelled as the ISS beam and the TTFSS beam. The ISS beam is directed into a light tight enclosure. This enclosure contains two photodiodes which are for measuring the intensity noise of the laser. One of these is an in-loop sensor; the other is an out-of-loop sensor. A QPD can be used to monitor the alignment of the ISS photodiodes. The reflected beam from the PMC is made up of the junk light in the beam from the second amplifier.

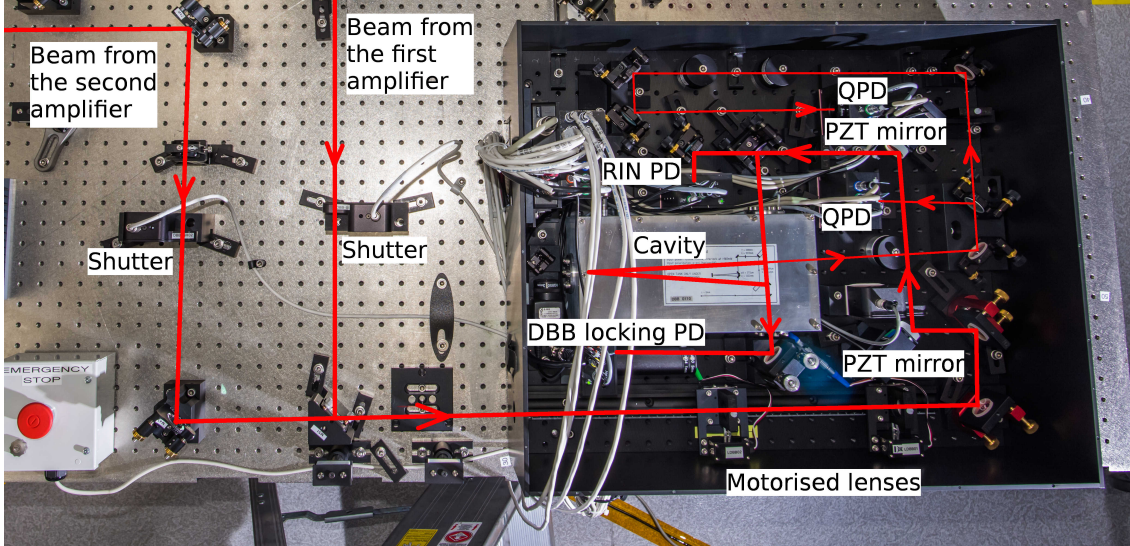


Figure 3.6: Photograph of the DBB with the beam path highlighted in red. Low power beams are used to measure properties of the laser produced by both amplifiers. The shutters control which amplifier will be characterised. The DBB contains motorised lenses and PZT mirrors so that remote mode matching and automatic alignment can be performed. The intensity noise of the beam can be measured with the RIN photodiode. The DBB locking photodiode is used to lock the cavity to the beam. This photodiode is also used during mode scans to measure the higher order content of the beam. QPDs separated by 90° of Gouy phase are used to measure the pointing noise of the incoming beam.

3.4 The Power Transmitted by the PMC

The PSL should deliver 125 W to the interferometer [35]. The power transmitted through the PMC is determined by the power in the beam generated by the amplifiers and the amplified beam's distortion. The beam's distortion is affected by the input alignment, mode matching and thermal lensing. Increasing the pump-diode current increases the power that is produced by the amplifier as well as the amount of thermal lensing. The flow rate of coolant can be increased to mitigate thermal lensing; however, this can generate jitter.

The prototype layout was able to deliver a maximum of 111 W through the PMC during a 55 hour trial run. This is shown in Figure 3.7. The power was measured with recently calibrated power meters. The power emerging from the PMC dropped to 107 W during the course of the trial run, while the power incident onto it remained ~ 125 W. Changes in the lab temperature and humidity caused slight alignment changes. This resulted in the power decreasing. As the laser will be in a controlled environment, this effect will be less severe when it is installed.

To increase the output power, the mode matching to the PMC could be improved and the amplified beam's distortion could be reduced. Mode matching between the final amplifier and the PMC could be improved at the percent level. Increasing the flow rate would reduce the beam's distortion and may allow for higher pump-diode currents, however concerns of jitter noise meant that the flow rate was kept low. With higher flow rates, a beam with 137W of power and a higher mode content of 4.6% was produced by the amplifiers.

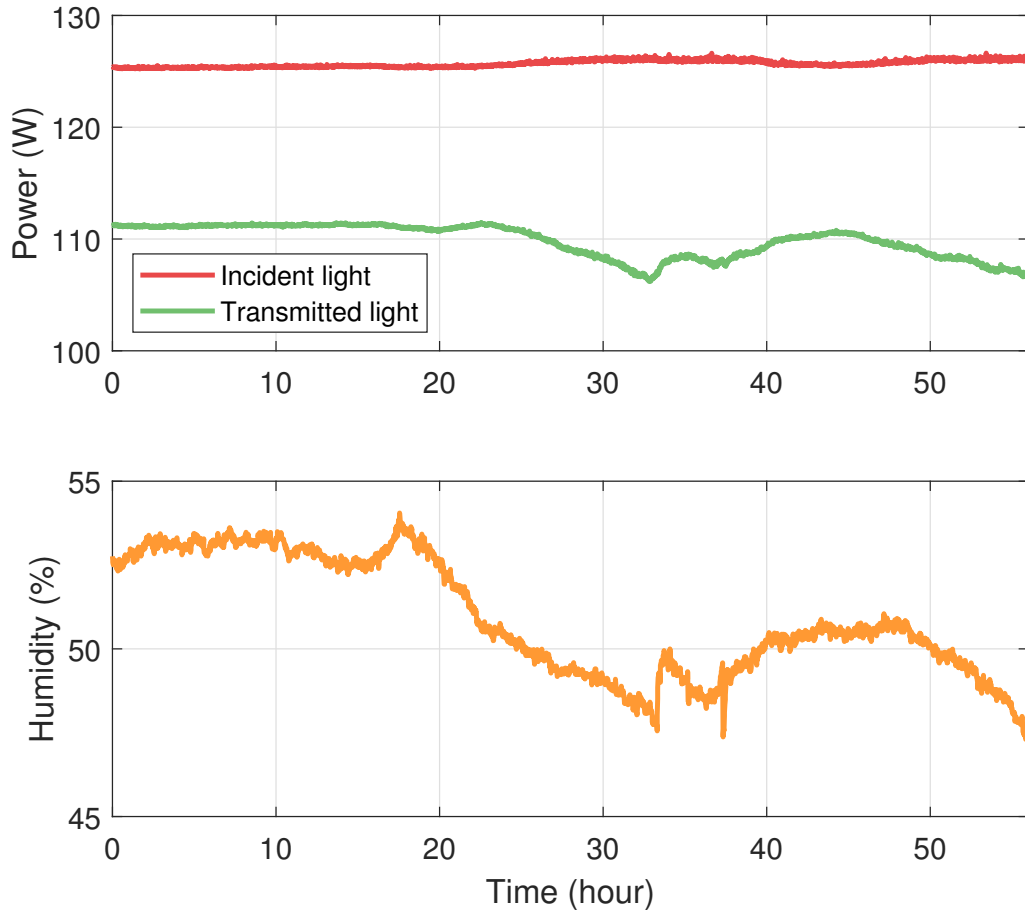


Figure 3.7: The top panel shows the power that was incident on the PMC and the power emerging from the PMC over a 55 hour trail run. The light produced by the amplifiers did not change significantly for the duration of this measurement, however the transmitted light through the PMC dropped from 111 W to, at its lowest, 107 W. This is likely due to the environmental conditions in the lab changing. The bottom panel shows the humidity of the lab. Changes in the humidity, which can be caused by a temperature change, coincide with changes in the power emerging from the PMC.

To actuate on the laser’s power, some of the incident light will need to be ‘dumped’ to provide headroom for the actuator to increase the beam’s power. This means that the full power will be reduced by $\sim 10\text{ W}$ when the intensity stabilisation is implemented.

Long term measurements should be taken, however the results in [46] show these amplifiers can be run continuously for at least 80 days. LIGO Hanford measured the input laser power dropping from 55 W to 52 W over the course of one year (cite Nina’s new paper); this is likely due to mode mismatches as a result of changes in the set-up’s thermal state.

3.4.1 The Effect of Cooling on the Amplifier’s Power and Beam Quality

As the quality of the beam emerging from an amplifier is affected by the temperature of the amplifier’s head, an experiment was performed to determine a suitable temperature to run the amplifiers at. Due to the jitter noise witnessed at the Hanford detector, the flow rates were set as low as possible before the amount of power in the fundamental mode dropped significantly. As the Nd : YVO₄ crystals are cooled non-uniformly, the thermal lens is non-circular. For small amounts of thermal lensing, the beam will become slightly astigmatic.

The cooling layout that was used to find a suitable amplifier head temperature to operate with is shown in Figure 3.8. The diode box and amplifier head were connected in parallel, and each amplifier was on a separate cooling loop so that the amplifier head temperatures could be independently controlled. The flow rate to the

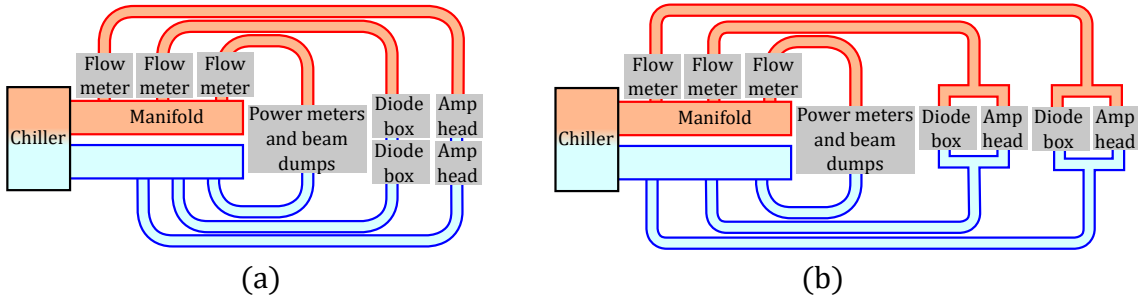


Figure 3.8: Cartoons of water cooling layouts for the prototype layout. Panel (a): The laser will be run with the amplifier heads cooled in series with each other to reduce jitter. Panel (b): To perform the amplifier head temperature experiment, the water cooling was arranged so that the amplifiers temperature could be independently controlled.

first amplifier was kept high to a high quality input beam for the second amplifier. The HOM content for the beam entering the second amplifier was 3.8%. The second amplifier and diode box flow rate was decreased and the temperature of the diode head was monitored. The higher order mode content was measured with the DBB by averaging over sixty mode scans. The power was measured with a high-power power meter.

The change in beam quality and output power as a function of amplifier temperature is shown in Figure 3.9. At 22.3°C the higher order mode content was $\sim 4.5\%$ and the amount of power in the fundamental mode was ~ 127 W. Figure 3.10 shows that changes in the second order mode make up the majority of the HOM content change. This is due to the beam becoming more astigmatic as the amplifier head heats up. It was decided that to keep the amount of power in the fundamental mode high, the diode heads would be operated such that their temperature did not

rise above $\sim 27^\circ\text{C}$.

To be able to keep the flow rate as low as possible on the table, the water cooling was set up so that the amplifiers heads were in series with each other. The diode boxes were on another loop to the heads. The cooling scheme that will be used is shown in Figure 3.8. Meters which monitor the flow rate of the coolant will be placed on the return side of the system rather than the supply side. By having the flow meters on the return side, they will measure a drop in flow if there is a leak, allowing for the water supply to be cut.

3.4.2 Amplifier Alignment Procedure

To obtain high beam power and low higher order mode content, the alignment of the input beam to the amplifier needs to be optimised. The output power of the amplifier may increase at the expense of higher order mode content, so the beam profile and the beam power must be monitored when aligning the amplifier. Optimising the alignment to the amplifier is an iterative process, so with each increase in power a small change in the amplifier's alignment is required.

Initially, the amplifier was switched off. This protects it against damage as well as preventing the possibility of stray light beams with several tens of Watts of power being in the lab. The beam path between the NPRO and the exit of the first amplifier were inspected to make sure the beam path was clear and terminated with dumps.

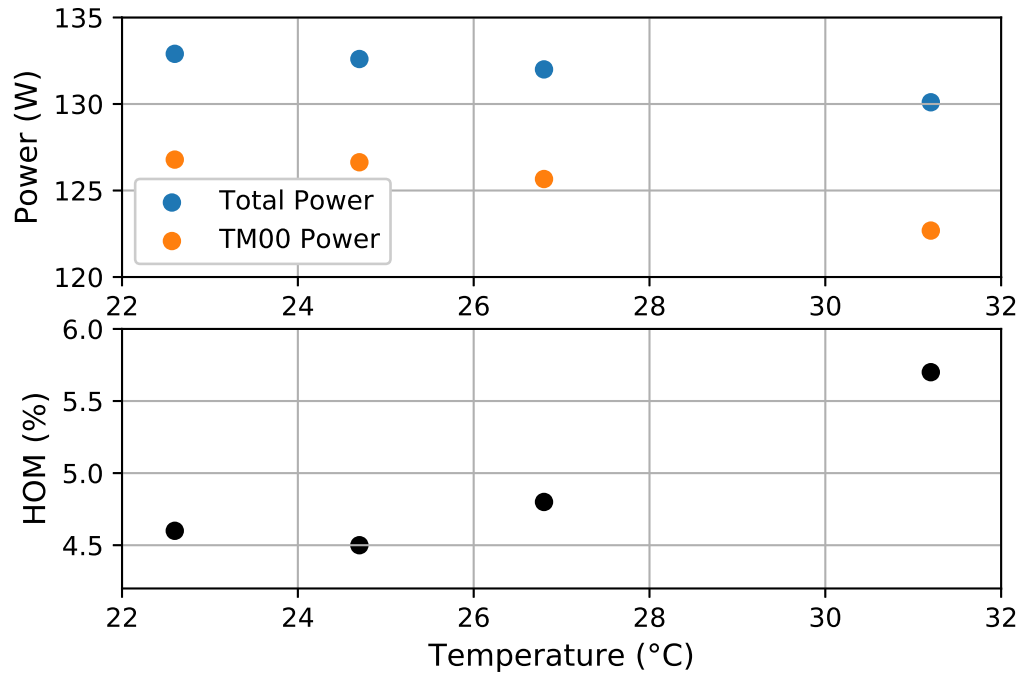


Figure 3.9: The top panel shows that as the second amplifier head's temperature increases the power produced by it decreases. The bottom panel shows that the higher order mode content increases as the temperature of the amplifier increases. This means that the power in the fundamental mode decreases as the temperature decreases due to two effects: less overall power and worse beam quality.

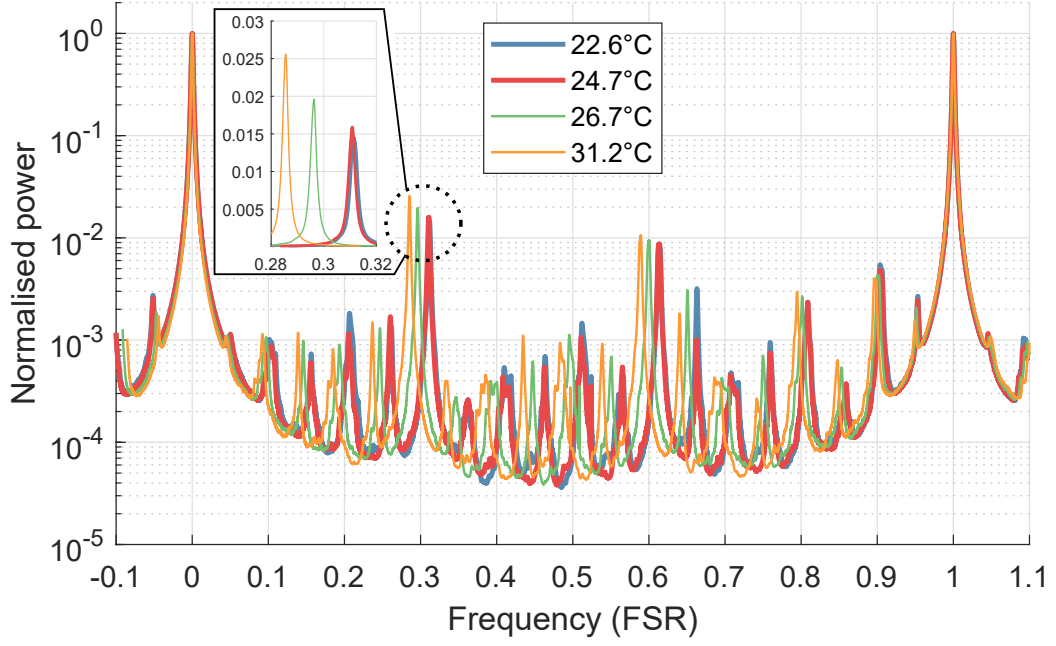


Figure 3.10: Mode scans were performed with the DBB for each amplifier head temperature. The second order mode for this cavity is located at ~ 0.3 FSR; its exact location drifts due to thermal expansion of the cavity. As the temperature of the amplifier head changes, the second order mode content of the beam changes at the percent level. This second order mode height is shown in the inset. The second order mode changes due to the beam's astigmatism changing.

Next, the NPRO was set to produce 1.3W of light. The power of the light was measured on the input side of the amplifier. The beam was roughly aligned so that the spot would hit the centre of the amplifier's input aperture using the two mirrors that immediately precede the amplifier. The power meter was placed on the exit side of the amplifier, and the input alignment was adjusted until the power emerging from the amplifier was the same as the input power. This indicates that the beam will be roughly aligned and some amplification will occur when the pump-diodes are turned on. The high power attenuator immediately after the amplifier was set to dump all the light. The initial mode matching was based on the manufactures specifications, and the input light was set to s-polarisation.

The pump-diode currents were increased until the amplifier 'switched on'. This was at 2 A of current through each of the pump-diodes. Amplification begins when there is sufficient pump light to cause enough population inversion that stimulated emission occurs. At this threshold, the output power was maximised by optimising the alignment. Then, the mode matching was adjusted in tandem with correcting for mis-alignments caused by non-perfect centring of the beam on the mode matching lens.

A fast low-power power meter was used to monitor the beam transmitted through the high power attenuator. Compared to the high-power meter, the low-power meter drifted less and responded to changes in power faster. This gave quicker feedback and allowed for easier alignment.

A Wincam was set up to monitor the beam profile. If the amplifier is significantly misaligned and mode mismatched, the beam profile will become distorted. Large

misalignment will cause the beam profile to look lopsided, and smaller misalignments will cause the beam profile to have shoulders.

Once the amplifier had been optimised for a low pump-diode current, the pump-diode current was incremented in small steps of roughly 1-2A. Each time this was done, the thermal lensing increased. Due to the non-uniform way in which the heating occurred, the input alignment was re-optimised each time as was the input mode matching. By going in small steps of diode current, the optimal alignment was safely achieved. The amplifier should be operated at high diode currents, $\sim 9\text{A}$, so at low currents it was sufficient to be within 1% of optimal.

To determine when the alignment was optimal, mode scans with the DBB were performed. Due to thermal lensing, the beam will be slightly astigmatic and so optimising the mode matching from the amplifier to the DBB was difficult. When optimising the pump-diode currents, the thermal lensing changes, so the mode matching to the DBB needs to be corrected for this.

3.5 Intensity Stabilisation of a 100 W Beam

The free running-intensity noise of the laser exceeds the level needed for the interferometer to reach its design sensitivity. Therefore, the laser must be stabilised. As actuators on the lasers intensity, a current shunt and an AOM were compared. Using the current shunt rather than the AOM enabled 10 W more power to be obtained in the beam emerging from the PMC, however the behaviour of the current shunt was highly dependent on the DC pump-diode current at 10 Hz where the laser noise requirement is most strict. If the amplifiers were required to operate under different

circumstances during the course of an observation run, the current shunt may prove to be highly unreliable. In contrast, the AOM has proved reliable during previous observation runs. While the AOM causes the beam to have less DC power, this may not be a problem for O4 as it is unlikely that the full 125 W will be used due to point defects on the test masses.

3.5.1 The Concept and Requirements for Intensity Stabilisation Servo

The intensity noise requirement for LIGO's laser is found in [35] and reproduced in Figure 3.11. At low frequencies, the requirement is driven by the need for the intensity noise to be less than the radiation pressure noise induced by the difference between the arm powers. At 10 Hz, the laser must be stabilised to a level of $\sim 1 \times 10^{-9} \frac{1}{\sqrt{\text{Hz}}}$. Above 10 Hz, the requirement on the intensity noise is relaxed as there is passive filtering of the light by the PRC. The light is passively stabilised by the PMC to reduce the intensity noise of the laser at RF frequencies.

The active intensity stabilisation of the laser is done in two stages; this is shown in Figure 3.12. First, the laser is stabilised by the inner loop to a level required to lock the IMC. The light used to measure the intensity noise for the inner loop is a low power beam that is transmitted by the PMC. There is ~ 1 mW of light in this beam, so the laser cannot be stabilised beyond the shot noise associated with a coherent state. For ~ 1 mW this is $\sim \frac{2 \times 10^{-8}}{\sqrt{\text{Hz}}}$. Once the IMC is locked, a pick-off from the beam entering the interferometer is used to measure the intensity noise of the laser. This beam has enough power, ~ 350 mW, to reach the level required, $\frac{1 \times 10^{-9}}{\sqrt{\text{Hz}}}$, during observation mode. The noise of the beam entering the interferometer must

be measured just after the IMC as this contains the intensity noise which cannot be measured by the Inner-loop diode e.g. the noise due to the periscope immediately after the PMC.

3.5.2 Free Running Noise of the Prototype Laser

To determine the open-loop gain needed for the servo to suppress the noise to the requirement, the free-running noise of the laser must be measured. The noise at different points along the beam path was measured to see the contribution of the NPRO, each amplifier and the PMC to the total noise.

The power noise of the NPRO was measured using the DBB RIN photodiode (Figure 3.13). The NPRO had a typical noise level for this type of laser [54]. This was recorded via CDS, so the measurements near 10 kHz are affected by the anti-aliasing filter. The noise between 100Hz and 1kHz had features associated with the motion of the table and mirror mounts.

The power noise of the first and second amplifiers was measured using the DBB RIN photodiode (Figure 3.13). The noise was measured with a spectrum analyser rather than CDS so that measurements up to 100kHz could be made. As the amplifiers were pumped with low noise diodes and were operating in their saturated regimes, the RIN is expected to decrease after each amplifier. When an amplifier is saturated, the amplifier will not behave linearly and instead will add a fixed amount of input power regardless of its seed power. As expected, at frequencies above 10kHz the RIN after the second amplifier was less than the RIN after the first amplifier. There was a peak at 20Hz; this can be attributed to the air conditioning.

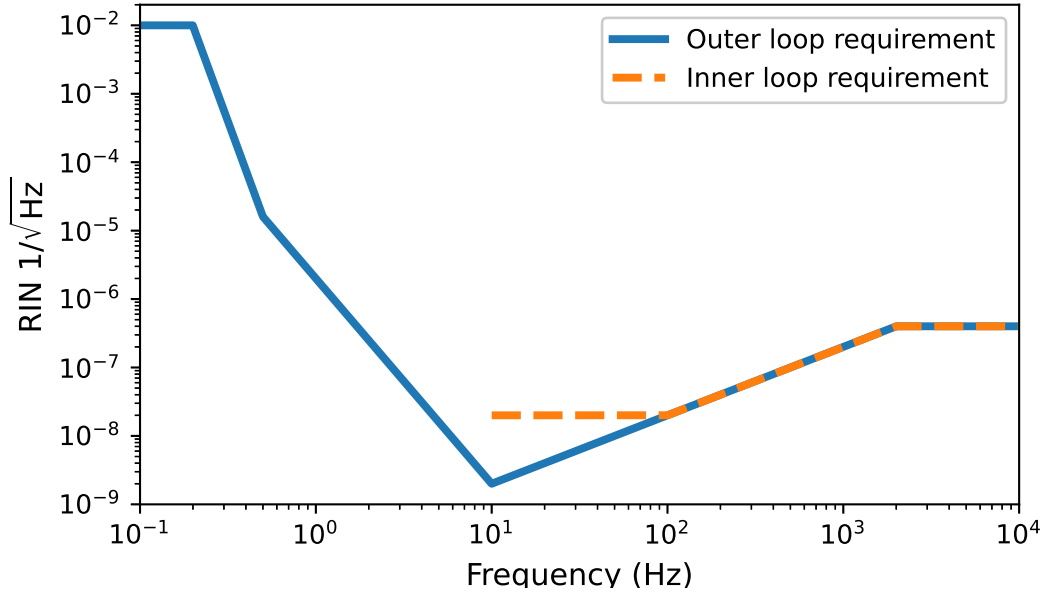


Figure 3.11: A reproduction of the intensity noise requirement from [35]. The outer loop requirement is shown in blue and the inner loop is shown by the dashed orange line. To lock the IMC, the inner loop requirement must be met. At frequencies below 10 Hz, the outer loop requirement is determined by the radiation pressure noise that is expected to arise due to imbalances in the arm power. The requirement is most stringent at 10 Hz. The requirement relaxes above 10 Hz due to the PRC cavity pole. The inner loop requirement is less extreme than the outer loop as it is only used to stabilise the laser in the initial locking sequence.

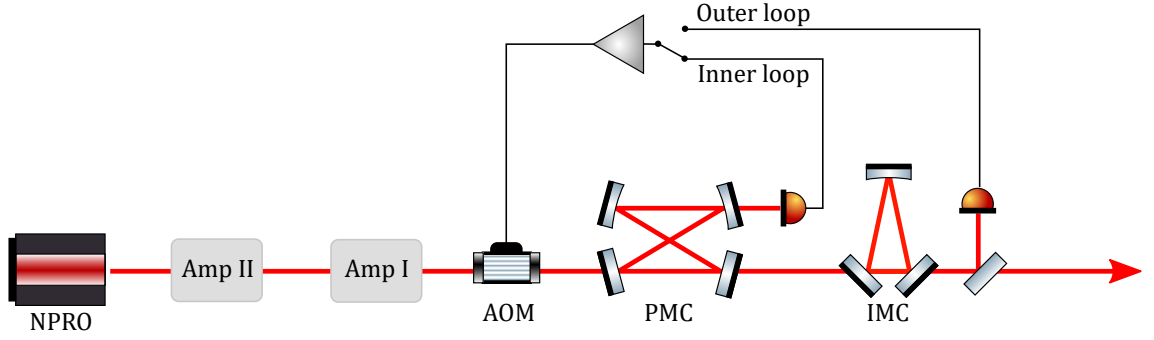


Figure 3.12: The intensity stabilisation concept involves two loops. The inner loop is responsible for stabilising the laser so that the IMC can lock. The outer loop is used to stabilise the laser for observation mode. The two loops sense the laser intensity noise at different points. The inner loop uses a low power beam, ~ 1 mW, transmitted by the PMC. The outer loop uses a higher power, ~ 350 mW, pick-off from the main beam. An AOM will be used to actuate on the laser's power.

The noise was covered in a forest of lines that were spaced 4Hz, starting from 4Hz and continuing up to high frequencies. These lines were found to originate in the Beckhoff slow-control computer which is used as an interface for the amplifiers. The laser control box needs better electrical isolation between its digital and analogue sections to remove the 4 Hz lines.

The noise of the beam emerging from the PMC was measured (Figure 3.13). The PMC introduces length to intensity noise coupling via the offset in the PDH loop. The offset needed to remove this coupling drifted due to changing levels of RF pick-up in the PMC servo electronics. The coupling from length noise of the PMC to intensity noise was decreased as much as possible.

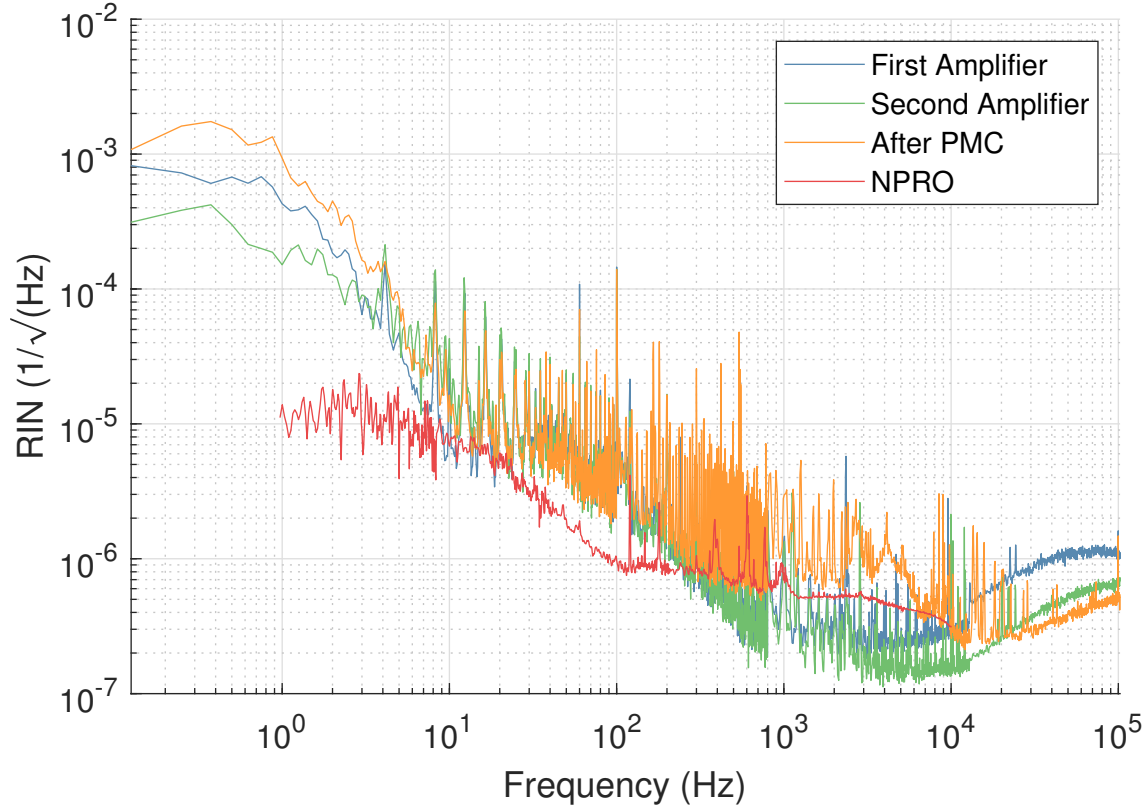


Figure 3.13: The free running noise of the NPRO and each amplifier is shown. The noise after the PMC was also measured. There is a forest of 4 Hz lines due to the Beckhoff control computer in the amplifier and PMC noise curves. Due to the non linear nature of the saturated amplifiers, at high frequencies the output beam intensity noise decreases relative to the input beam. The PMC couples length noise to intensity noise between 100 Hz and 10 kHz. This excess can be tuned away in the PDH servo.

3.5.3 Power Control Actuators

The actuator that will be used to provide intensity stabilisation will be an AOM. The use of a current shunt integrated into one of the pump-diodes in the second amplifier was investigated. The current shunt had the potential to reduce the number of transmissive optics in the beam path and increase the total power. Despite this, the current shunt may be an unsuitable choice for stabilising LIGO's laser. The current shunt transfer function depends on the DC current, at 10 Hz it changed by ~ 10 dB when the DC current was changed by ~ 1 A (see Figure 3.17), and so this may make the intensity stabilisation servo unreliable if the DC pump-diode currents needed changing over the course of an observation run.

3.5.3.1 Acousto-Optic Modulator

An AOM uses acoustic waves to scatter light into higher order modes. The amount of light that is Bragg scattered depends on the intensity of the acoustic wave. The AOM used was a Crystal Technology 3080-199. An RF driver, gooch and housego r21080-2am-rfx, was used to control the intensity of the acoustic wave. This driver produced an 80 MHz signal whose power can be modulated. The AOM's response was linearised with an IC (AD734). By modulating the power of this signal, the power in the zeroth order beam can be controlled. The diffraction efficiency of this AOM was found to be 74%. This means that the AOM can be used for coarse and fine power adjustment. To modulate the intensity of a laser beam, some of it must always be scattered into the first order beam. To operate linearly, the DC level of the light that is dumped was ~ 10 W.

The transfer function, in relative intensity units, between the AOM control electronics to the intensity of the zeroth order beam as measured by the ISS photodiode is shown in Figure 3.15. The schematic for this measurement is shown in Figure 3.14. This transfer function is flat from 1 Hz to 100 kHz, and, as the servo's unity gain point will be between 10-100 kHz, it is worth noting that above 10 kHz there is some phase lag.

3.5.3.2 Current Shunt

Modulating the current through a laser diode will modulate the intensity of the laser. This is a standard way of implementing intensity control at the 10W level [58–62]. Using a current shunt as an intensity actuator has been demonstrated in a similar system at the 100W level [47]. The current shunt technique allows for one to have fewer transmissive high power optics in the beam path, which potentially can reduce noise and thermal distortions of the beam. It requires a smaller DC offset compared to an AOM in order for it to work, therefore higher beam power can be achieved.

The amplifier's pump-diode current determines the amount of power produced by the amplifier. By modulating the current through one of the pump-diodes, the power of the laser can be stabilised. The current shunt consists of a power MOSFET and a resistor. It diverts current from the pump-diode by pulling it through the resistor instead. The amount of current that is pulled through the resistor depends on the voltage at the gate of the MOSFET (see [63, Chapters 2 and 3]).

The transfer function between the current shunt input and the ISS photodiode in transmission of the PMC was measured for a variety of different DC pump-diode

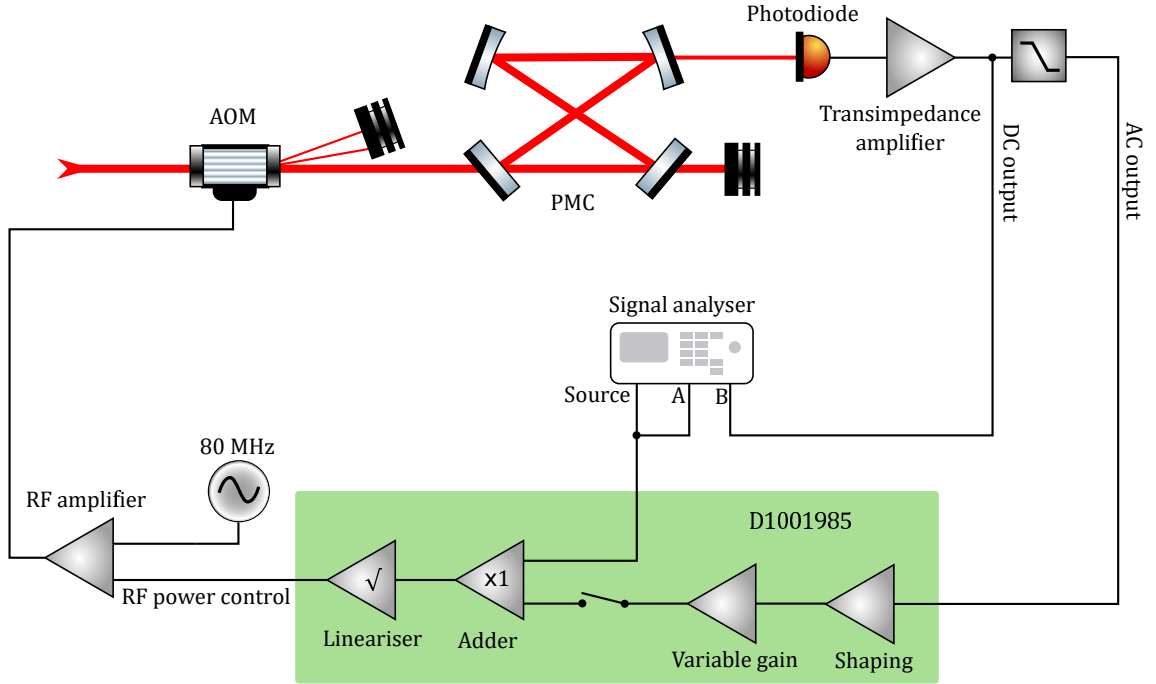


Figure 3.14: A sketch of how the AOM transfer function was measured. The components of the intensity stabilisation servo has also been shown. The PMC was locked and a signal analyser was used to perform a swept sine measurement. An IC (AD734) that performs a square root operation to an input signal was used to linearise the AOM's effect on beam power to an input signal. The transfer function was measured between the input to the lineariser and the photodiode used for the intensity stabilisation. The servo electronics are detailed in D1001985 and D1001998.

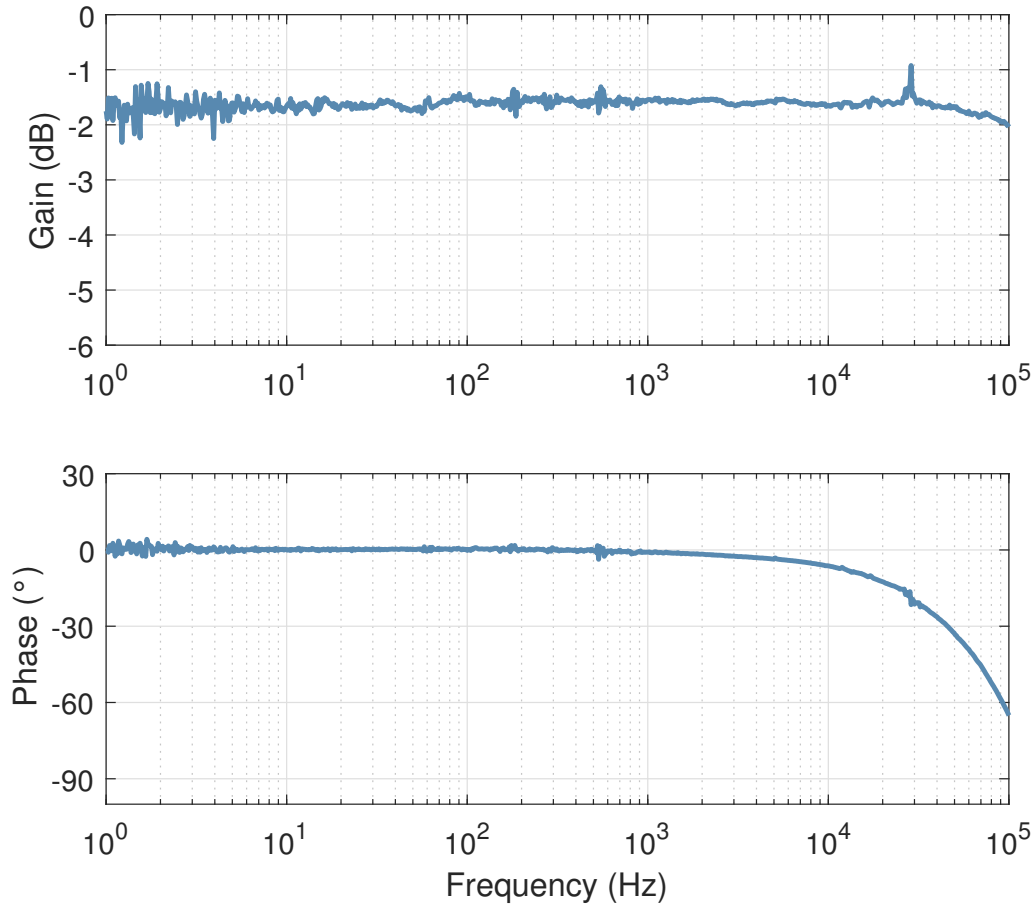


Figure 3.15: Using the layout shown in Figure 3.14, the transfer function between the input to the linearisation stage before the AOM to the light power change on the ISS photodiode was measured. The response was normalised by dividing by the DC power on the photodiode. The transfer function is flat over the region where the open loop gain needs to be greater than one (~ 10 kHz). There is a some phase lag between 10-100 kHz; this is where the unity gain point of the servo will be and so this needs to be accounted for when determining if the loop will be stable.

currents. This is shown, in relative intensity units, in Figure 3.17. A schematic of the measurement is shown in Figure 3.16. The transfer function was found to have significant amounts of variation in gain and phase between 1 Hz-100 kHz. Above 10 kHz the transfer function did not vary much with DC pump-diode current. This is good, as the unity gain point of the intensity stabilisation servo will be between 10-100kHz. The AOM is a much stronger actuator than the current shunt; this can be compensated for by changing the servo electronics. It was found that the behaviour of the actuator at 10 Hz, where the servo requirement is hardest to meet, strongly depended on the DC pump-diode current.

3.5.3.3 The Reason for Choosing the AOM Instead of the Current Shunt

While the current shunt offers some advantages over the AOM, it cannot reduce the power at the many tens of Watts level. This is a functionality that LIGO would likely need as the detectors have never operated at full power, at present the LIGO detectors operate with between 25-50W, due to point absorbers on the test masses. The pump-diode DC current would have to be reduced to reduce the power by many tens of Watts. This would affect the noise of the laser and the open loop transfer function of the servo. These amplifiers work best in their saturated regimes and the RIN may be increased if the pump-diode currents are decreased significantly. There may be other circumstances in which the pump-diode currents need altering; towards the end of their life-time the diodes need to be frequently adjusted. Alternatively, high power attenuation optics would be required. Unlike the current shunt, the AOM's response depends on the DC light level in a trivial way. The AOM causes a decrease in total power, but this downside is outweighed by the reliability of the AOM compared to the current shunt. When the LIGO detectors are able to operate

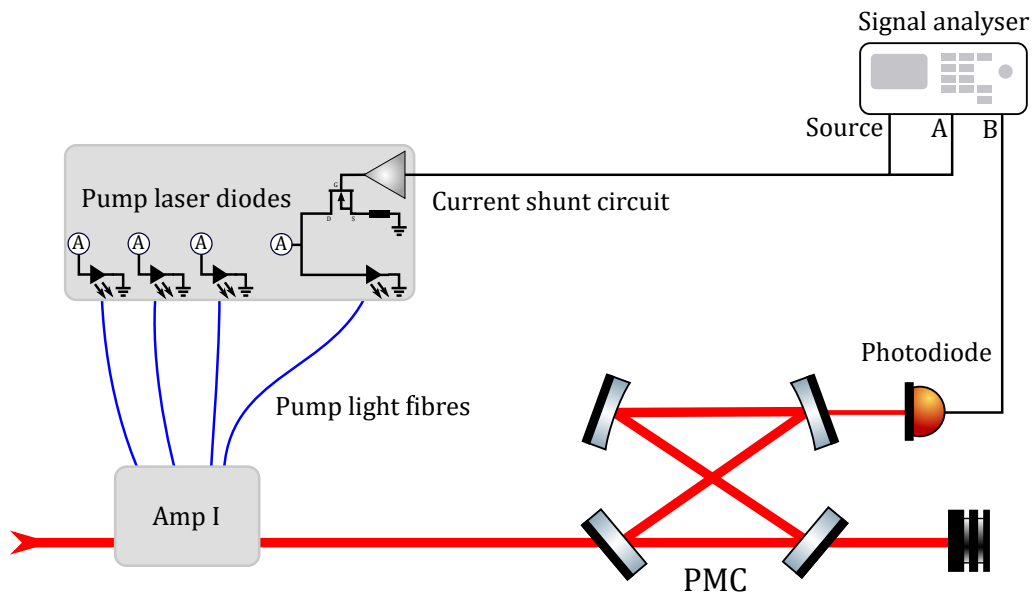


Figure 3.16: A sketch of how the current shunt transfer function was measured. The PMC was locked and a signal analyser was used to perform a swept sine measurement. The current of one of the pump-diodes in the second amplifier was modulated by applying a signal to a current shunt. The transfer function was measured between the input to the current shunt electronics and the photodiode used for the intensity stabilisation.

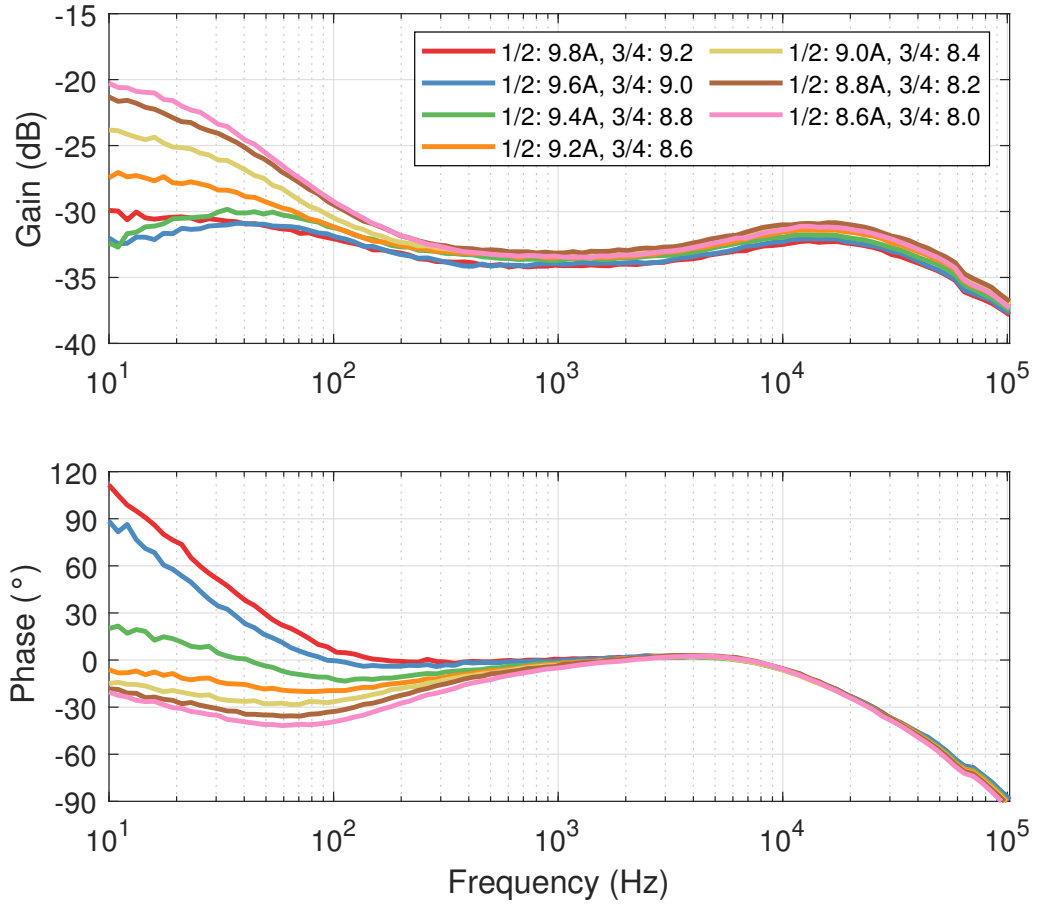


Figure 3.17: Using the layout shown in Figure 3.16, the transfer function between the current shunt input electronics and the intensity stabilisation photodiode was measured. The transfer function was normalised by dividing by the DC power on the photodiode. At 10 Hz, where the noise requirement is hardest to meet, the gain of the transfer function changes significantly depending on the pump-diode's DC current. The phase at this frequency does not matter as it is not near the unity gain point of the servo.

at full power, then the current shunt actuator would be worth revisiting

3.5.4 Open Loop Transfer Function of the Intensity Stabilisation Servo

The free running noise of the laser transmitted by the PMC is shown in Figure 3.13 and the requirement is shown in Figure 3.11. Based on the noise and the requirement, the unity gain point of the servo must be above 10 kHz. At 10 Hz, where the requirement is hardest to meet, the noise requirement is $2 \times 10^{-8} \frac{1}{\sqrt{\text{Hz}}}$ and the noise is $1 \times 10^{-5} - 1 \times 10^{-4} \frac{1}{\sqrt{\text{Hz}}}$, so there needs to be at least 60 dB of open loop gain at this frequency. If the noise of the laser was not dominated by the 4 Hz lines, the amplitude stabilisation could be achieved with a single pole low pass filter. However, this servo uses additional shaping to ensure that there is enough gain to suppress these peaks.

A sketch of the servo is shown in Figure 3.18. The servo features two circuits that add shaping; these are found built into the photodiode circuit [64] and within the rack electronics [65]. The servo model is described in detail in Section 3.5.4.1. The inner loop ISS features two photodiodes, one for in-loop measurements and one for out-of-loop measurements. The rack electronics allow for the servo to be controlled via CDS. This gives the user the ability to turn the servo on/off, to vary the overall gain and activate extra integrators.

A swept sine was applied at a times one adding stage of the servo to measure the open loop transfer function. The signal at the other input to this stage and the signal immediately after this stage was measured. The ratio of these two gives the

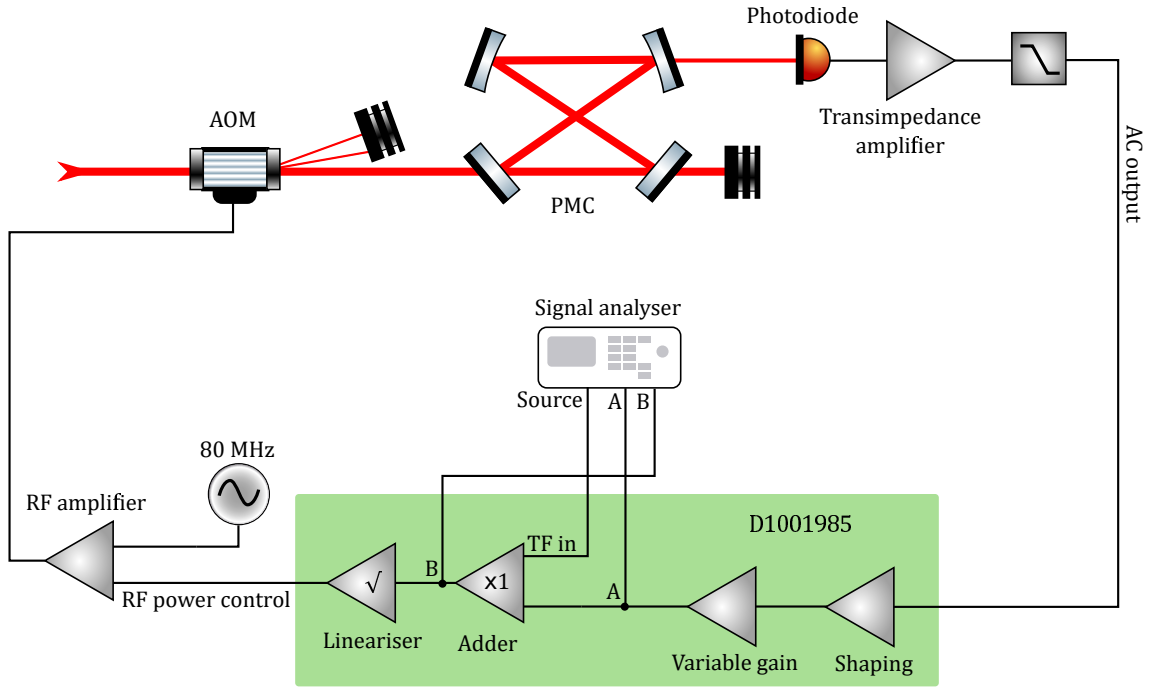


Figure 3.18: Sketch of the amplitude stabilisation servo. The open loop transfer function was measured by injecting a signal into a $\times 1$ adding stage and measuring the ratio of the signal immediately after and just before this stage.

open loop transfer function of the servo. For this measurement, the gain in CDS was set to 20 dB. Once the intensity stabilisation was activated, the integrators were engaged to increase the open loop gain at low frequencies. The light transmitted through the PMC was 100 W. The open loop transfer function is shown in Figure 3.19. The unity gain point is at 48 kHz with $\sim 70^\circ$ of phase margin. This servo satisfies the requirements.

3.5.4.1 Model of the Open-Loop Transfer Function

To understand the behaviour of the servo at frequencies where the measurement of the open loop gain saturated, a model is required. The overall gain of the servo is de-

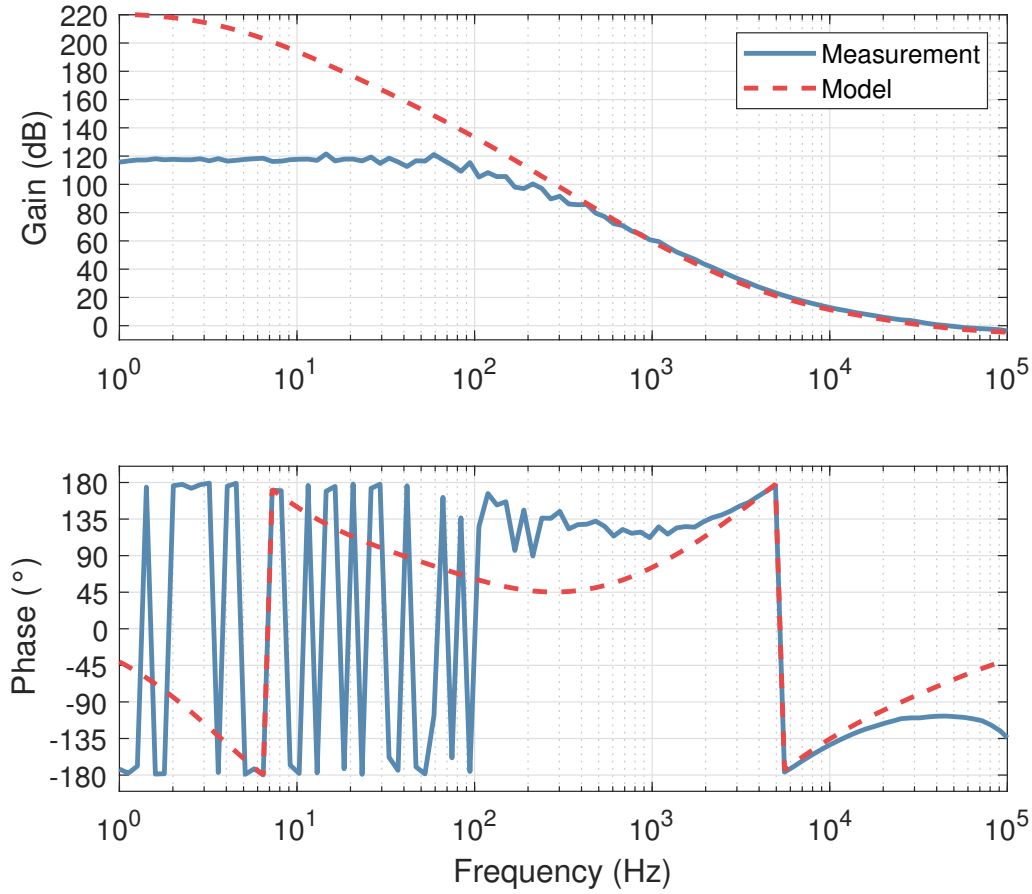


Figure 3.19: The open loop transfer function was measured using the layout shown in Figure 3.18. The unity gain point is at 48 kHz with 70° of phase margin. The shaping of the servo is $1/f$ at the unity gain point, but to achieve higher gain at low frequency, the servo has a $1/f^3$ slope below the unity gain point. The measurement saturated below 100 Hz. The model and measurement do not agree well below a few kHz due to the measurement saturating.

terminated by how much light there is on the in-loop photodiode and the setting of the variable gain. The following description puts all the gain that is distributed over all the components of the servo into one block. The photodiode transimpedance amplifier has a low-pass corner frequency of 5.9 Hz. This passes through a filter which boosts the photodiode's response to AC signals; this filter consists of two transitional differentiators with corner frequencies at 0.05 Hz and 3 Hz. The rack electronics feature two transitional integrators and one transitional differentiator. The first transitional integrator has corner frequencies at 1.06 kHz and 4.83 kHz. The second transitional integrator has corner frequencies at 1.06 kHz and 4.42 kHz. The transitional differentiator has corner frequencies at 59 kHz and 2.9 MHz. Activating the extra integrator stages removes the zeros in the transitional integrators at 1.06 kHz. The AOM introduces phase lag via a delay between the input and the output of 2.5 μ s. This model is summed up in Table 3.1 and shown in Figure 3.19.

3.5.5 Measurement of the Intensity Stabilisation of a 100 W Beam to the Inner Loop Requirement

The PMC was set to transmit 100 W of intensity stabilised light. The stabilised laser noise is shown in Figure 3.20. A noise budget was created using the signal measured on the in-loop photodiode and the shot noise level of the light used to stabilise the laser. This agrees with the out of loop noise measurement above 10 Hz. Below 10 Hz, there is excess noise in the out of loop measurement compared to the noise budget. Possible sources of this noise could be scattered light or dust falling through the ISS beam. Near the unity gain point, due to the open loop transfer function phase and gain, there is a small increase in the stabilised noise compared

Component	Transfer function
Photodiode low pass	$\frac{36.945}{s+36.95}$
Photodiode filtered output	$\frac{3600(s+0.3142)^2}{(s+18.85)^2}$
Rack transitional integrator I	$\frac{0.21946(s+3.035e04)}{(s+6660)}$
Rack transitional integrator II	$\frac{0.023982(s+2.777e04)}{(s+666)}$
Rack transitional differentiator	$\frac{49.153(s+3.707e05)}{(s+1.822e07)}$
Rack additional integrators	$\frac{1 \times 10^{-8}(s+6660)^2}{(s+0.666)^2}$
AOM phase lag	$\exp(-2.5 \times 10^{-6} s)$
Overall gain	2.6775×10^{10}

Table 3.1: Model of the intensity stabilisation servo expressed in terms of the complex angular frequency $s = \omega + i\sigma$. Note that in reality the gain is distributed amongst each block; in this model all the components have unity gain at DC. The additional integrators have been expressed as a separate block to the transitional integrators with a pole that cancels a zero in the transitional integrators; the zero in this additional integrator transfer function was chosen to be below any frequency of interest and does not exist in the servo. It was included here to make modelling the servo more clear.

to the free running noise, however this is out of the requirement frequency band and so this is not an issue. This set-up will be able to deliver at least 100 W of intensity stabilised light, which will likely be more than is required at the time of writing, to the interferometer.

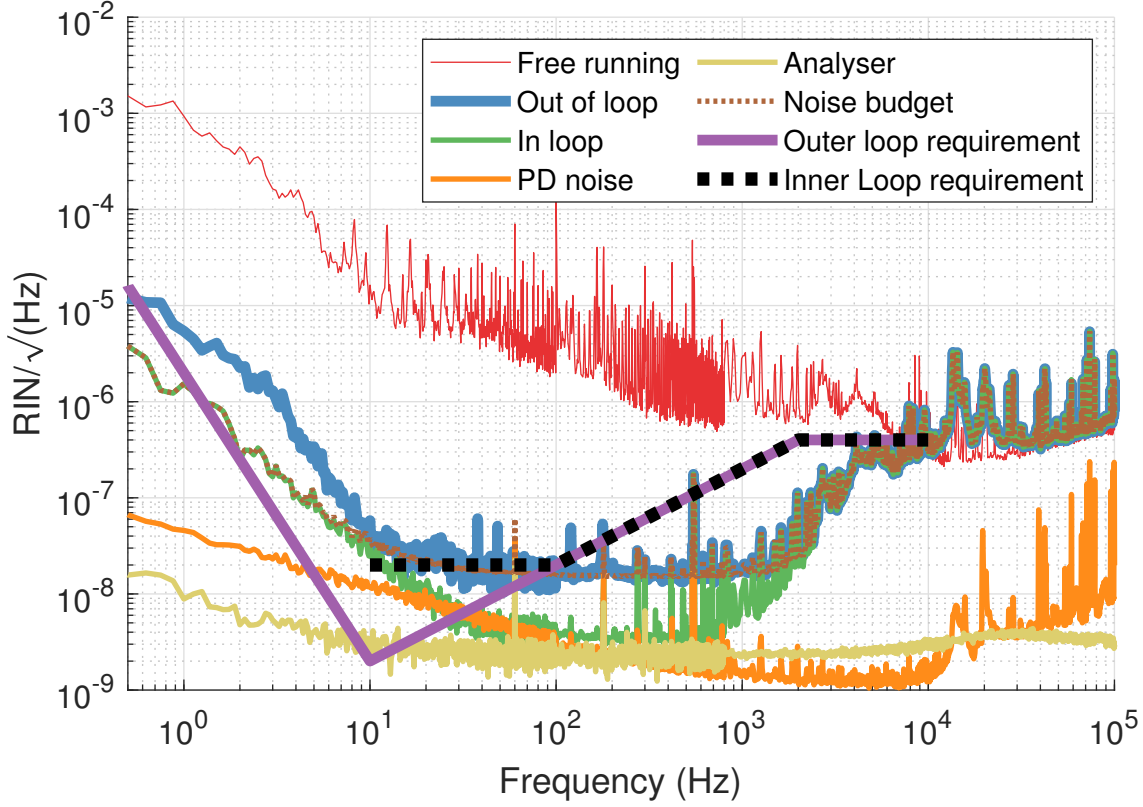


Figure 3.20: The free-running noise of the laser is shown in red. The noise of the light transmitted by the PMC is shown in blue; this meets the requirement that is shown by the dashed black line. The noise level is determined by the in loop noise, shown in green, the shot noise and other technical noise sources. The sum of these is shown by the dotted brown line. Below 100 Hz, the noise is shot noise limited. There is a discrepancy between the noise budget and the measured noise. This is likely due to dust falling through the beam or scattered light. Near the unity gain point of the servo, the noise is enhanced by the servo due to gain peaking.

3.6 Pointing Noise of the Beam Entering the PMC

The pointing noise witnessed at LHO is one of the reasons why single-pass amplifiers were chosen as the replacement for the HPO. Pointing noise manifests as intensity noise since a change in beam direction can be modelled as the introduction of higher order modes. These new amplifiers are cooled in such a way that the jitter on the PSL beam is reduced. Pointing is passively stabilised by the PSL via the PMC. The pointing noise requirement is based on the transfer function of the HG10 mode between the input of the interferometer and the dark port [66].

The pointing noise was measured using the DBB. The pointing noise is shown in Figure 3.21. To measure pointing noise, the laser was locked to the DBB cavity. The reflected light from the cavity was made up of the fundamental mode of the cavity (some of this mode leaks out) and the first order modes associated with misalignment. The beat between these two fields can be used to determine the pointing of the beam; this is known as differential wavefront sensing [67]. The electronic noise floor is at $1 \times 10^{-7} \frac{1}{\sqrt{\text{Hz}}}$. The pointing is expressed in units of $p = \delta x/w_0 + i\delta\alpha/\Theta$, where Θ is half the divergence angle of the beam, α is the angle of the beam, x is the beams position and w_0 is the beams waist.

The pointing noise measurement shows that this layout is close to fulfilling the requirement. More time could be used to reduce turbulence in the water cooling as this can introduce jitter. The noise below 100 Hz may be coming from the environmental control in the TnT lab. The LIGO PSL table is different, so this data may not reflect how quiet this layout will be in the main lab.

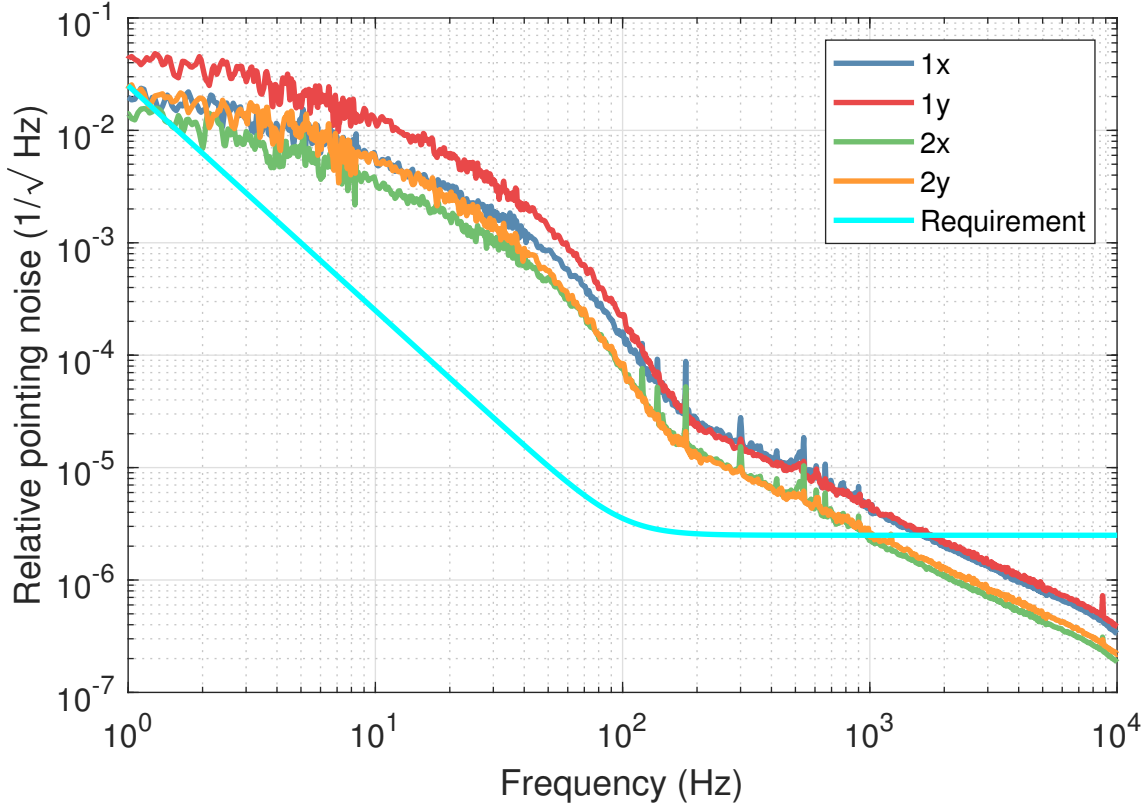


Figure 3.21: The four relative pointing noise degrees of freedom for the beam into the PMC was measured with the DBB. The requirement [66] is shown in light blue. The pointing noise is above the requirement, however the TnT lab has a higher amount of environmental noise; this pointing noise will be less in the PSL. Measurements below 100 Hz may be affected by scattering.

3.7 Free Running Frequency Noise of the Prototype Laser System

The LIGO detectors employ a multi-stage frequency stabilisation system. Ultimately, the arm cavity is used as a frequency reference since $\delta f/f = \delta l/l$, however the initial stabilisation of the laser uses an ultra low expansion fused silica reference cavity located on the laser table [68]. The free-running frequency noise must be measured to design the frequency stabilisation servo.

The PMC can be used to measure the frequency noise of the beam emerging from the PMC. The beam's frequency will be determined by the length noise of the PMC and the frequency noise of the NPRO. The signals used to control the PMC can be converted into a measurement of the systems frequency noise. The control signal which is sent to the PZT in the PMC can give the frequency noise of the system where there is sufficient open loop gain. Around the unity gain point, the servo's response needs to be factored in. Alternatively, the error signal at the mixer and the open loop gain can be used to measure the frequency noise of the system.

The NPRO is locked to the PMC by controlling the length of the PMC. The length of the PMC is controlled with two actuators. A PZT is attached to one of its mirrors to provide fast control, and a heater which determines the thermal expansion of the PMC provides slow control. The PDH technique is used to sense the length of the cavity. RF frequency side-bands are applied to the laser via an EOM. The light that is reflected from the PMC is demodulated at the same frequency as the RF sidebands. When the cavity is near resonance the demodulated signal is proportional

to the difference between the laser frequency and the frequency corresponding to the cavity length. A sketch of the PMC length control servo is shown in Figure 3.22.

To see how the error signal used to lock the cavity and the open-loop transfer function gives the frequency noise, consider Figure 3.23. The difference between the laser frequency and the cavity frequency is obtained via the PDH technique. The PZT signal, c , is then obtained via the transfer function of the optics, i.e. the frequency to voltage transfer function corresponding to the slope of the demodulated PDH signal, and the transfer function of the electronics. Together this transfer function is G . First, define the error signal $c = a - b$. Next, express the control signal in terms of the error signal, $cG = d$. Expressing the signal b in terms of d allows for signal a to be written in terms of the error signal c ,

$$a = c(1 + GH). \quad (3.1)$$

Equation (3.1) shows that to measure the frequency noise of the laser, a , the error signal should be, approximately, multiplied by the open loop transfer function, GH .

3.7.1 Calibrating the Mixer Signal, the PZT and Modelling the Open Loop Transfer Function for Locking the PMC

As the cavity is short compared to the frequencies of interest, the transfer function representing that converts a difference in NPRO and PMC frequency, the signal due to the PDH scheme, to a voltage is a flat gain factor. A measurement of this can be obtained by measuring the slope of the PDH signal as the frequency of the cavity

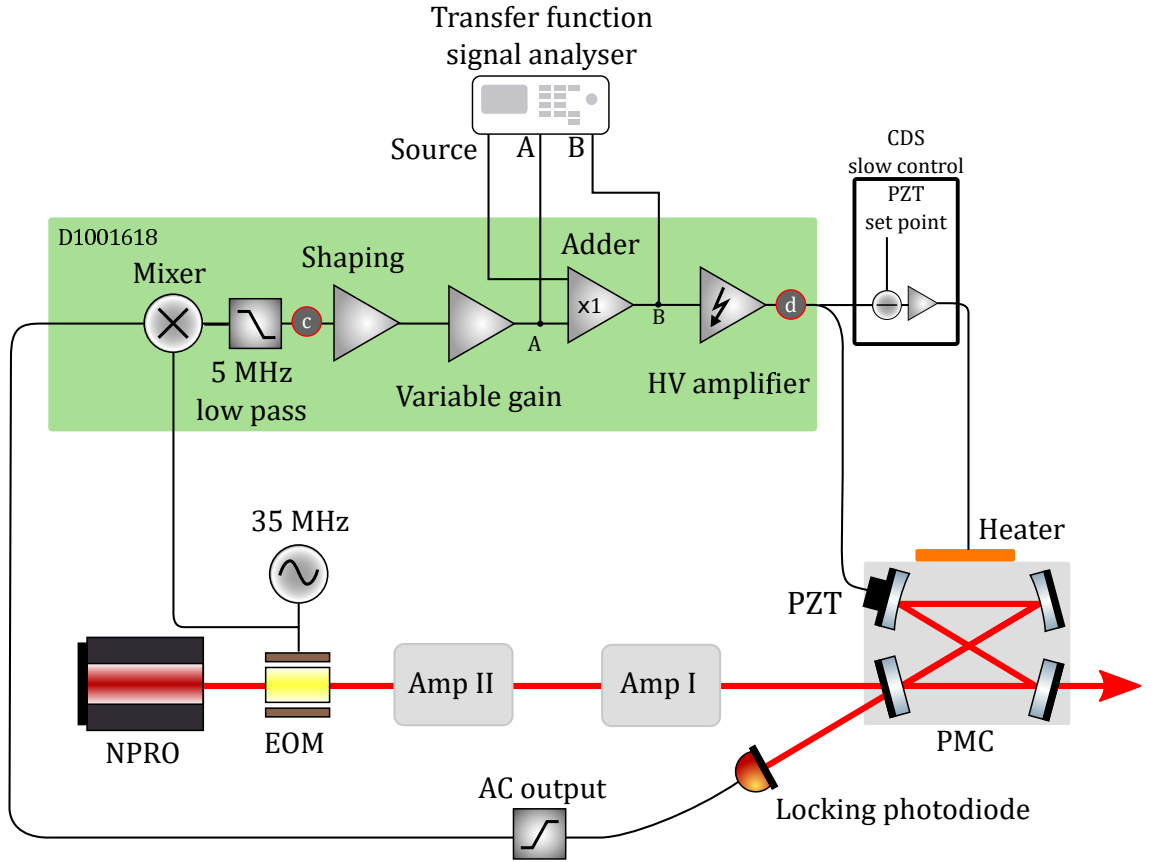


Figure 3.22: A sketch of the frequency stabilisation servo. An EOM creates 35 MHz sidebands which are used to produce a PDH locking signal. A PZT controls the length of the cavity at high frequencies. A heater and a digital loop is used to provide slow control that corrects for frequency drifts associated with change in the NPRO temperature and the room temperature. This is required as otherwise the PZT would run out of range. The open loop transfer function was measured using signals at A and B. The frequency noise was measured using the error point and control signals marked by *c* and *d*, with this notation corresponding with the notation in Figure 3.23 and Equation (3.1).

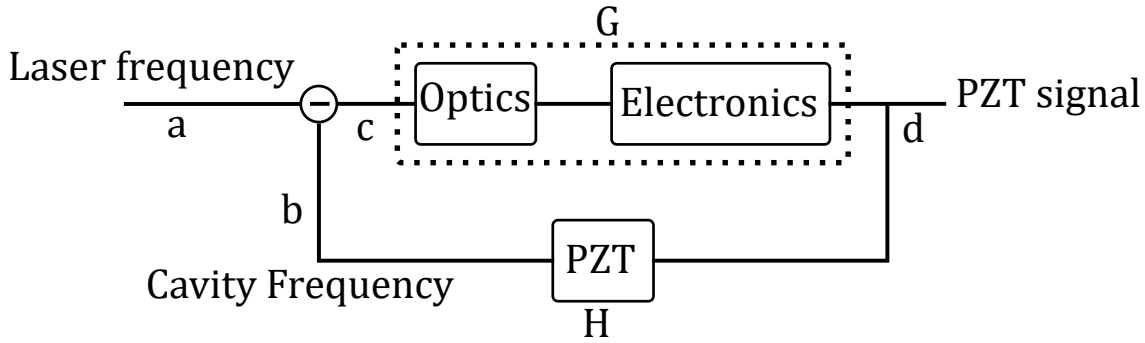


Figure 3.23: The servo can be considered as being made of three parts: the optics, the shaping electronics and the PZT. The transfer function of the optics represents how a difference in frequency between the laser and the cavity transforms into a signal after the demodulator. When at resonance and below the cavity pole, this is a flat transfer function that is proportional to the difference in frequency. Since this cavity pole is in the MHz region and this measurement will only go to the kHz region, the cavity pole can be ignored. The transfer function of the electronics includes everything after the demodulator up to and including the final high power OP amp. The PZT transfer function converts a voltage to a cavity frequency. The PZT acts as a low pass filter due to it having capacitance.

passes through a resonance. By driving the PZT with a ramp and simultaneously measuring the signal before and after it is demodulated, the transfer function of the optics can be obtained. Figure 3.24 shows this measurement. The time axis can be converted to frequency as the full-width-half-maximum (FWHM) of the reflected light is determined by the optical properties of the PMC. The design value of the FWHM for the PMC is 1.19 MHz. The slope of the demodulated signal as the cavity passes through the resonance can be used to give the conversion between difference in frequency of the cavity and laser to voltage after the demodulation electronics. This factor is 0.1364 MHz/V.

A ramp was applied to the PZT and the spacing between two resonances was measured to convert the PZT voltage to the cavity frequency at low frequencies. The difference in ramp voltage between the two resonances corresponds to the FSR of the cavity. This is shown in figure 3.25. The nominal FSR for the PMC is 148.529 MHz, so the PZT to frequency transfer function is 42.2870 MHz/V. The PZT forms a low pass filter as it has capacitance; this must be included when modelling the servo. The corner frequency of the PZT was estimated to be at 1700 Hz. The corner frequency of the PZT is above the unity gain frequency of the PMC locking servo.

The measurement and model of the open loop transfer function, i.e. the transfer functions of the optics, electronics and the PZT multiplied together, is shown in Figure 3.26. The model is within 2 dB of the measurement. A sketch of how the open loop transfer function was measured is shown in Figure 3.22. The unity gain point of the servo was measured to be at approximately 800Hz. The difference between the model and the measurement may be due to the use of nominal values

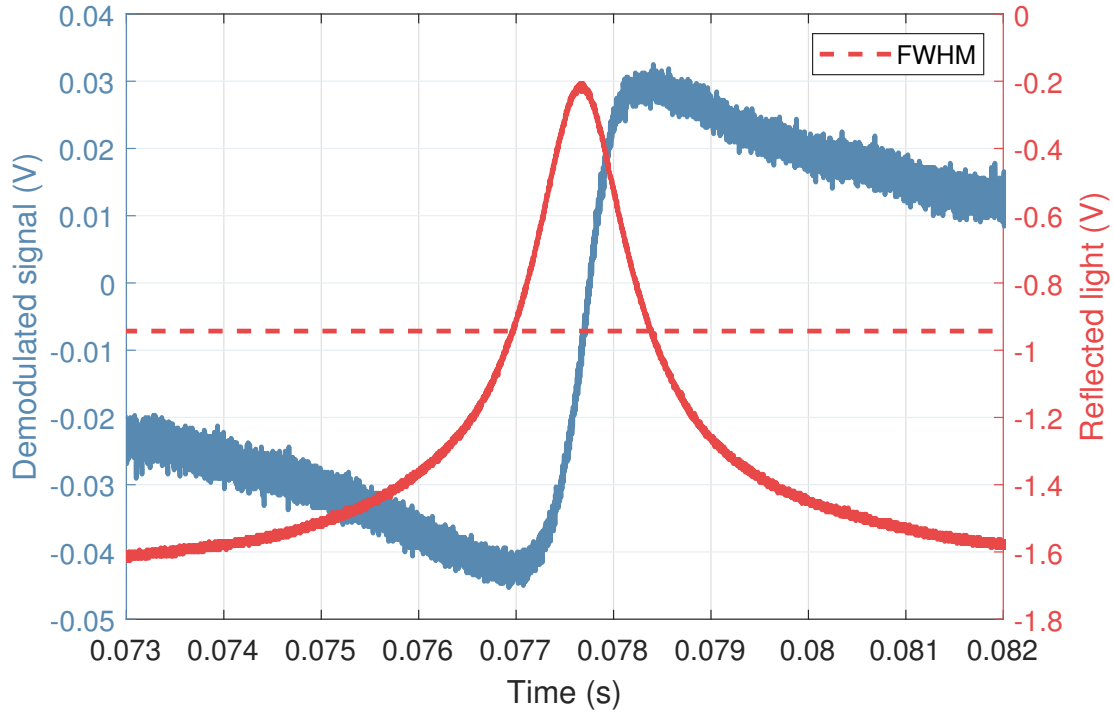


Figure 3.24: A ramp was applied to the PZT and the photodiode signal was measured before and after demodulation. The demodulated photodiode signal is shown in blue. The photodiode signal is shown in red. Around the peak of the photodiode signal, the slope of the demodulated signal gives the gain of the ‘optics’ transfer function. This can be expressed in units of MHz/V by converting the time axis to a frequency axis. This is done by using the FWHM of the photodiode signal as a reference frequency. The level of the FWHM is indicated by the dashed red line.

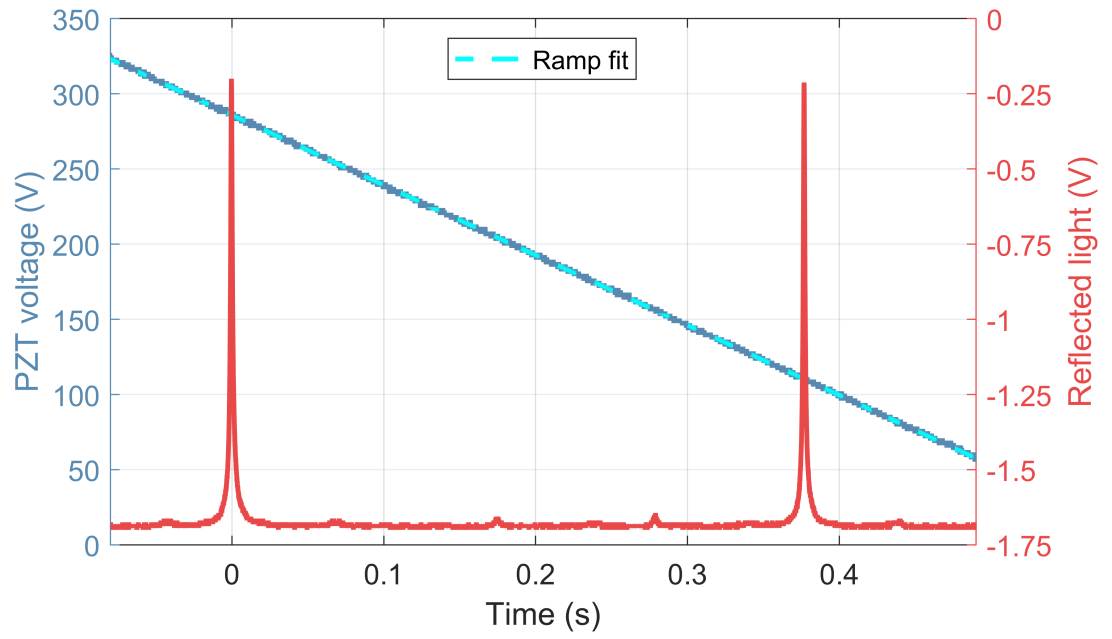


Figure 3.25: A ramp was applied to the PZT and the reflected light from the cavity was measured. The FSR of the cavity is known, so the time in this measurement can be converted to frequency. This gives the PZT transfer function at low frequencies. The ramp signal is shown in blue and the reflected light is shown in red.

for the properties of the PMC.

3.7.2 Frequency Noise Measurement

The measurement of frequency noise is shown in Figure 3.27. Despite imperfect calibration, this measurement provides useful information about the frequency noise of the laser. As the PMC was not vacuum sealed, between 100 Hz to 1 kHz, there was a bulge in the frequency noise. This was due to the air conditioning generating PMC length noise combined with non perfect offsetting in the PDH loop. Below 100 Hz and above 1 kHz, the frequency noise of the laser was determined by the NPRO. Near the UGP, there was gain peaking which will enhance the noise.

Below the unity gain point, as the PZT response is flat, the PZT signal can be converted into frequency noise. The mixer signal and the open loop transfer function $c(1 + GH)$ gives the frequency noise. These two measurements are within a factor of 2 of each other. This noise is higher than the expected level for the NPRO. There is uncertainty in the calibration as nominal values for the optical properties of the PMC were used.

At frequencies much above the unity gain point of the servo, a signal applied to the PZT will not be suppressed by the loop. Near the unity gain point, a signal applied to the PZT will be attenuated. A peak at 1111 Hz was used to check the calibration of the loop. The closed loop gain at 1111 Hz was modelled to be -5.2 dB, so the peak measured at the mixer should be reduced by a factor of $1 - 10^{-5.2/20} = 0.45$. The peak generated at the PZT was measured at the mixer to be reduced by a factor of 0.70; the difference between the prediction and measurement would be fixed with

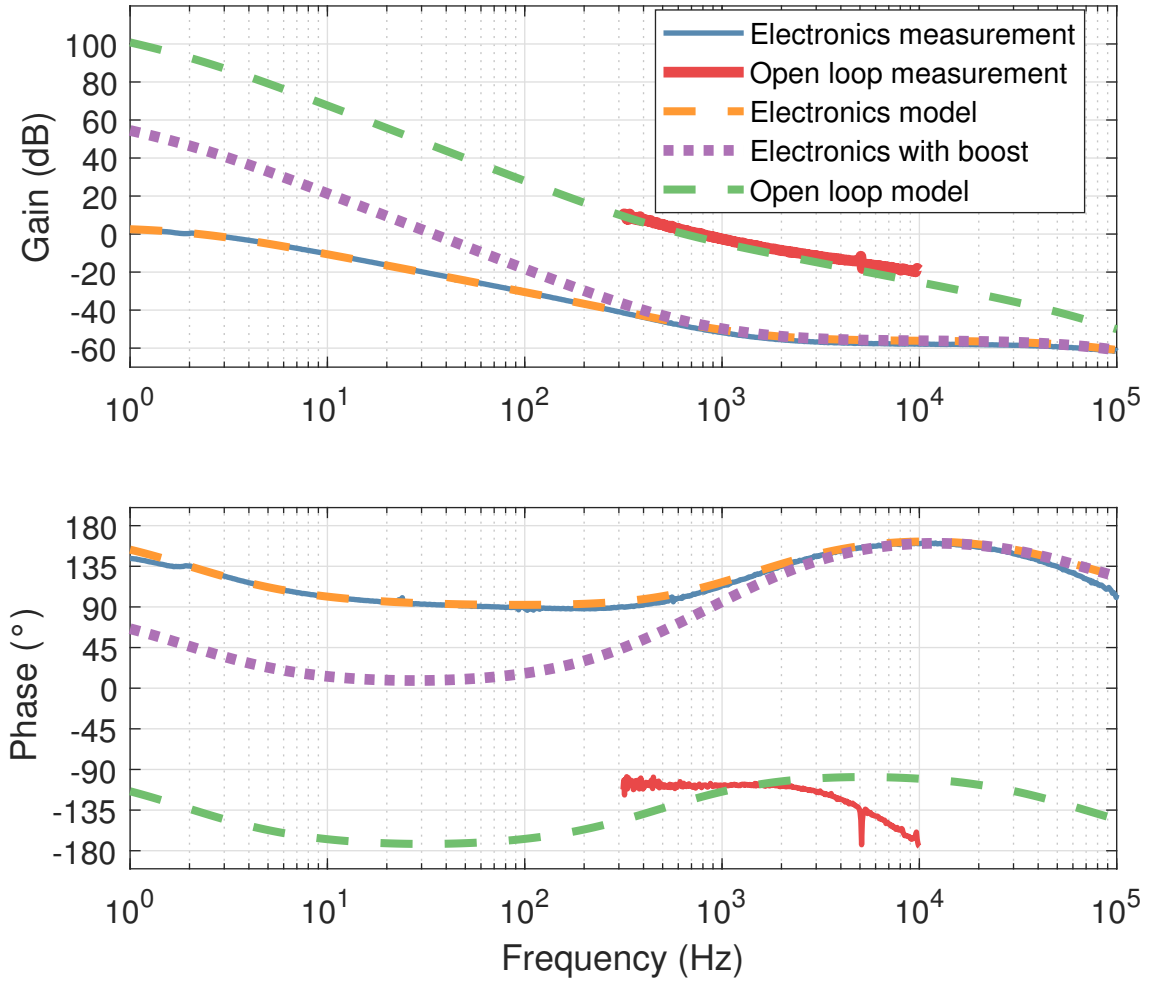


Figure 3.26: The open loop transfer function was measured using the layout shown in Figure 3.22. This is shown in red. The transfer function of the electronics without the ‘boost’ provided by an extra integrator stage was measured and is shown in blue. The model of this is shown in orange. The boosted electronics model is shown in purple. The green line shows the expected servo gain based on the boosted electronics model and the measurements shown in Figures 3.24 and 3.25. This model is within 2 dB of the measurement. The unity gain point was measured to be at ~ 800 Hz.

better characterisation of the PMC.

The discrepancy between the blue and red traces in Figure 3.27 may be due to the optical values used for the PMC, as these were not measured. If the cavity is detuned, a frequency to amplitude conversion occurs. This conversion does not happen at the FSR of the cavity. Measuring the transfer function between a frequency modulation generated by a broadband EOM and the amplitude of the light transmitted by the cavity will give an accurate measurement of the length of the cavity [69]. To measure the finesse of the cavity, an RF amplitude modulator could be used. The transfer function between the input modulation and the transmitted light depends on the FSR and the finesse of the cavity [69]. The AOM in the layout cannot be used as its bandwidth is too low. In Figure 3.27, the red trace below the unity gain point is better calibrated since the FSR does not depend on the cleanliness of mirrors.

3.8 Conclusion

The PSL has undergone a major redesign due to the jitter noise introduced by the cooling for the HPO. The new layout consists of an NPRO and two solid state amplifiers followed by the PMC. A prototype of this new design was constructed and tested. The set-up procedure was established and a cooling scheme was trialled.

The frequency noise of the beam transmitted by the PMC was investigated using signals in the PMC control loop. The laser is dominated by NPRO noise at frequencies below $\sim 100\text{Hz}$ and above $\sim 1\text{kHz}$. As the PMC is not shielded from acoustic vibrations, the PMC introduces noise between these two frequencies.

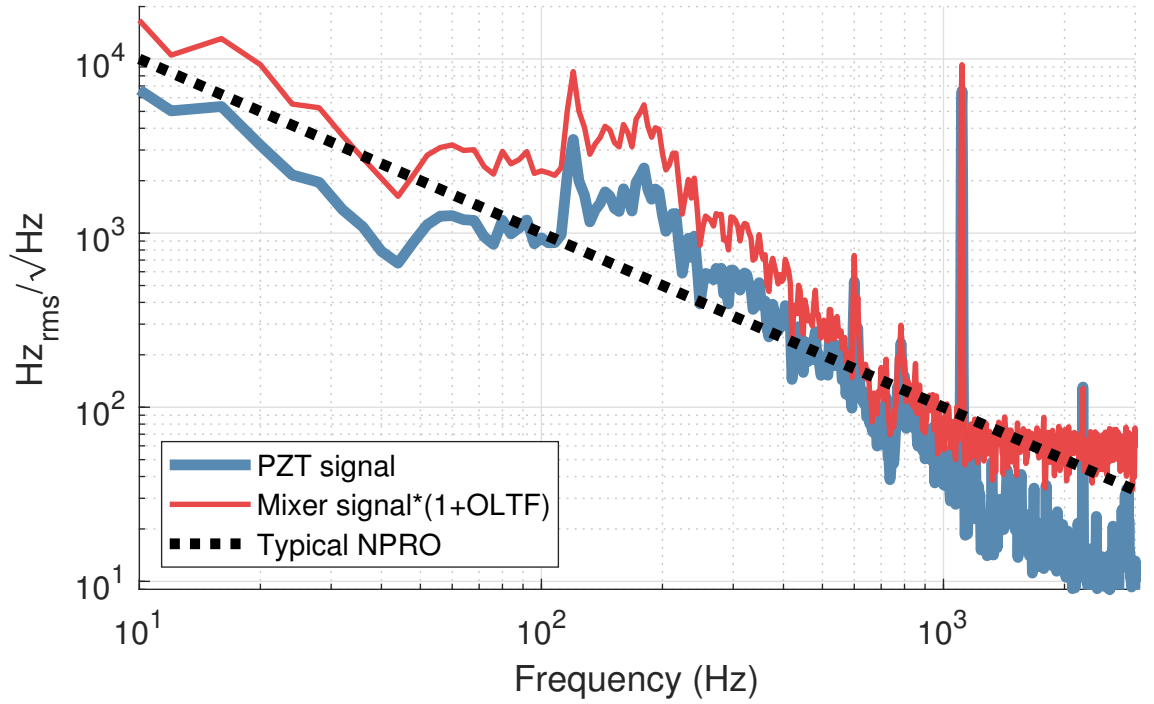


Figure 3.27: The mixer signal combined with the open loop transfer function is shown in red. This is equivalent to the frequency noise of the system. The PZT signal should agree with this measurement below the unity gain point (~ 800 Hz). A signal was applied to the PZT at 1111 Hz. This is below the corner frequency of the PZT and above the unity gain point for the servo. This should match with the mixer signal. There is a factor of two between these two measurements. This can be attributed to imperfect calibration due to nominal parameters of the PMC being used. The typical noise expected from an NPRO is shown by the dashed black line. Between 100 Hz and 1000 Hz, there is a bulge in the frequency noise. This is likely due to acoustically driven length noise of the PMC as the PMC is not vacuum sealed.

The jitter was found to be close to the requirements. As the PSL is in a quieter environment than the lab this layout was tested in, this may not be a problem. To reduce the jitter noise further, the flow rates being reduced may help, however this reduces the output power. Active alignment could be implemented using the light that is reflected from the PMC. Turbulence generated by bends in the water cooling pipes could be reduced to lower the jitter noise.

The output power was measured over two days and was found to drop from a peak of 111 W to 107 W. This was due to changes the lab temperature affecting the alignment. Higher output power could be achieved if more pump-diode current and cooling was used. Additionally, more power could be coupled to the PMC if more time was spent on optimising the mode matching between the second amplifier and the PMC.

100W of intensity stabilised light was transmitted through the PMC. The free running intensity noise of the beam transmitted by the PMC was measured and the inner loop intensity stabilisation was tested. Two actuators were considered, a current shunt and an AOM. The AOM was chosen as this offers the most flexibility in terms of loop gains and operating power. The inner loop noise requirement was met with this set-up.

Future iterations of the PSL may use fibre lasers and fibre amplifiers [44, 45] as there is rapid development in this technology. These have been demonstrated at the power levels required for LIGO. Coherent power addition is another technique that can be used to obtain high beam powers. When long term operation has been demonstrated, these technologies may be a viable replacement for the current laser

design.

Chapter 4

The Consequence of Radius of Curvature Errors within the SRC on Modes at the SRM.

4.1 Introduction

A gravitational wave interacting with the LIGO detectors creates a differential signal in the arm cavities, and as the signal is differential it is directed to the Signal Recycling Cavity (SRC). The SRC tuning shapes the frequency response of a differential arm length change to light power transmitted through the antisymmetric (AS) port, with there being two modes of operation: signal recycling (SR) and resonant sideband extraction (RSE). The beam is then directed to the Output Mode Cleaner (OMC) to filter away ‘junk’ light. Junk light is comprised of higher order modes (both spatial and temporal), and some of this junk light comes from mode mismatches between the arm cavities and the SRC. The light transmitted by the

OMC ideally will only contain the signal generated by the differential arm length change, which is a signal that originated in the arm cavities. This will be explained in the ‘detector’ chapter, so does this need references?

Losses between the arm cavities and the readout photo-diodes result in less signal as well as more noise. The decrease in signal is straight forward to understand: if there are less photons to measure, then the signal will be smaller. Since the power is lower, the signal to noise ratio will be larger since shot noise goes as $\sqrt{\text{Power}}$. As gravitational wave detectors implement squeezing to decrease the quantum noise in the detection quadrature, it is important to preserve the quantum state of the signal field. This means that losses contribute additional noise, as losses pollute the squeezed quantum state with unsqueezed vacuum. This type of loss will be featured earlier in the thesis in the ‘detector’ chapter, so does this need references?

It is known that 20% of the light is lost between the AS port and the readout photo-diodes, as the shot noise can be compared to the expected shot noise for a given fringe slope, input power and known losses. Roughly half of the loss is accounted for e.g. some comes from the Faraday isolator, some comes from the OM3 pick off, however the other half is not fully understood e.g. see [70]. The level of squeezing can also be used to cross check the loss [71]. Some of the unaccounted for loss may come from mode mismatch between the arm cavities and the OMC.

Mode matching losses between the squeezed light source, the interferometer and the OMC act coherently. Mode mismatch can be modelled as a modal beam splitter that transfers some light from the HG00 mode to the HG20+02 mode. The amount of light transferred between the HG00 and HG20+02 mode is determined

by the magnitude of the mismatch, and the phase difference between the HG00 and HG20+02 mode depends on the nature of the mismatch. An error in waist size is in-phase (real), whereas an error in waist location is out-of-phase (imaginary). In general the mismatch introduces a complex phase. A sequence of mode mismatches can be modelled in the same way as a Mach-Zehnder interferometer. The difference in phase between the HG00 and HG20+02 mode is analogous to a difference in path length in a Mach-Zehnder interferometer. It differs from a typical Mach-Zehnder interferometer as the HG00 to HG20+02 splitting ratio is not well balanced. As the mode matching losses act coherently, the degradation of the squeezed state can be greater for mode matching losses than for other types of loss e.g. the loss due to photo-diodes with a quantum efficiency less than one. If the squeezer is perfectly matched to the OMC, then the mode mismatch between the squeezer and the interferometer is four times the loss you would expect from treating a mismatch loss as a regular loss. This is explained in detail in [72].

Since the SNR of the gravitational wave signal is directly related to mode-matching between the arm cavities and the OMC, knowledge of the beam parameter at the SRM is highly important. The mode in the arm is well defined since the arm cavities are high finesse. Uncertainties in the RoC of optics in the SRC telescope (SR2 and SR3) results in an uncertainty of the beam parameter of the arm mode at the SRM, as well as the eigenmode of the SRC. A fair amount of mode mismatch can be tolerated by the SRC as it is a low finesse cavity, so this leads to an uncertainty in the beam parameter that is transmitted through the SRM.

Junk light will be generated if the arms are perfectly matched to the OMC but not to the SRC. If the arms are matched to the OMC but not to the SRC, then actuating on the beam parameter with the ITM thermal lens and the SRM thermal lens would not increase the signal as these two actuators are separated by enough Gouy phase that a large portion of the beam parameter phase space can be covered with them. This means the mode matching between the arm and the OMC can be optimal but the light can still contain HOM. Additionally, if the beam is astigmatic, then optimal mode matching would not correspond to perfect mode matching.

The aim of this work is to see how the errors in the focal length of SR3 and SR2 effect the beam parameter of the arm mode at the SRM and higher order mode (HOM) content of the beam transmitted through the SRM. A LIGO-like interferometer was simulated using FINESSE. The simulated interferometer has properties very close to the the as built LLO LIGO detector, however some simplifications were made to make the effect that a geometry error in the SRC has on the beam transmitted through the SRM clearer. The difference in the HOM content of the beam transmitted through the SRM between an ITM single bounce measurement and a measurement which involves the arm cavities was investigated. The effect that the interferometer being in signal recycling and resonant sideband extraction (RSE) mode had on the HOM content of the beam as the focal length of SR3 deviated from its nominal value was also simulated.

It was found that the dependence of the ratio of HG00 light to the total light depends mainly on the focal length of SR3 and is not effected by the presence of arm cavities. This is because the arm cavities are well matched to the input

laser [73]. When the arm cavities are on resonance, the ITMs act as highly reflective mirrors with frequency dependent behaviour. This means that there is no difference in HOM content of the beam transmitted through the SRM between a single bounce measurement and an arm cavity measurement.

The HOM content of the light does depend on the detuning of the SRM. It does not depend on whether the interferometer is in signal recycling mode or RSE mode (as the arm cavities do not matter). When the SRC is on resonance, it is much more sensitive to an error in the SR3's focal length than when the SRC is anti-resonant. The light is less sensitive to an error in the focal length of SR2.

The beam parameter of the arm mode at the SRM behaves in a predictable way since the arm mode is well defined, so the Gouy phase of a beam at the SRM can be used to acquire information about the arm mode at the SRM. The beam width and RoC were simulated for a range of SR3 focal length errors. By analysing where the SRC becomes unstable, the beam size cannot exceed 2.2 mm. These results can be used to constrain design tasks related to active wavefront control. The Gouy phase has been measured to be $21.3 \pm 0.3^\circ$ [74]. If the thermal lensing in the ITM is roughly at design, then the focal length of SR2 is -3.205 ± 0.003 m and the focal length of SR3 is 17.971 ± 0.002 m. The arm mode would be 1.85 mm and a curvature of -3.8 m and there would be a few percent HOM in OMC scans. **Double check values, need extracting from the plot.**

4.1.1 Measurements of the Mode Matching Between the Arm Cavities and the OMC at LLO Between 2018-2019.

The mode matching between the arm Cavities and the OMC, and the input light to the OMC has been studied at LLO. The following subsection describes measurements made at LLO between 2018-2019. Techniques for determining the mode matching between the arms and the OMC include: actuating the waist size and waist location of the beam coming out of the arm and measuring what effect this has on the mode matching to the OMC; direct beam profile measurements; the application of different DARM offsets to determine what portion of light in an OMC scan corresponds to the arm mode and what portion of the light comes from elsewhere in the interferometer (e.g. from contrast defect); and use of a beacon signal to distinguish light from the signal-carrying arm cavity light from junk light. Modelling work relies on the modematching of the PRC and the arms to be $> 99\%$ (estimated from interferometer reflectivity [73]), as this allows singlebounce and full interferometer measurements to be compared. The upper limit on mode matching from the arm to the OMC is 99% , and this was obtained from actuation on the beam parameter and the use of a beacon signal (meaning this must be arm cavity light). The entire collection of measurements find the mode matching to be between 87% and 99% . Since this is quite a large discrepancy, the measurements are briefly discussed below. *Analysis of these results is being developed (cite Carl Blairs work)?.*

Mode scans for an input power of 10W find the mismatch to be 11% [75]. The DARM offset was changed in order to determine what light was coming from the arm mode. An input power of 10W is fairly low, the design is 125W , so the thermal

state of the interferometer may be different in this measurement compared to when the interferometer is in observing mode. This method was repeated for 40W of input power [76]. In this measurement, it was found that there was 4% mode mismatch.

A direct measurement of a beam profile finds there to be A 15% mode mismatch between the OMC waist and the beam transmitted through the SRM while the interferometer was in an ITM single bounce [77] [78] configuration. This measurement also shows a fair amount of astigmatism, with the waist in the x and y directions being separated by ≈ 0.5 m. The x direction data is discarded in the analysis as there was trouble orienting the beam profiler. Since there is good mode matching between the input and the arms ($> 99\%$), there should not be much difference between this beam and the beam when the interferometer is locked as long as the thermal lensing in the ITM was similar to how it would be in observation mode. This type of measurement cannot be performed while the arms are locked as they require commissioner to be working nearby the arm cavities. This analysis is in fairly close agreement with the mode scan measured in [75].

Altering the beam parameter using the ITM thermal lens actuator has shown that the mode matching between the arm cavity and the OMC can be improved by 1% or 2% [79]. Using the SRM CO2 heater increased the optical gain [79], and the amplitude of the beacon signal increased by 10%. The ITM and the SRM are separated by enough Gouy phase that actuating on them covers a sufficient amount of beam parameter phase space in determine if the mode matching was optimised. does [79] agree with [75] and [77] [78]. The interferometer was in single bounce mode, but since the mode matching between the PRC and arm Cavity is high, this

is similar to a locked-arm cavity measurement. There was 25W of input power, so thermal effects may be different between this measurement and observation mode. This type of measurement was repeated using 40W of input power [80]. Assuming the ITM thermal lens was optimal, then the mode matching from the arm to the OMC was found to be optimal with no SRM heating applied. This suggests that the mode matching is $> 99\%$. A beacon signal was used to determine the mode matching from the arm to the OMC.

Modelling of the arm mode finds that a 3.5% mode matching improvement between the arm cavity and the OMC can be made using the ITM thermal lens and the SRM thermal lens [81]. This modelling is compared to the measurement made with data from [78]. It was found experimentally that a 4% improvement in the arm to OMC mode matching could be achieved for a single bounce measurement with 25W of input power. This evidence is in disagreement with [80] which shows that the mode matching is already high.

Just after the interferometer had lost lock, a series of mode scans were used to create a model which predicts 92% mode matching [82] between the beam transmitted through the SRM and the OMC. The input power was 20W and the lock lasted 1 hour and 41 minutes, so the interferometers thermal state immediately after this lock loss was the same as it would be for a typical time during observing mode when the input power was 20W. The thermal lens in the ITM was measured using the Hartmann wavefront sensors, and this was used to model the interferometer at the time of unlocking. The Hartmann data is very noisy, which could cause uncertainty in the mode matching prediction made by the model, although no error on the fit

parameters is given in [82].

The mode matching between the squeezer and the optics within HAM6 was studied [83] [84]. Measuring the beam profile between the squeezer and the OMC allows for the layout in HAM6 to be modelled. This means that there is less uncertainty in where the mismatch between the arms and the OMC may come from. The mode matching between the squeezer and the OMC was measured to be 98%. The model of HAM6 that was developed from this data differs significantly from the nominal values of the optics in HAM6, a 15% error in the RoC of OM2 is required to explain the beam profile measurements. This may not be unreasonable, as accurate measurements of focal lengths for mirrors with $f = 2\text{m}$ are hard to make (ref?).

4.2 Modelling Interferometers With Mode Mismatches

Using FINESSE

Mode decomposition and mode mismatching will be described earlier in thesis. This section is specific to FINESSE.

FINESSE is software for simulating optical layouts. Optical layouts are modelled with components such as mirrors, lenses, and spaces. The way these components are connected is defined by nodes. FINESSE performs a nodal analysis to calculate the steady state amplitude of light fields within the optical layout, similar to how a circuit can be analysed. PyKat is a python wrapper for FINESSE which provides many quality of life improvements such as the ability to programatically updating and re-running simulations [85].

The light fields that are calculated in a FINESSE simulation can be measured by using detectors such as a photodiode, `pd`, or an amplitude detector, `ad`. These can be configured to detect only certain modes using the `mask` command. The `bp` detector is used for measuring the beam parameter, and the `gouy` command computes the Gouy phase accumulated along a set of spaces. These detectors were used to find the ratio of HOM to HG00 light, the beam parameter of the arm mode that is transmitted through the SRM, and the Gouy phase accumulated along the SRC.

The beam parameter at nodes in a FINESSE simulation are determined by `cav` and `gauss` commands. The `cav` command is used for defining linear and circular cavities. The beam parameter is found by solving for the cavity eigenmode. Once these nodes have had a beam parameter defined at them, the components connected to them have their beam parameter calculated based on the set of linear equations that define the component. If there are more than one `cav` commands used in a simulation, the latter `cav` command overwrites the previous one. The `gauss` command executes after the `cav` commands, and sets the beam parameter of the laser used in the simulation.

FINESSE can perform simulations using HG modes, allowing for the effects of mode mismatches to be modelled as long as the problem satisfies the paraxial approximation. A coupling between the HG modes happens whenever a beam described by q_1 enters a segment described by q_2 . This could happen if the `gauss` command used on the laser does not correspond to the eigenmode of a cavity defined by the `cav` command.

A laser beam can be modelled as a linear combination of Hermite Gaussian modes. Ideally an infinite number of these modes would be used in a calculation, however since it is not feasible to simulate an infinite number of TEM modes the maximum number of TEM modes needs to be defined in a FINESSE simulation. This is done using the `maxtem` command. FINESSE will calculate the coupling of the light amongst these modes. If an insufficient number of TEM modes is used in the simulation, the output will be nonsense as power from the higher order TEM modes that would be required to accurately model the optical layout will be ‘aliased’ into the ones defined in the simulation. When a cavity becomes unstable, FINESSE will not generate physical results since the cavity will not be mathematically describable.

The distance between components, L , in a FINESSE simulation is defined with a macroscopic distance, l , which is modelled with the `space` command, and a microscopic distance which is modelled with the tuning parameter, ϕ , belonging to components such as mirrors (`m`) and beamsplitters (`bs`). The macroscopic distance l is always an integer number of wavelengths, whereas the microscopic tuning, ϕ , determines the sub-wavelength accuracy of the length of the space. FINESSE uses the convention that 1 wavelength corresponds to 4π rad phase. It is useful to split the total length, $L = l + \phi$, in this way since the l determines things like Gouy phase, whereas ϕ determines whether a cavity is on resonance/which mode is resonant. When simulating mode mismatches, the command `phase 2` is required to give physically correct results. This sets the phase accumulated by the HG00 mode in a `space` to 0° .

4.2.1 Thin and Thick Beamsplitters

Thin beamsplitters are made with the `bs` command. Care needs to be taken when using these components since incorrect usage can yield erroneous results. To illustrate this two simulations of Michelson interferometers were run, one with a thin beam splitter and the other with a thick beamsplitter. The FINESSE code for these simulations can be found in appendix A.3.

For energy to be conserved, each reflection from a dielectric coating where the refractive index is lower on the reflecting side than on the transmitting side, a light field gains 180° of phase. In a Michelson, there is one path with 2 of the π phase change reflections, while in the other path there is only one such reflection. This means when the arms have equal lengths, destructive interference occurs at the AS port. The intensity, I , at the AS port of the Michelson as a function of the tuning of one of the arm mirrors, $\Delta\phi$, including the FINESSE convention for phase, is

$$I = I_0 \sin^2(\Delta\phi) \tag{4.1}$$

where I_0 is the input power.

The effect of using a thin beamsplitter versus a thick beamsplitter in a FINESSE simulation is shown in figure 4.1. The thin beamsplitter causes there to be constructive interference at the AS port for equal arm lengths, whereas equation 4.1 shows that it should be destructive interference. Using a thick beamsplitter fixes this ‘bug’, as the phase at the beam splitter is properly accounted for.

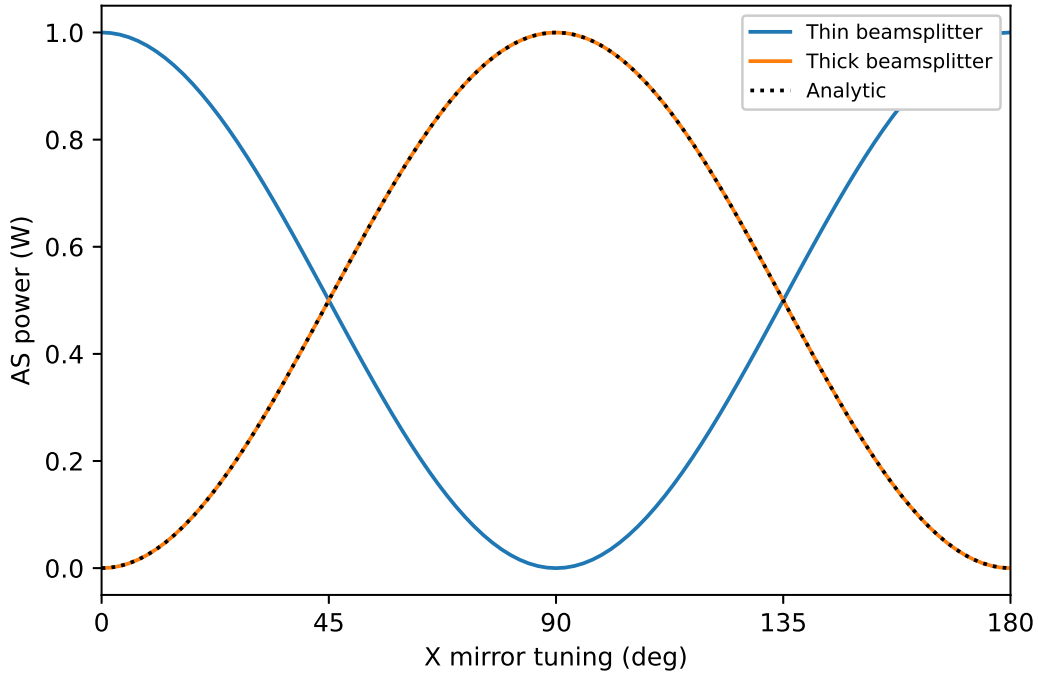


Figure 4.1: The simulation of a Michelson interferometer with thin a beamsplitter is shown in blue. The same simulation was done with a thick beam splitter, this is shown in orange. The analytic behaviour of a Michelson interferometer (equation 4.1) is shown by dashed black line. The tuning of one of the arm mirrors was changed and the power at the AS port was measured. Comparing the outputs of the simulation to equation 4.1 shows that thick beamsplitters should be used.

4.3 Modelling of Michelson Interferometer With Fabry-Perot Arm Cavities and A Signal Recycling Cavity

To investigate what effect a mode-mismatch between the SRC and the arm cavities has on the HOM content of the beam transmitted through the SRM, a ‘LIGO-like’ interferometer was simulated in FINESSE. This interferometer uses identical, design specification arm cavities and the LLO SRC. The reason for using identical, design value arm cavities is to eliminate contrast defect. The OMC, IMC, and PRC were not included in this simulation as these do not effect the beam in the SRC. To perform ITM single bounce measurements, the ETMs are misaligned resulting in the ITM having an effective power reflectivity of 1. To replicate this in FINESSE, the R for each ITM was set to 1 minus the loss of the ITM HR coating. This is valid as long as the mode-matching between the PRC and the arms is close to 100%. The interferometer was set up in DC readout mode. The tuning of the arm mirrors was set so that there was 30 mW of light emerging from the AS port with nominal spacings and optic RoC. A schematic of the simulation is shown in figure 4.2. Nominal parameters in the simulation are listed in table A.7.

Geometry errors in the SRC cause couplings between the HG00 mode and higher order modes. There are two sorts of these errors: RoC errors and spacing errors. RoC errors can be found in SR3, SR2, SRM and the off-design thermal lensing in the ITM and compensation plate, and spacing errors between each of the optics in the SRC. The largest geometry errors in the SRC come from RoC errors in SR3 and

SR2, with both errors affecting the beam in a similar way as there is not much Gouy phase between them. The thermal lensing in the ITM is assumed to be 50 km, and it would require significant deviation, in the direction of more heating, for this lens to have a large effect. The radius of curvature of SRM is well enough known that it is not the dominant cause of junk light in the SRC. The spacings between the optics in the SRC is known to a great enough precision that errors in their locations have a negligible effect. Therefore, to investigate how geometry errors effect the beam transmitted through the SRM, the RoC of SR3 and SR2 will be varied.

All simulations used the same spacings, RoC and focal lengths, apart from the focal lengths of SR3 and SR2 which was a variable for most simulations. The ETM and ITM both have 10 ppm loss. The arms are both made to be identical so there is no contrast defect, and are based off the nominal design. A 50km thermal lens is assumed in all simulations. The lengths in the SRC are ones directly obtained from the Zemax model of the LLO interferometer. To simplify the modelling, the SRC telescope is constructed using `lens` rather than `bs` with `attr` commands (although its not much more complex to use `bs`). The focal lengths in the SRC are the measured values for the LLO optics.

For simplicity, no Schnupp asymmetry was included. In a real interferometer, a Schnupp asymmetry is required to make the interferometer controllable. Since spaces are by default on resonance in FINESSE, no Schnupp asymmetry is needed to control the inner michelson length. The Rayleigh legnth of the beam at the ITM is of the order of several hundred metres which is huge in comparison to the Schnupp asymmetry, which is 8 cm. The Schnupp asymmetry will have a small effect on the

HOM content of the beam transmitted through the SRM, but since the dominant source of HOM comes from RoC errors in SR3 and SR2, the behaviour of the beam in the SRC can be largely understood without it.

The beam parameter of the input laser was defined so that there is no difference between the single bounce and the locked arm simulation. To define the beam parameter, the **gauss** command was used. In this simulation, the mode matching from the laser to the arm cavities should be perfect, the actual amount used in this simulation is 99.99999%. See appendix A.1 for details on how w_0 and z were determined for the **gauss** command used on the laser. The power of the input laser was set to 5 MW.

The **cav** command was used to set up the arm cavities, although the **gauss** command used on the laser means the **cav** command is not required (on the fly testing showed this to be the case). A **cav** command was not used on the SRC since this forces the beam parameter at the SRM to be determined by the SRC geometry. An SRC cannot be defined in FINESSE using the **cav** command as one of the end mirrors cannot be properly defined.

The detectors used to analyse the mode-matching between the nominal arm mode (which matches the laser mode) and the SRC mode were **pd** and **ad** with various **mask** commands applied to them. To detect the total amount of light, no **mask** is applied. To detect the amount of higher order light a **mask [pd] 0 0 0** command is used. This sets the sensitivity of the photodiode to the HG00 mode to 0. The mode matching is then determined as $1 - \text{HOM}/\text{total light}$. **maxtem** is set to 6. As a cross check, 2 amplitude detectors were used to detect the HG02 and HG20 modes:

```
ad ad_HOM_2_y 2 0 0 nSRMb
ad ad_HOM_2_x 0 2 0 nSRMb
```

When plotting the power of the HG20+HG02 mode, the square of the `ad` detector output needs to be taken. The `bp` detector was used to measure the beam parameter. To measure the gouy phase accumulated in the SRC by the HG00 mode, the `gouy` command was used.

Both the signal recycling and RSE modes of operation were investigated. The tuning of the SRM determines whether the interferometer is in RSE mode or in signal recycling mode. If the interferometer has arm cavities, the SRM detuning which corresponds to RSE is 90° while the detuning which gives signal recycling is 0° . For an interferometer without arm cavities, to operate in RSE mode the SRM detuning should be set to 0° , while to be in signal recycling mode the detuning of the SRM should be set to 90° .

4.3.1 Determining Tunings for DC readout

To maintain the same SRC finesse seen by the differential mode for each simulation, the interferometer set up so that it was on the dark fringe with 30 mW of light being transmitted through the SRM. This was achieved by applying a differential tuning to the ETMs for simulations with the arm cavities, and a differential tuning to the ITM for single bounce simulations. The light transmitted through the SRM for an interferometer with arm cavities as a function of differential ETM detuning is shown in figure 4.3. For an interferometer without arm cavities, the power transmitted through the SRM as a function of differential ITM detuning is shown in figure 4.4.

The tunings used to operate with 30 mW of light being transmitted through the SRM are shown in tables 4.1 and 4.2.

Mirror	SRM Tuning ($^{\circ}$)	Mirror Tuning ($^{\circ}$)
ETM _x	0	$+2.76\text{e} - 06$
ETM _y	0	$-2.76\text{e} - 06$
ETM _x	90	$+2.64\text{e} - 05$
ETM _y	90	$-2.64\text{e} - 05$

Table 4.1: The tunings used for the arm-cavity simulation to get 30mW of light at the dark port.

Mirror	SRM Tuning ($^{\circ}$)	Mirror Tuning ($^{\circ}$)
ITM _x	0	$+0.00709$
ITM _y	0	-0.00709
ITM _x	90	$+0.00074$
ITM _y	90	-0.00074

Table 4.2: The tunings used for the no-arm-cavity simulation to get 30mW of light at the dark port.

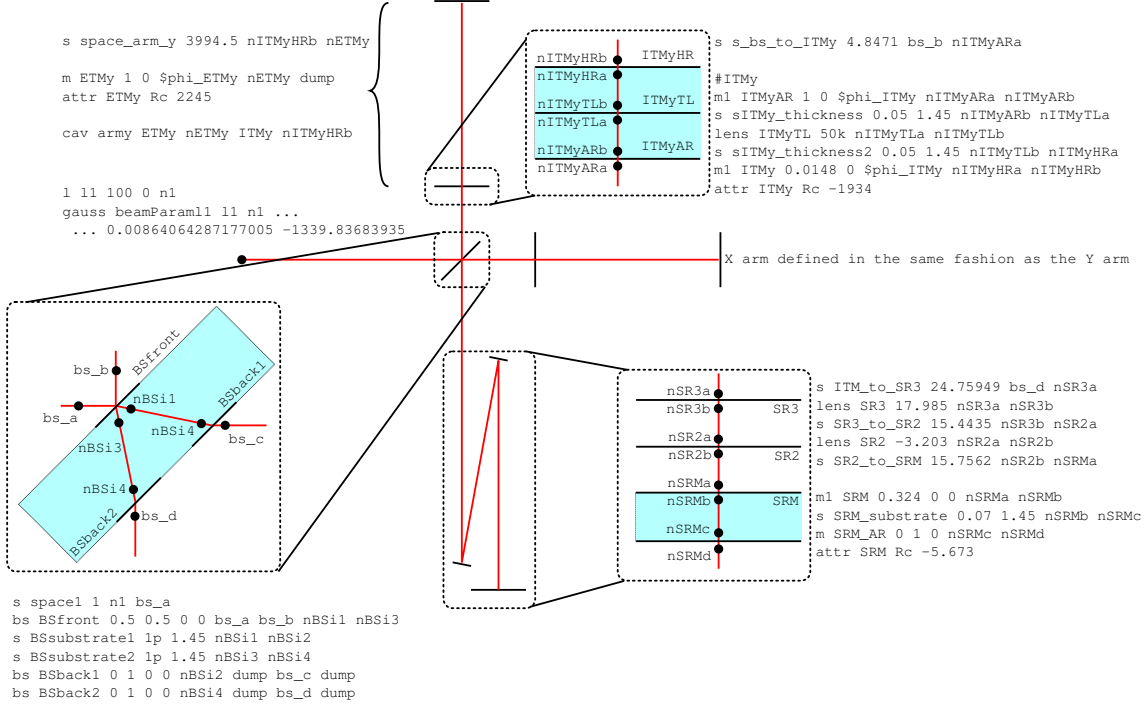


Figure 4.2: Schematic of the simulation. The dimensions of the schematic are not to scale. The laser light is shown as a red line. Black lines with red beams interacting with them corresponds to components such as mirrors and lenses. Important nodes are highlighted with black dots and the associated FINESSE commands are shown nearby. Zoom-ins provide more detail for the main beam splitter, ITMy and the SRC. The elongated curled bracket highlights the arm cavity, and the text with it shows how it is implemented in FINESSE.

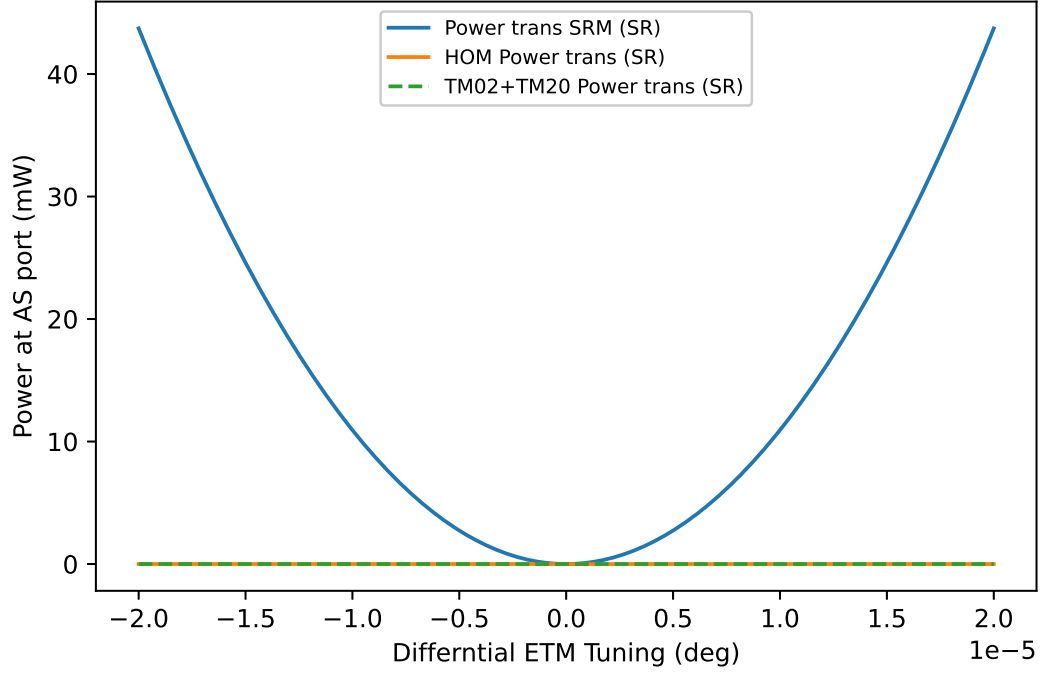


Figure 4.3: The minimum of the dark fringe is at 0° differential ETM detuning, as shown by the blue line. A slight offset causes light to be as the AS port. This result was used to determine tunings for simulating what effect focal length errors in SR3 and SR2 have on the beam transmitted through the SRM. The orange and green dashed line show the HOM content of the beam.

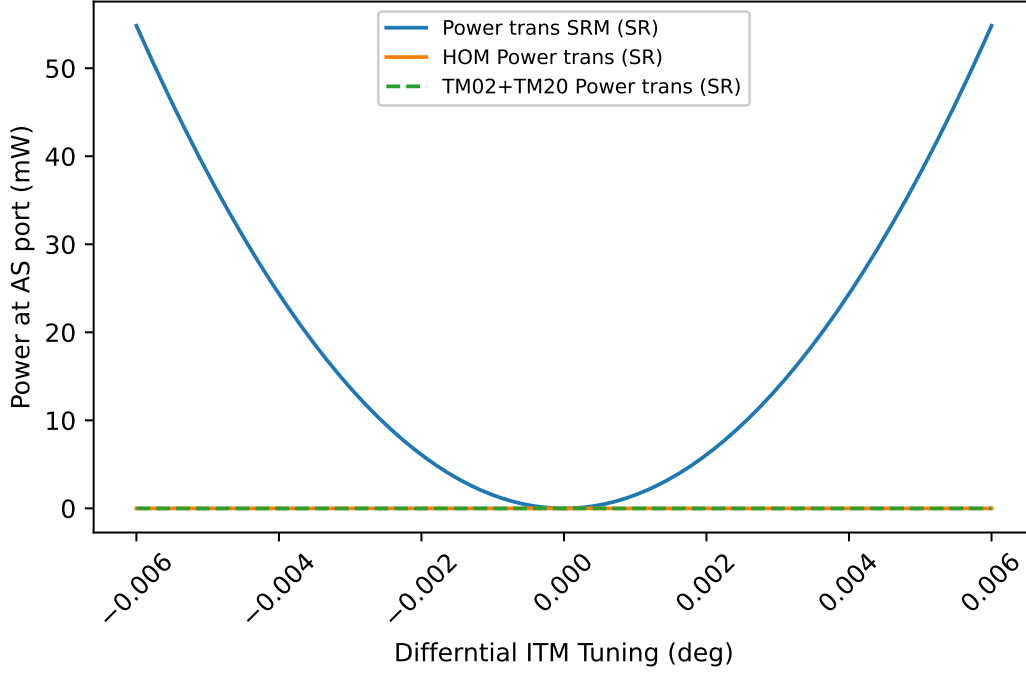


Figure 4.4: The minimum of the dark fringe is at 0° differential ITM detuning, as shown by the blue line. A slight offset causes light to be as the AS port. This result was used to determine tunings for simulating what effect focal length errors in SR3 and SR2 have on the beam transmitted through the SRM. The orange and green dashed line show the HOM content of the beam.

4.4 The Beam Parameter of the Arm Mode at the SRM

The mode defined by the arm was measured at the SRM. The beam parameter at the SRM depends on macroscopic quantities such as the distances between mirrors and the RoC of the optics, rather than microscopic tunings. Therefore, the beam parameter is not affected by the interferometer being in RSE or signal recycling mode. The beam parameter depends only on the geometry of the cavities. The focal length of SR3 and SR2 were varied around 3 standard deviations of the measured values, and the effect this had on the beam transmitted through the SRM was measured with a **bp** detector.

The effect of a focal length error in SR3 and SR2 on the width and RoC of the arm mode at the SRM is shown in figure 4.5, and 4.6. These figures show that the beam parameter depends mainly on the focal length of SR3, although SR2 has a noticeable effect. When the focal lengths of SR3 and SR2 is at the nominal values, the simulated beam has a width of 2.1 mm, a RoC of -4.0 m and the Gouy phase accumulated between the ITM and the SRM is 18.7° . There is a slight difference between the RoC found here and the design value of -3.88 m, this is because the focal lengths of the LLO SRC optics is not the same as the design values. Measurements of the single bounce beam profile [78] suggests that the beam is much larger at the SRM than the design value. This measurement also found the beam to be astigmatic, with $w_y = 2.6$ mm and $w_x = 2.4$ mm. This would correspond to a positive error in the focal lengths of SR3, as shown in figure 4.5. If the beam was this large it is unlikely that the SRC would be stable, as shown in figure 4.9.

The Gouy phase as a function of focal length error in SR3 and SR2 is shown in figure 4.7. Measurements of Gouy phase of the SRC find it to be 22.81° [86] and 21.3° [74]. These measurements of Gouy phase suggests that the beam at the SRM is smaller than the design value, and has a lower curvature. This amount of Gouy phase also corresponds to the same order of magnitude of HG00 loss that is measured and simulated (figure 4.12). The Gouy phase accumulated in the SRC is almost entirely dependent on the RoC of SR3 and SR2, positioning errors of up to 1 cm of SR3 and SR2 do not effect the Gouy phase significantly. Figure 4.7 shows that it is highly unlikely that the Gouy phase could be 22.81° in a state where the arm has greater than 90% mode matching to the SRC, as the RoC of both SR3 and SR2 would have needed to have been three standard deviations from the measured value. This could be accounted for by an much shorter thermal lens in the ITM (see figure 4.8). The measurement of 21.3° of Gouy phase is plausible with the 50 km thermal lens. The single pass thermal lens was measured to be 38.92 uD or ≈ 25.7 km [82], which corresponds to the design value and the value used in this simulation.

The focal length of SR3 and SR2 can be estimated from the Gouy phase measurement. A normal probability density function was used to model the uncertainty in the focal length of SR3 and SR2. This was combined into a 2-D normal distribution. Combinations of SR3 and SR2 that result in 21.3° of Gouy phase where found and the value of the 2-D normal distribution for those combinations was calculated. The maximum value of the resulting probability density function (which is normal) corresponds to the most likely combination of SR3 and SR2, and one standard deviation corresponds to the error. Figure 4.10 shows the that focal length of SR2 to

be -3.205 ± 0.003 m and the focal length of SR3 to be 17.971 ± 0.002 m.

The beam parameter at the SRM is determined by the focal length of SR3 and SR2. Each beam width and RoC that was computed in this simulation has a probability density associated with it (as this depends on SR3 and SR2, see figure 4.10). Therefore, the probability density function for the beam parameter of the arm mode at the SRM can be computed. Based on the measurement of Gouy phase, the beam is likely to have width 1.84 mm and a defocus of -2.80 Dioptre . This is shown in figure 4.11. This is a slight simplification as the beam parameter cannot be such that the Gouy phase of the cavity becomes too small ($< \sim 17.5^\circ$). IF the Gouy phase is too small the cavity will be unstable, so beam widths of more than 2.25 mm are very unlikely. The shape of the distribution of the arm mode beam parameter at the SRM is driven mostly by the focal length error of SR3.

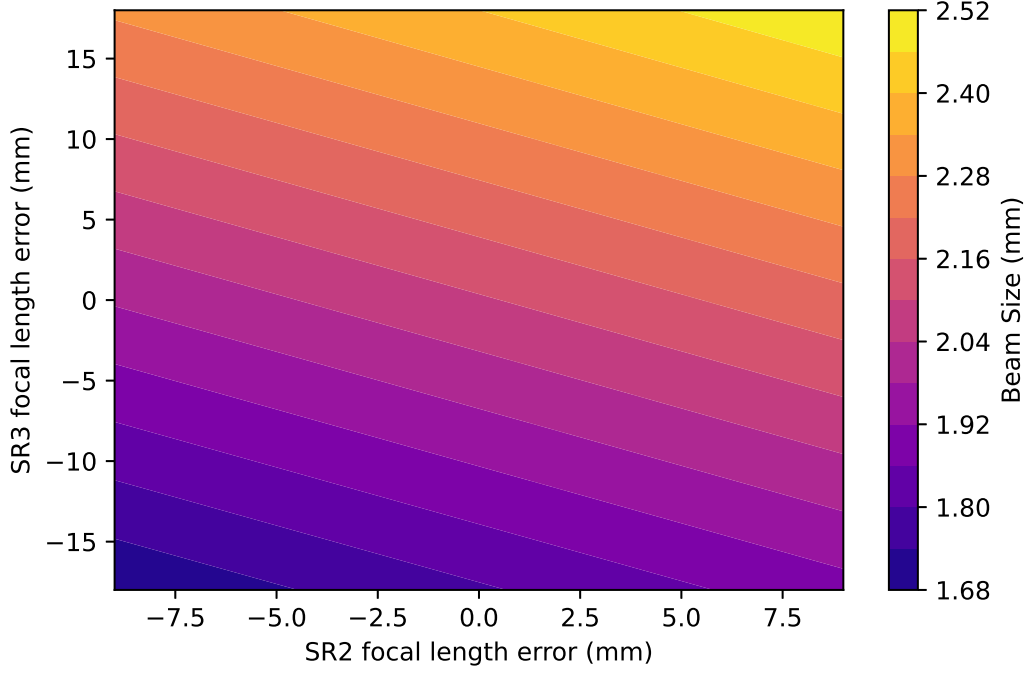


Figure 4.5: The Gaussian width of the beam that is transmitted through the SRM as a function of the error ($\pm 3\sigma$) in focal length of SR3 and SR2. The Gaussian beam width is strongly dependent on the error in SR3's focal length, whereas SR2's focal length error is less important. When the SR3 focal length error > 6 mm and SR2 error is zero the SRC cavity is unstable (see figure 4.9), so this area of the parameter space is unlikely to correspond to the beam at the SRM at LLO.

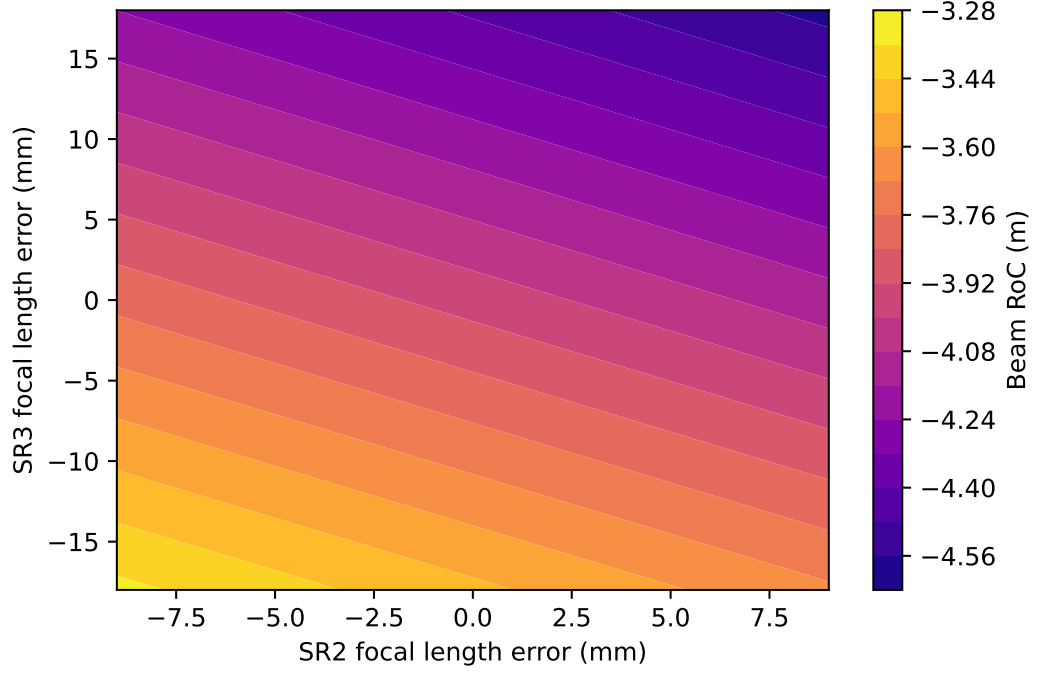


Figure 4.6: The RoC of the beam that is transmitted through the SRM as a function of the error ($\pm 3\sigma$) in focal length of SR3 and SR2. If the arm cavities and the SRC are well mode matched, then the RoC of the beam transmitted through the SRM would be $\frac{5.67\text{m}}{1.45} = 3.91\text{m}$. When the SR3 focal length error > 6 mm and SR2 error is zero the SRC cavity is unstable (see figure 4.9), so this area of the parameter space is unlikely to correspond to the beam at the SRM at LLO.

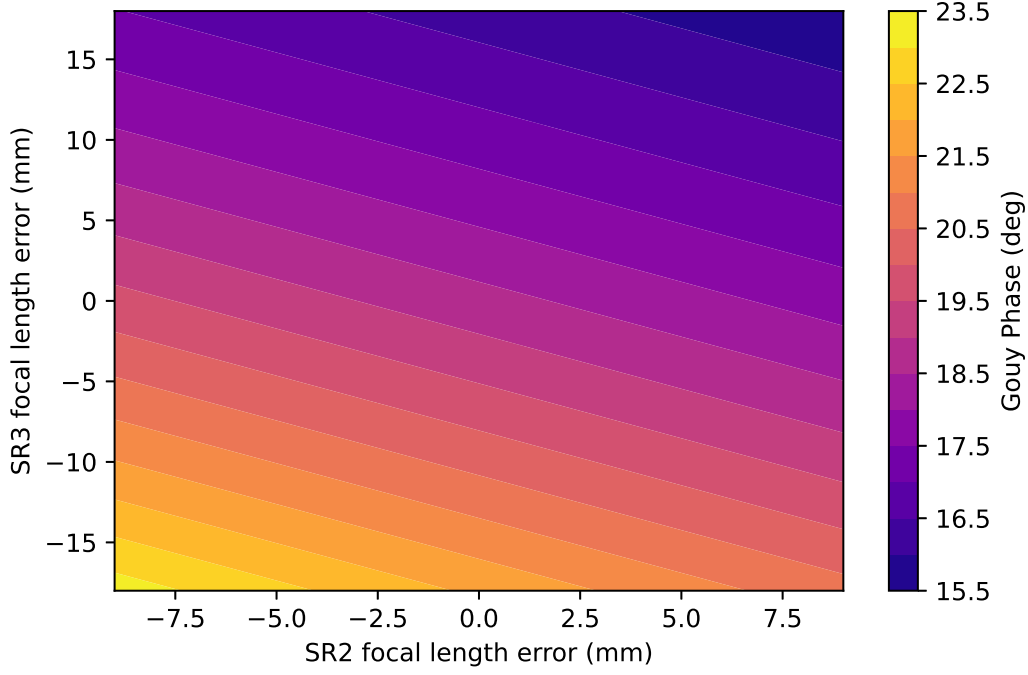


Figure 4.7: Gouy phase between the ITMs and the SRM as a function of the error ($\pm 3\sigma$) in SR3 and SR2. When the SR3 focal length error > 6 mm and SR2 error is zero the SRC cavity is unstable (see figure 4.9), so this area of the parameter space is unlikely to correspond to the beam at the SRM at LLO.

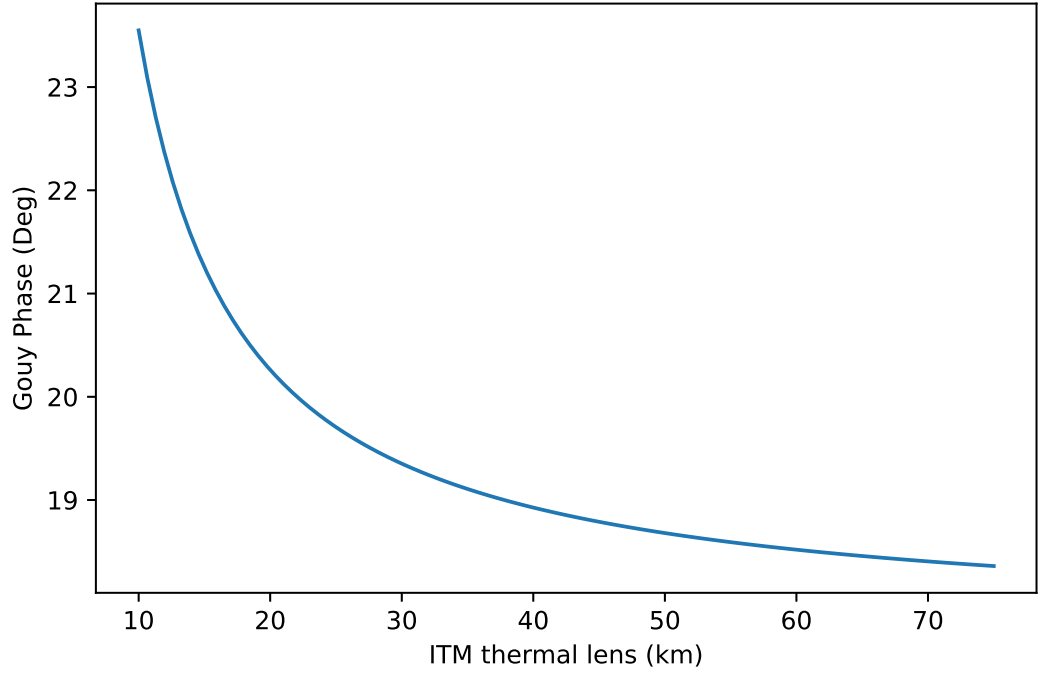


Figure 4.8: Gouy phase between the ITMs and the SRM as a function of the thermal lens in the ITMs for nominal values of SR3 and SR2. The nominal ITM thermal lens is 50 km, and this corresponds to about 19°. More heating results in a greater thermal lens, however at LIGO this is compensated for by the thermal compensation system (TCS).

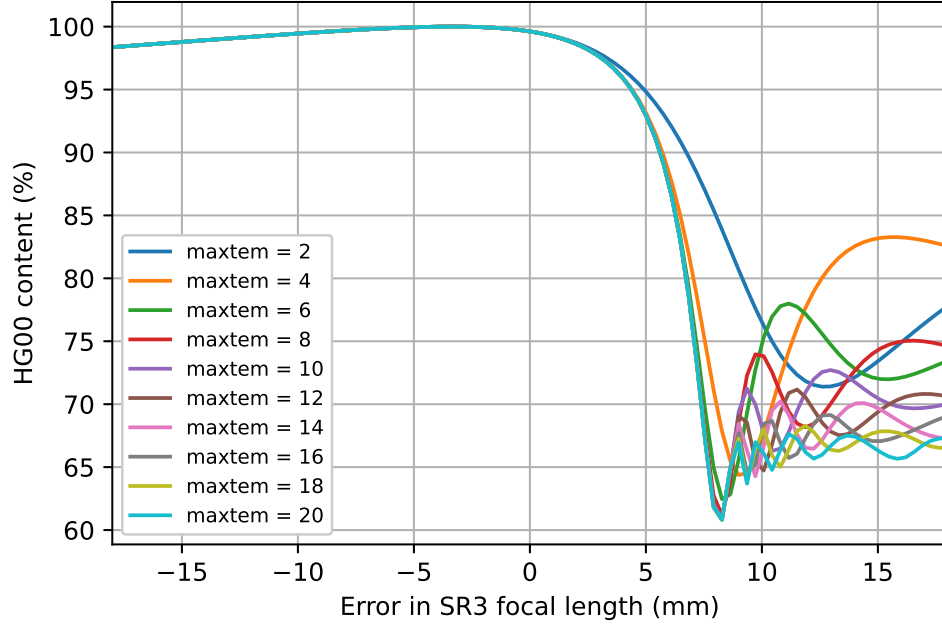


Figure 4.9: The percentage of HG00 light transmitted through the SRM as a function of SR3 focal length for the Michelson interferometer with Fabry-Perot cavities in its arms was simulated. Each simulation used a different `maxtem` command. The focal length of SR3 was changed. For focal lengths with a positive error greater than 7.5 mm, the cavity becomes unstable and FINESSE does not produce reliable results. For a stable SRC, the minimum number of TEM modes for sub 1% accuracy is six.

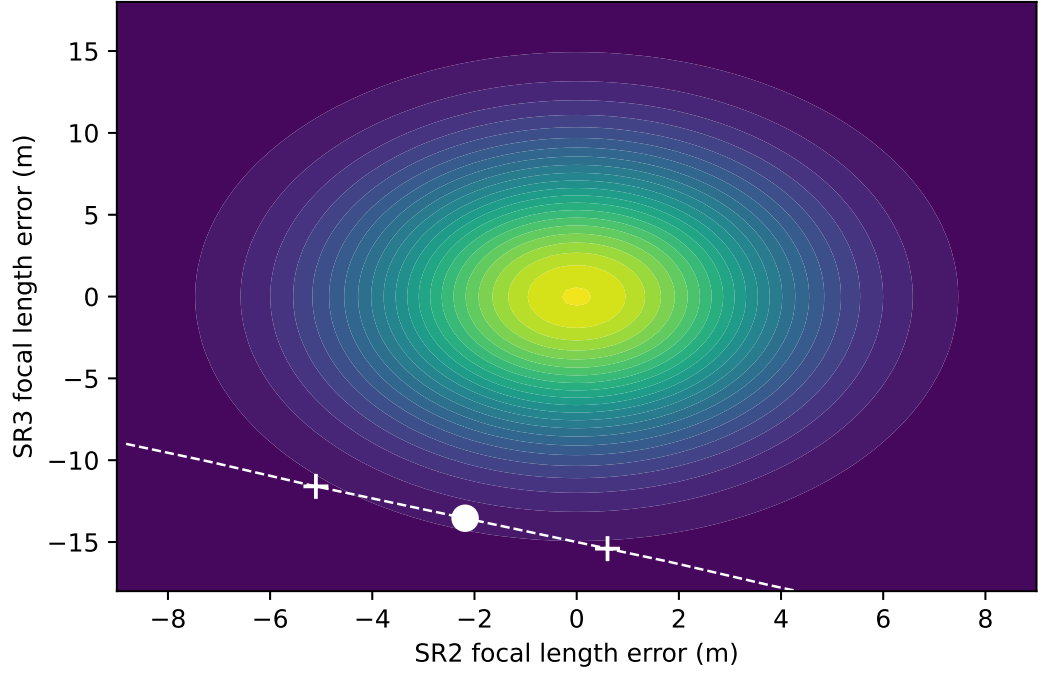


Figure 4.10: The measurement error of SR3 and SR2 follows a normal distribution. These were combined to make a 2-D normal distribution. Dark blue corresponds to the least likely combinations of SR3 and SR2, and bright yellow corresponds to the most likely. The measured Gouy phase was 21.3° [74], this is indicated by the white dashed line. The white circle marks the most likely combination of SR3 and SR2 which gives 21.3° of Gouy phase, and the white crosses show 1 standard deviation from this. The white circle and crosses indicate that the focal length of SR2 is -3.205 ± 0.003 m and the focal length of SR3 is 17.971 ± 0.002 m.

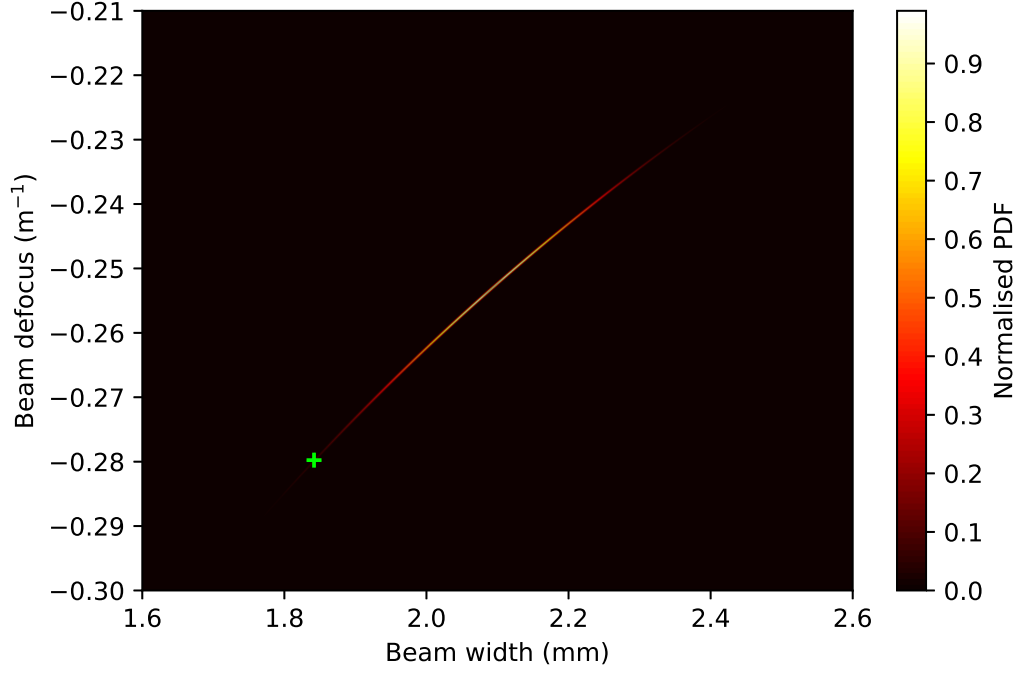


Figure 4.11: The beam parameter at the SRM was calculated for combinations of SR2 and SR3 focal lengths. The associated probability density for these combinations can be calculated (see figure 4.10), so the probability of the beam parameter having a certain width and defocus can be calculated. The probability density for each pair of beam width and defocus is indicated with the colour bar. The measured Gouy phase of 21.3° [74] is indicated with a green cross (1.84 mm, -2.80 Dioptre). This is a slightly simplified approach since the beam parameter cannot be such that the SRC is unstable. This means certain combinations of beam width and beam defocus are not possible. Beam parameters with a width $> \sim 2.25$ mm result in an unstable SRC, therefore these beams are very unlikely to be present at the SRM at LLO.

4.5 Power Transmitted Through SRM for a Mode Mismatch Between the Arm Cavity and the SRC

The HOM content of the beam as a function of the error in focal length in SR3 was investigated. As the arm cavities become more mismatched to the SRC, less HG00 mode is transmitted. The mode mismatch can be seen in figure 4.6. As the focal length of SR3 becomes further from its nominal value, the radius of curvature of the beam at the SRM stops matching the radius of curvature of the SRM. Since the SRC is a low finesse cavity, a significant amount of mismatch can be tolerated before there is a significant drop in the HG00 power transmitted through the SRM. This is shown in figure 4.12. Figures 4.12 and 4.13 show that the nominal value of the focal length SR3, 17.985 m, is slightly longer than optimal. When the SRC is on resonance, the peak in transmission is -3 mm from the nominal focal length of SR3. This is shown by the blue and green lines in figure 4.13.

The power of the HG00 light transmitted through the SRM as a function focal length error is shown in figure 4.13. In signal recycling mode, shown as blue and green in figure 4.13, the transmitted power has a peak shape. This is because an error in the focal length of SR3 moves the cavity off resonance. In RSE mode, shown as red and orange in figure 4.13, a change in focal length of SR3 does not have much effect on the amount of HG00 transmitted by the SRM since an anti-resonance is a much broader feature than a resonance.

In a single bounce measurement, the beam incident onto the ITM from outside the cavity is reflected straight back. This is equivalent to the reflectivity of the ITM

$R = 1$. When the light is resonant in the arm cavity, the ITM acts as a highly reflective mirror with some frequency dependent behaviour. This means that there is no significant difference between a single bounce measurement and a measurement with arm cavities. This is shown in figure 4.12. Higher order mode beams generated by errors in the SRC are not suppressed by the arm cavity as they are reflected by the ITM.

The ratio of the amount of light in the HG00 mode versus the total light depends on the detuning of the SRM, rather than the presence of cavities in the arm. This is shown in figure 4.12. When the light enters the arm cavities, it is reflected back with an extra π rad phase, however the HOM that does not enter the cavity does not acquire this phase. Since the HOM only interact with the SRC, the ratio of HOM to HG00 depends on the detuning of the SRM, rather than whether the interferometer is in signal recycling or RSE mode. The amount of higher order mode generated in the SRC is proportional to the HG00 power in the SRC, so the percentage of HOM beam is dependent only on the tuning of the SRM and the focal length error of SR3.

As the SRC gets closer to being unstable, more TEM modes are required to simulate the behaviour of the cavity. The effect of using more TEM modes in the simulation is shown in figure 4.9. For positive errors above 6 mm in the focal length of SR3, the cavity becomes unstable. As the error in SR3 focal length approaches this, a greater number of TEM modes is required to accurately simulate the behaviour of the interferometer. It was found empirically that six TEM modes is sufficient to simulate the interferometer while it is stable. The greyed out regions of figures 4.12 and 4.13 show where the simulation is not accurate.

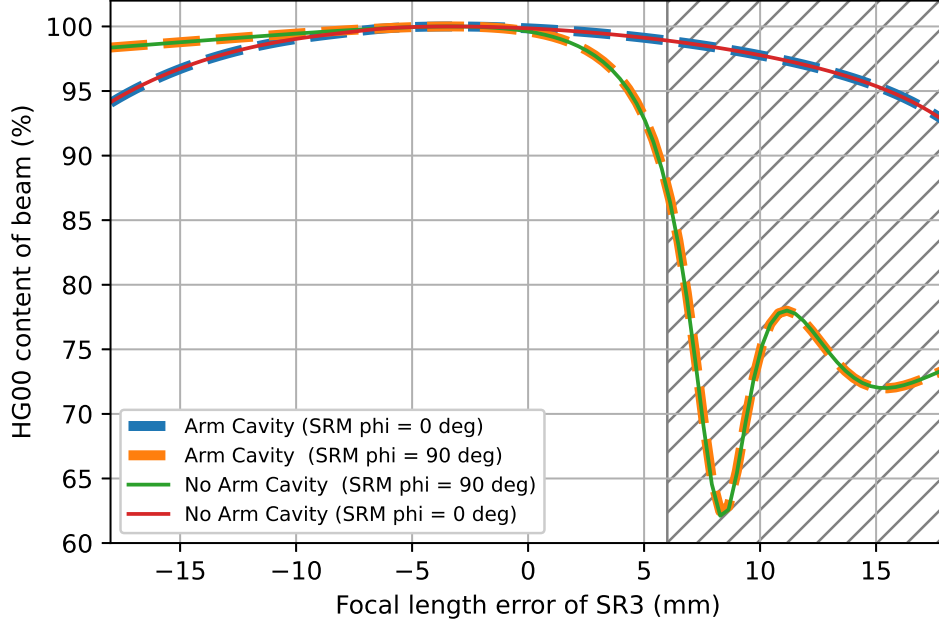


Figure 4.12: The percentage of light which is in the HG00 mode as the focal length of SR3 changes compared to the total light power does not depend on whether there are arm cavities. It is determined by the detuning of the SRM. The blue dashed line shows the behaviour of the beam with arm cavities present and with the SRC in signal recycling mode (SRM $\phi = 0^\circ$). The red line shows how the beam changes with focal length error for a single bounce measurement and with the SRC in RSE mode (SRM $\phi = 90^\circ$). The orange dashed line corresponds to the interferometer having arm cavities and being in RSE mode (SRM $\phi = 90^\circ$). The green line shows behaviour of the beam when the interferometer is configured for a single bounce measurement and is in signal recycling mode. The hatched region indicates where the SRC becomes unstable (see figure 4.9), so the behaviour of the interferometer cannot be accurately simulated for these values of focal length.

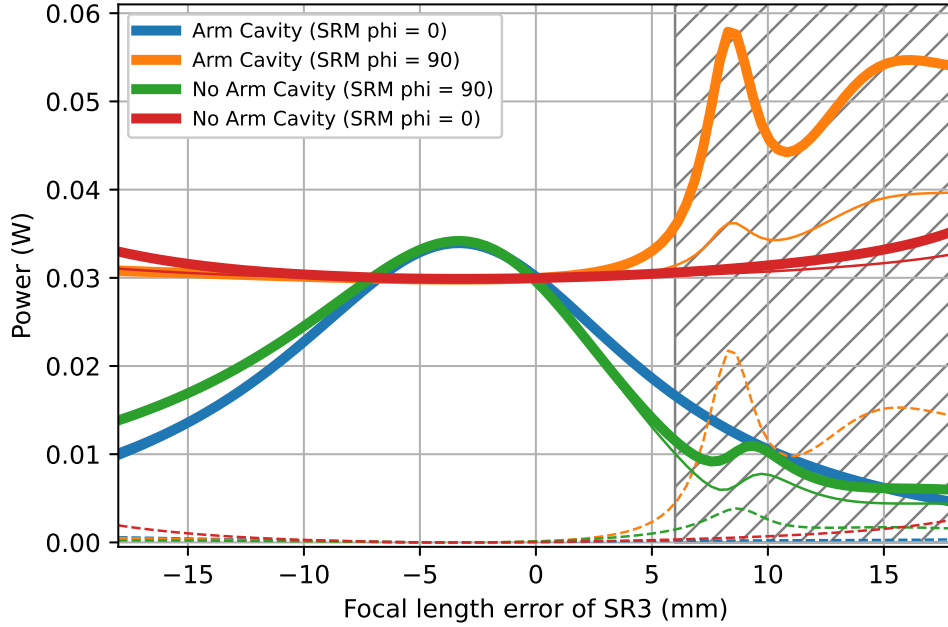


Figure 4.13: The total power transmitted through the SRM for each simulation, is shown in the thick pale lines. The thin solid lines corresponds to the power in the HG00 mode, and the thin dashed lines correspond to the power in the higher order modes. Blue lines show the behaviour when the interferometer has arm cavities and is in signal recycling mode, and the orange lines shows when it is in RSE mode. Green lines show how the power changes when the interferometer is configured for a single bounce measurement and is in signal recycling mode, and the red line shows when its in RSE mode. The hatched region indicates where the SRC is becomes unstable, and so the behaviour of the interferometer cannot be accurately simulated for these values of focal length.

4.5.1 Change of finesse of SRC

As the transmission of the SRM decreases, the finesse of the cavity increases. A FINESSE simulation was run to see the effect of increasing the power transmission of the SRM. The HG00 mode power decreases with lower SRM transmission slightly slower than the power transmitted in the HG20+02 mode. This is shown in figure 4.14.

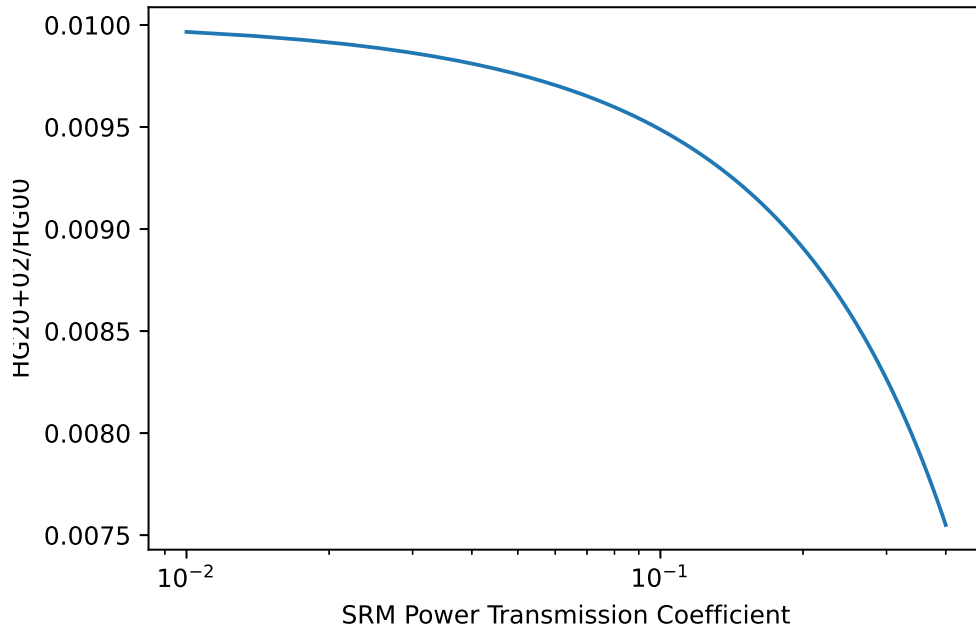


Figure 4.14: Ratio of HG20+02 to HG00 as the transmission of the SRM is increased for a mode mismatch between the arm cavity and the SRC. The interferometer is in RSE mode. A mode mismatch was introduced by setting the focal length of SR3 to be 12 mm shorter than the nominal amount.

4.6 Conclusion

Mode-mismatching between the arm cavities and the SRC results in the scattering of HG00 mode into higher order modes. Losses in HG00 usually reduces the SNR by $\sqrt{\text{Power}}$, however since there are multiple points between the squeezer and the OMC where mode-mismatches can happen, the quantum noise is enhanced by more than the usual amount for a loss. There exists a collection of measurements which show the arms are matched between 92% and 98% to the OMC.

FINESSE simulations were used to model mode-mismatches between the arm and the SRC caused by RoC errors in SR3 and SR2. The simulation used as built parameters for the SRC at the LLO site and design specification arm cavities. This allows us to study the effect of non-ideal mode-matching in a close to real life situation while reducing the complexity of the model. The interferometer was modelled in signal recycling mode and RSE mode for and locked arm cavities and single bounce configurations. It was found that there is no significant difference between single bounce and locked arm cavity configurations as there is very high levels of mode matching between the PRC and the arm cavity. In both single bounce and locked arm modes, the ITM forms a highly reflective mirror as almost all the light is reflected back. The amount of HOM is proportional to the HG00 mode in the SRC. The ratio of HOM to HG00 depends on the detuning of the SRM. As HOM do not enter the arm cavity, they cannot distinguish between the signal recycling and RSE modes of operation.

The uncertainty in the beam parameter of the arm mode at the SRM and the eigenmode of the SRC is driven by the uncertainty in SR3. This is shown in figures

4.5 and 4.6. The single pass Gouy phase of the SRC for different combinations of SR3 and SR2 was simulated, and is shown in figure 4.7. Using the measured single pass Gouy phase of the SRC, the focal length of SR2 was found to be -3.205 ± 0.003 m and the focal length of SR3 was found to be 17.971 ± 0.002 m. This is shown in figure 4.10.

Future work could involve using a heater to change the focal length of SR3. This would produce a predictable change in the focal length of SR3. This could be compared to the Gouy phase of the SRC to check that this changes the in mode matching the expected way. As the lengths between all the optics is well known, and the focal length of the ITMs is well known, the squeezer beam could be used to probe the radius of curvature of SR3 and SR2. By setting up a single bounce measurement, the incoming and outgoing beam profile could be measured and compared to a model. Having greater confidence in the beam parameter of the arm mode at the SRM would be of great benefit for the design of future upgrades to LIGO.

Chapter 5

Balanced Homodyne Detection for LIGO A+

A major change between aLIGO and LIGO A+ will be the replacement of DC readout with Balanced Homodyne Detection (BHD). BHD plays a key role in many experiments that utilise non-classical states of light (for a recent example of such an experiment relating to gravitational wave interferometry, see [33]). The transition from aLIGO to LIGO A+ involves the redesign of many aspects of the interferometer, and elements of this effort aim to exploit quantum effects in order to increase the SNR of the detector. Therefore, the implementation of BHD is an attractive prospect for those who want to harness the quantum nature of light to enhance the detector.

It was found that, aside from the potential quantum mechanical benefits, BHD offers the potential to increase the SNR by reducing Signal Recycling Cavity (SRC) length noise coupling to the differential arm motion (DARM) signal. Even if LIGO were to never operate in a novel, quantum-mechanically enhanced way, the reduction

of SRC length noise is enough motivation to implement BHD. The longitudinal motion requirement for the optics that deliver the signal and local oscillator (LO) beams to the combining beam splitter was found to be $\leq 5 \times 10^{-17} \text{m}/\sqrt{\text{Hz}}$ at 100 Hz in order to avoid increasing the noise floor of the detector. Therefore, triple suspensions will be used to hang these optics.

This Chapter begins by exploring the fundamentals of BHD. Section 5.2 contrasts BHD with DC readout within the specific setting of LIGO A+ and Section 5.3 derives the phase noise requirement for the LO beam for A+. Section 5.4 concludes this Chapter.

5.1 What is BHD

BHD is a method of measuring information encoded by a light field, and it forms a key part of many quantum optics experiments. By combining a signal beam with a much stronger LO beam, phase modulations in the signal beam become amplitude modulations in the combined LO-signal beam (e.g. [34]). BHD can be configured to measure any quadrature of the signal beam which is vital for inferring the quantum state of a system and is required for more complex topologies of interferometer [27, 87–89]. BHD has been implemented to detect 11.6 dB of shot noise suppression to below 1 Hz due to the squeezing of vacuum fluctuations. [90].

All readout schemes for gravitational wave detectors rely on a LO field of some kind. The first generation of detectors used heterodyne readout (e.g. [91]). Heterodyne readout involves a modulation-demodulation scheme whereby the signal is mixed with a RF LO to a frequency where technical noise is no longer a concern. It

is then demodulated at the same modulation frequency to recover the original signal. This method comes with the draw back that the quantum shot noise is enhanced by the modulation-demodulation process [92].

The LO in a BHD readout scheme is a light field with the same optical frequency as the signal beam. The LO beam has a higher power than the signal beam and acts as a stable reference frequency. Ideally the LO would have no phase noise, however in reality there are fluctuations in the phase between the LO and the signal beam. The interference of the LO and signal beam causes this phase noise to become an amplitude modulation in the combined LO-signal beam [93]. This phase noise must be kept below the size of the phase fluctuations in the signal beam that are to be measured, which in A+ are determined by the shot noise of the light within the arm cavities. Unlike Heterodyne readout, BHD does not cause a fundamental increase in the noise floor of the measurement.

BHD is implemented by injecting the LO and signal beams into different input ports of a balanced beam splitter and placing photodiodes in the remaining output ports. The information encoded in an arbitrary quadrature of the signal beam can be selected by correctly setting the relative phase between the LO and signal beams. The relative phase between the LO and signal beam is also known as the *homodyne angle*. Subtracting the two photo-currents results in a measurement which is proportional to the selected quadrature of the signal beam. A simple BHD layout is shown in Figure 5.1.

Due to the balanced nature of the beam splitter, the classical intensity noise of the LO is not present in the BHD signal. This and the ability to set the homodyne

angle make BHD ideal for measuring quantum states of light. For example, BHD can be used to detect the level of squeezing in a quadrature of the signal beam by setting the homodyne angle to the quadratures angle (e.g. [94]).

BHD is useful for inferring the quantum state of two-photon phenomena, such as squeezing, as without it the phase quadrature cannot be measured. A two-photon process can be described in terms of the amplitude and phase quadrature operators, denoted as $\hat{\mathbf{X}}$ and $\hat{\mathbf{Y}}$ respectively (e.g. [30]). These operators are related to the annihilation and creation operators of the electromagnetic field, $\hat{a}^\dagger(t)$ and $\hat{a}(t)$, by

$$\hat{\mathbf{X}}(t) = \frac{1}{2} (\hat{a}(t) + \hat{a}^\dagger(t)) \quad (5.1)$$

$$\hat{\mathbf{Y}}(t) = \frac{1}{2i} (\hat{a}(t) - \hat{a}^\dagger(t)) \quad (5.2)$$

$$\hat{\mathbf{X}}_\phi = \hat{\mathbf{X}} \cos \phi + \hat{\mathbf{Y}} \sin \phi. \quad (5.3)$$

Consider a LO field $\hat{a}$ and a signal field $\hat{b}$. It is arranged so that these fields interfere on a beam splitter and are detected on two photodiodes, c and d . The fields these diodes detect are $\hat{c}$ and $\hat{d}$. The photo-currents are then

$$\hat{\mathbf{N}}_c = \hat{c}^\dagger \hat{c} = \left(\sqrt{R} \hat{a}^\dagger + \sqrt{T} \hat{b}^\dagger \right) \left(\sqrt{R} \hat{a} + \sqrt{T} \hat{b} \right) \quad (5.4)$$

$$= R \hat{a}^\dagger \hat{a} + T \hat{b}^\dagger \hat{b} + \sqrt{RT} (\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}) \quad (5.5)$$

$$\hat{\mathbf{N}}_d = \hat{d}^\dagger \hat{d} = T \hat{a}^\dagger \hat{a} + R \hat{b}^\dagger \hat{b} - \sqrt{RT} (\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}). \quad (5.6)$$

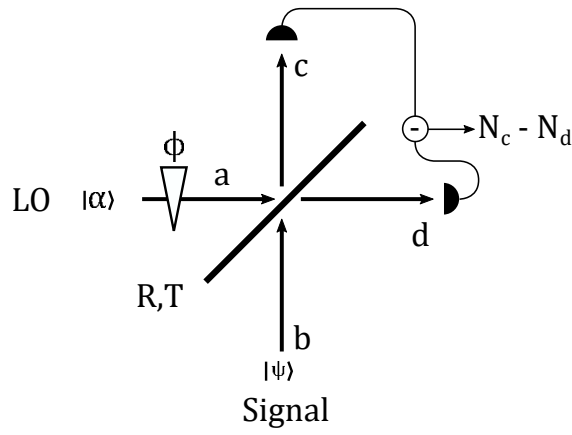


Figure 5.1: A coherent state $|\alpha\rangle$, enters port a of a beam splitter. A signal $|\Psi\rangle$ enters port b . These two beams interfere, resulting in photons being detected at the c and d ports. The two photo-currents are subtracted, and the signal that is produced is proportional to an arbitrary quadrature of the state $|\Psi\rangle$. The quadrature that is measured is determined by the homodyne angle, ϕ . This is expressed mathematically by equation (??).

For BHD, the beam splitter must be balanced so $R = T = 1/2$. This results in the difference in the photo-currents being

$$\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d = \hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}. \quad (5.7)$$

The operators $\hat{a}$ and $\hat{b}$ can be ‘linearised’, $\hat{a} \rightarrow \alpha + \delta a$, [?] in order to separate the quantum behaviour and classical behaviour. Furthermore, the phase between the signal LO field can be expressed in terms of the homodyne angle with $\hat{b} \rightarrow \hat{b} \exp(i\phi)$. By neglecting terms of the order $\delta a \delta b$, this allows one to rewrite equation (5.7) as

$$\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d = 2\alpha\beta \cos \phi + 2\alpha\delta \mathbf{X}_{-\phi}^b + 2\beta\delta \mathbf{X}_\phi^a. \quad (5.8)$$

The variance, $\sqrt{V(X)} = \Delta X$, of this measurement would then be

$$V(\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d) = 4\alpha^2 V(\delta \mathbf{X}_{-\phi}^b) + 4\beta^2 V(\delta \mathbf{X}_\phi^a). \quad (5.9)$$

If the LO is much stronger than the signal field, $\alpha \gg \beta$, then the measurement is

$$\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d = 2\alpha\beta \cos \phi + 2\alpha\delta \mathbf{X}_{-\phi}^b, \quad (5.10)$$

which is proportional to general quadrature of the signal.

5.2 BHD versus DC readout for A+

There are some advantages for BHD over DC readout [95], although for A+ a quantifiable difference is hard to make. The main attraction for implementing BHD is the ability to reduce the dark fringe offset that is required to produce the LO for DC readout. This offset gives rise to most of the limitations associated with DC readout, such as the coupling between the SRC length noise and the DARM signal. BHD also provides the ability to select which quadrature of the signal beam is measured which is a requirement for novel interferometer configurations.

5.2.1 Perks of DC readout

The current detection scheme is known as DC readout [96]. A differential offset in the arm lengths produces the LO in DC readout. The LO beats with a signal created by a gravitational wave in order to produce intensity modulations at the AS port. The size of the LO can be controlled with the DC offset, and the homodyne angle can be controlled with a differential arm power. This method of readout has some key positive outcomes, such as there being no phase noise between the LO and the signal, and the LO and signal being perfectly aligned and mode matched. The problems with this readout technique are associated with requiring the DC offset as this light gives rise to noise couplings.

For BHD to be worthwhile it must be at least as good as DC readout. Thus, BHD should be considered in relation to the perks of DC readout. Unlike DC readout, BHD may introduce phase noise between the LO and signal [93]. Therefore the relative stability between the LO and signal beams has to be considered. Control

over the alignment and mode matching between the LO and the signal is also important. LO light that is scattered into higher order modes, by misalignment or mode mismatch, may backscatter from the OMC into the interferometer and could result in noise in the DARM signal. The implementation of BHD will only be successful if it provides a stable enough LO beam; sufficient mode-matching between the LO and signal beam, and active alignment of the LO and signal beams.

5.2.2 BHD Advantages

BHD has some practical benefits over DC readout [95]¹ as well as some interesting prospects in relation to the quantum noise limit of the detector. The practical benefits arise due to being able to eliminate the DC light required for DC readout to work. The primary benefit is in the reduction in SRC length noise coupling to the gravitational wave signal. Another benefit is being able to reduce the scattering from the septum window between HAM5 and HAM6. Quantum noise shaping is, along with other interferometer enhancements (e.g [27]), related to the complete free choice of homodyne angle that a BHD readout scheme permits.

¹One ‘advantage’ that is often talked about in relation to BHD is based on the electronic noise of the readout photodiodes. This is not a real advantage. aLIGO currently uses DC readout and is not limited by the electronic noise of the transimpedance amplifier. As the BHD signal is obtained through subtracting the signals from two photodiodes, it is possible to obtain a high SNR when designing the transimpedance amplifier by subtracting the currents produced by the photodiodes rather than having two separate transimpedance amplifiers and subtracting the signals made by these later. This is not really an advantage, as low enough electronic noise can be obtained using other construction techniques that allow for small DC signals while still having signal shaping that gives a large enough SNR in the detection band. Having separate transimpedance amplifiers for each photodiode also has useful diagnostic advantages, such as being able to see the sum of the two photodiode signals. Therefore, direct subtraction of the photodiode signals should not be considered an advantage of BHD over DC readout.

5.2.2.1 Quantum Noise Improvements

Different gravitational wave sources carry their information in different frequency bands and the detector may be optimised to reflect this [27, 97–99]. For instance, BNS signals continue to higher frequencies than BBH signals so it may be of interest to optimise the detector around these frequencies to improve our understanding of the properties of BNS systems. To optimise the sensitivity of the interferometer to different sources the detectors noise would need shaping, a task that can be achieved, along with other upgrades, BHD.

The current generation of gravitational wave detectors are going to be limited by a combination of radiation pressure noise and shot noise in their detection bands. These two noise sources are intrinsically linked by the Heisenberg uncertainty principle and can both be understood as arising due to the number of photons used to probe the position of the end test masses. This limit in the noise floor is known as the standard quantum limit [27]. To circumvent this limit the homodyne angle must be free to choose; a functionality that is provided by BHD. In principle, DC readout gives the ability to set the homodyne angle [95], although in practice it is much less flexible than a BHD configuration.

A squeezed light field can be used to surpass the standard quantum limit, and to detect these squeezed states the homodyne angle must be correctly set. The exact way in which the quantum noise is changed is controlled by the squeezing angle. Currently the squeezing angle is set to be at the same angle as the readout quadrature, rather than the preferred option of setting the readout quadrature to the squeezing angle. The effect that controlling the squeezing angle has on the quantum

noise floor has been demonstrated at LLO [33]. However, [33] used a diagnostic BHD set-up which was not intended for providing the low noise required to make gravitational wave detections.

More advanced upgrades, e.g. see [27], could alter quantum noise substantially, and they rely on the implementation of BHD. These changes to the interferometer are nontrivial, and would require significant redesign of many parts of the layout, so the implementation of BHD for A+ does not necessarily correspond to a suitable BHD system for one of these more advanced interferometer configurations. However, BHD for A+ would provide experience with implementing BHD on a large scale interferometer, which would help when considering how to implement these more novel interferometer configurations.

5.2.2.2 Consequences of a Lower DC Offset

The dominant coupling due to the power entering the SRC can be intuitively understood by considering a Fabry-Perot cavity. For small changes in cavity length, the light transmitted by a cavity that is on resonance or anti-resonance will be modulated by a term proportional to the square of the length change. This signal due to the length of the SRC is noise. The amplitude of this noise is linearly dependent on the amount of light entering the cavity which is a result of the DC offset. By decreasing the DC offset, this noise is decreased. As this noise is in the amplitude quadrature of the signal beam, unlike the gravitational wave signal which is in the phase quadrature, it will not be amplified by the LO if the homodyne angle is set to readout from the phase quadrature.

The second route SRC length noise may couple to the signal is via the optical springs between the SRC and arm cavities. Due to this optical spring effect, a change in length of the SRC will cause the arm cavity lengths to change as well [100]. The effect of optical springs on the SRC length noise coupling strength is much smaller than the first coupling, so the estimate for the reduction of SRC length coupling will only considering the first effect.

SRC length noise is currently too high to meet the A+ requirements [101]. To estimate the reduction in SRC length noise that BHD may provide, the amount of DC offset that is currently used should be compared to the amount that is proposed for BHD. aLIGO has 23 mW of light due to the DC offset and 1 mW of residual light (light that cannot be removed) at the AS port [102]. The AS power for A+ is between 1 and 5 mW. Therefore, a conservative estimate for the factor that the SRC length noise coupling is reduced by is four.

Strictly speaking, these are not properties of BHD readout as scatter from the septum window and the SRC length noise could be reduced by other means. For instance, the septum window could be removed and suspensions with additional stages could be used. However, these solutions are not trivial to realise, and they are no more practical than implementing BHD. The septum window is a key part in being able to carry out commissioning tasks within HAM6. The number of times HAM5 and the other connected chambers are vented should be kept low as each time this is done gas and water vapour enters the system (this contributes to the noise floor of the interferometer). This residual gas and vapour will not be removed as this is a large engineering challenge. The design and install of three new suspensions is

also a difficult task, which ultimately gives less benefit compared to a BHD readout scheme. Realistic upgrades that will reduce to the SRC length noise coupling are currently being developed [103].

5.2.3 Summary

In summary, BHD has a couple of key advantages over DC readout. BHD allows for the reduction in SRC length noise coupling and, in conjunction with other upgrades, the ability to shape the quantum noise floor of the interferometer. In DC readout, the SRC length coupling comes from the DC offset required to provide a LO beam for the signal generated by a differential modulation in arm length to beat against. This DC offset is not required for BHD. Most interferometer configurations that beat the SQL assume that the homodyne angle can be freely set. The drawbacks of BHD when compared DC readout are all engineering challenges; nothing fundamentally favours DC readout over BHD.

5.3 Local Oscillator Path Length Stability Requirement for A+

This Section will go through the steps needed to derive the phase noise requirement for the beam path of the A+ LO. This uses a semi-classical approach, and the end result is the same as if a full quantum mechanical treatment was used. The quantum mechanics of this problem is included in the A+ sensitivity curve and the expected power in the signal beam. The phase noise of the LO must be kept low enough so that it does not result in intensity fluctuations greater than those due to the shot

noise of the light in the signal beam. This requirement was also derived in [104], [105] and [106], however in this work the most current layout was considered.

Phase noise arises due to phase fluctuations in the LO beam being indistinguishable from phase fluctuations in the signal beam. A gravitational wave produces a phase modulated signal in the signal beam, and the LO field phase is modulated by the motion of the mirrors in the LO path. When these beams are combined, it is the phase difference between the two fields that produces a signal, so the phase noise of the LO path must be low enough that the signal that is generated by the interference of the LO and signal beams is predominantly due to the phase modulation of the signal beam.

The relationship between the differential phase in the arms and the power in the signal beam was used to find the power noise for the beam exiting the SRM in A+. The linear response function due to a differential phase change in the arms was computed with FINESSE. This was used to deconvolve the strain sensitivity curve to produce a power noise curve for the light exiting the AS port of the interferometer. Since the interferometer is linear, noise is transformed linearly. This means that the quantum nature of the noise is irrelevant to the calculation.

This light noise curve was then used to set the maximum allowable intensity fluctuations due to phase noise between the LO and the signal beams. This phase noise can then be converted into a distance, and this produces a stability requirement for the steering optics used to combine the LO and signal beams.

The following Sections will go through the aforementioned steps. First the LO path will be described. Then a linear response function between a differential strain and signal at the AS port will be obtained via FINESSE. Next, The sensitivity curve for A+ [107] will be expressed in units of $\frac{W}{\sqrt{Hz}}$. This will be converted into a longitudinal motion requirement. Finally, there is a short discussion about the proposed optics that fulfil this requirement.

5.3.1 Local Oscillator Path

The beam path for the LO and signal beams up to the point of recombination will be described in this Section. The LO will be picked off from the power recycling cavity and guided by one steering mirror, BHDM1, to the BHD combining beam splitter, BHDBS2. The signal beam will exit the OFI and be steered by one mirror, OM0, onto BHDBS2. OM0 and BHDM1 will be used to align the beams so they overlap on BHDBS2. The angle of BHDBS2 will be used to make the LO and signal beams overlap in the far field.

The LO could have been derived from a separate laser, but to acquire laser light with low frequency and intensity noise the LO will be picked off from a point within the interferometer. Two options for the LO pick off were considered [104]: a beam picked off from the main beam splitters AR coating (BSAR), or the beam transmitted through PR2 (POP). The determining factor in choice of pick off point was the effect of backscatter. The POP port was chosen as it is far superior to the BSAR choice at all frequencies and scatter phases. This is because backscattered light from the BSAR enters only one arm, thus making it a differential affect. Any backscatter that enters the POP port is common-mode, and therefore DARM is far less sensitive

to this source of backscatter.

The layout for the optics which combine the LO and signal beam is shown in Figure 5.2. The beam transmitted by PR2 is used for control of the interferometer, so BHDBS1 is needed to pick off some of this light. The LO beam is steered towards the combining beam splitter BHDBS2 with BHDM1. Two lenses (need names for these) are used for mode matching the LO beam to the signal beam. The signal beam is transmitted through the SRM and OFI, and then it is steered towards BHDBS2 with OM0.

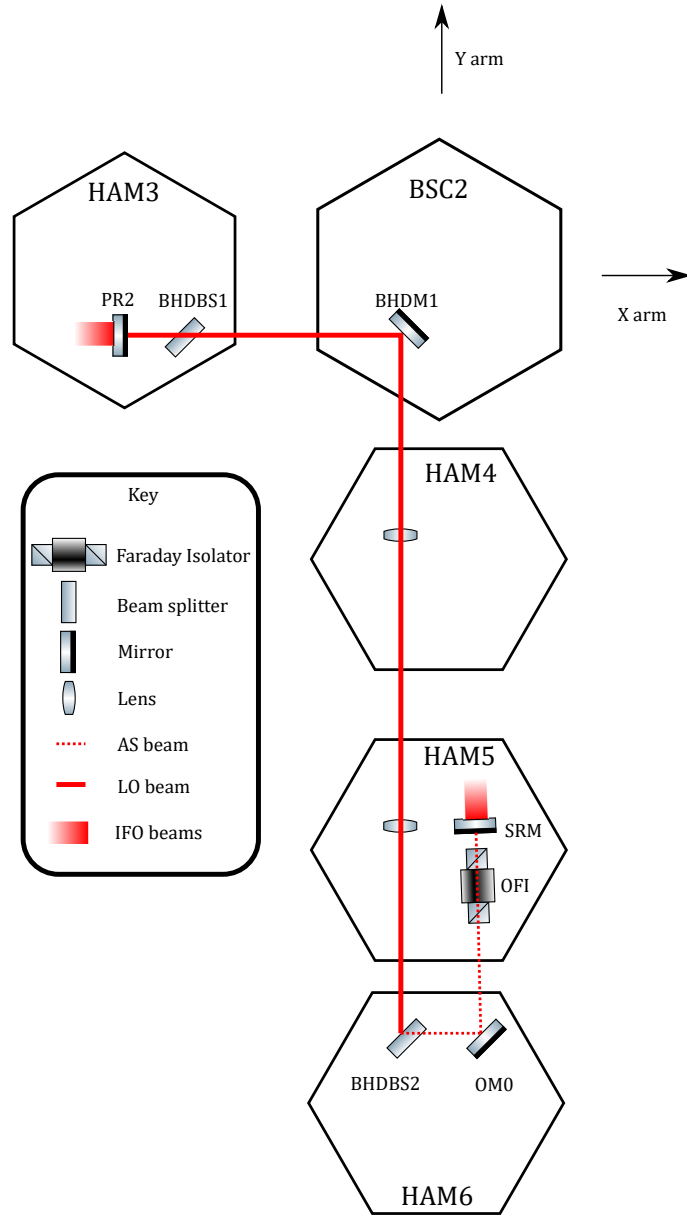


Figure 5.2: Sketch of the LO and signal beam paths. The HAM tables are shown by the hexagons. Most of the interferometer has not been shown for clarity. The main interferometer beam is indicated by a thick, gradated red line. The thinner red line indicates the LO beam, and the signal beam is shown with a dashed red line.

5.3.2 The Linear Response Function Between Differential Strain and Power at the AS Port

The linear response function of the A+ detectors was obtained using FINESSE. Since the core optics of the interferometer will not be changing in a way relevant to this calculation, the LLO model [108] was used to calculate the response between differential motion of the ETMs and the power transmitted through the OMC. The linear response function at the frequencies contained in the detection band are not influenced by the OMC. This means no major modifications to the FINESSE model were needed, and so the model represents a state that the interferometer would be in for observing gravitational waves.

FINESSE allows one to measure linear response functions by applying a ‘swept sine’ signal to a component and demodulating the light at the frequency of the applied signal. To create a differential arm length signal with FINESSE, `fsig` commands were used to apply two signals in anti-phase with each other to the ETMs,

```
fsig sig1 ETMXHR 1 0
fsig sig2 ETMYHR 1 180.
```

The frequency of this modulation was swept, and the signal produced at the AS port was measured by demodulating the light at the modulation frequency. The commands

```
xaxis sig1 f log 5 5000 2999
put P_DC_OMC_probe f1 $x1
```

do this.

For this calculation, the linear response function measured in FINESSE needs to be converted into different units. To get the linear response function between arm motion and AS power into units of $\frac{\text{Power}}{\text{Strain}}$, or $\frac{\text{W}}{\text{h}}$, several factors are required. Since the input modulation is doubled by moving the mirrors differentially, the response needs dividing by two. To account for the ETMs retro-reflecting the light, the response needs dividing by two again. To convert this from phase units to strain units, a factor of $\frac{\lambda}{2\pi L_{\text{arm}}}$ is needed. Therefore, the output of the FINESSE simulation needs to be multiplied by $\frac{1}{2} \frac{1}{2} \frac{2\pi L_{\text{arm}}}{\lambda}$ to be in units of $\frac{\text{W}}{\text{h}}$. The linear response function between a differential arm strain and the power at the AS port is shown in Figure 5.3.

5.3.3 Converting the Sensitivity Curve for A+ into a Longitudinal Motion Requirement

To convert the A+ sensitivity curve into a path length stability requirement for the LO, a few steps are required. First, the sensitivity curve must be expressed in units of $\frac{\text{W}}{\sqrt{\text{Hz}}}$. Second, the intensity noise of the signal beam needs to be converted into a phase noise. In order for the phase noise of the LO to not limit the performance of the gravitational wave detector, the phase signal in the signal beam must be greater than the phase noise of the LO beam. Finally, a factor of ten safety margin is required. The HRTS requirement is then obtained by including the number of reflective optics in the LO path and signal path.

The A+ sensitivity curve was multiplied by the function shown in 5.3 to convert it from units of $\frac{\text{h}}{\sqrt{\text{Hz}}}$ to $\frac{\text{W}}{\sqrt{\text{Hz}}}$. This is shown in Figure 5.4. At frequencies above 1 kHz the requirement is flat as this is the region dominated by shot noise. At lower

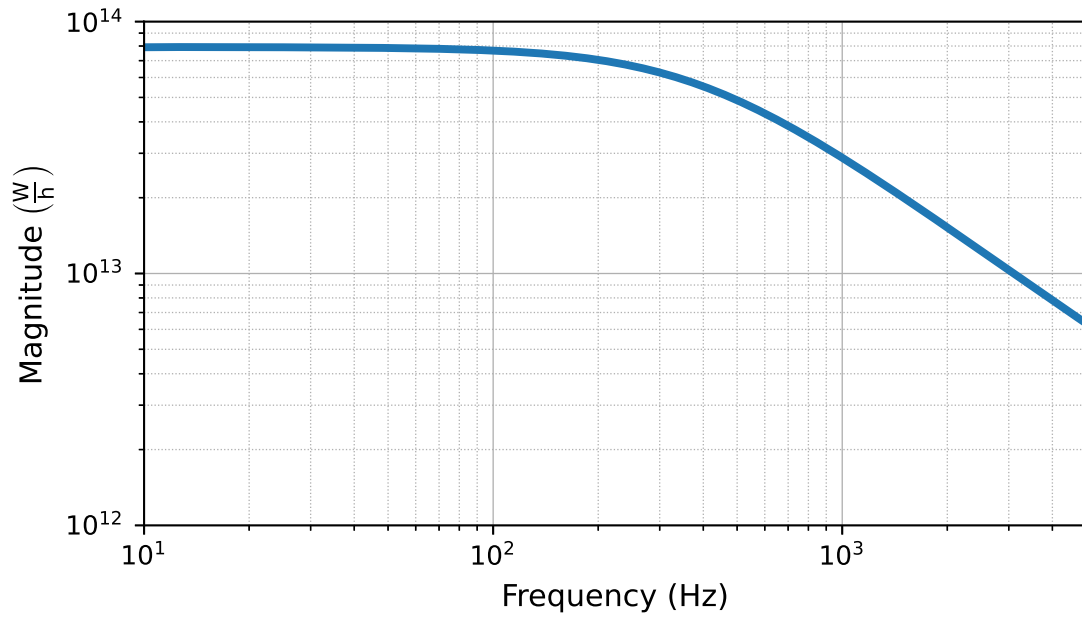


Figure 5.3: The linear response function between strain and light power, expressed in units of Light Power/gravitational wave strain, at the AS port was obtained from a FINESSE simulation of the LLO detector. The corner frequency at ≈ 300 Hz comes from the pole of the coupled SRC-arm cavity.

frequencies the requirement is relaxed as the sensitivity curve becomes dominated by other noise sources.

Next, a change in power measured in the combined LO/signal beam due to a phase modulation, i.e. path length modulation, in the LO or signal path needs calculating. When the LO and signal beams interfere at BHSBS2, the exiting beams have power P_c and P_d . These are given by

$$P_c = \left| \frac{1}{\sqrt{2}} E_s e^{i\omega t} + \frac{1}{\sqrt{2}} E_{\text{LO}} e^{i\omega t + \phi} \right|^2 = \frac{|E_s|^2}{2} + \frac{|E_{\text{LO}}|^2}{2} + |E_s E_{\text{LO}}^\dagger| \cos(\phi) \quad (5.11)$$

and

$$P_d = \left| \frac{1}{\sqrt{2}} E_s e^{i\omega t} - \frac{1}{\sqrt{2}} E_{\text{LO}} e^{i\omega t + \phi} \right|^2 = \frac{|E_s|^2}{2} + \frac{|E_{\text{LO}}|^2}{2} - |E_s E_{\text{LO}}^\dagger| \cos(\phi), \quad (5.12)$$

where E_s and E_{LO} are the (complex) field amplitude of the signal and LO beams; ω is the laser frequency; t is time and ϕ is the homodyne angle. The difference in these signals gives BHD signal

$$P = 2|E_s E_{\text{LO}}^\dagger| \cos(\phi). \quad (5.13)$$

Differentiating with respect to ϕ gives the relationship between a change in power and a change in phase between the two beams,

$$dP = -2|E_s E_{\text{LO}}^\dagger| \sin(\phi) d\phi, \quad (5.14)$$

and since the homodyne angle is 90° this becomes

$$dP = -2|E_s E_{\text{LO}}^\dagger| d\phi. \quad (5.15)$$

To convert ϕ to distance, use

$$\frac{\lambda}{2\pi} d\phi = dx. \quad (5.16)$$

The light noise curve shown in Figure 5.4 should be multiplied by $\frac{\lambda}{2\pi} \frac{1}{2|E_s E_{\text{sig}}^\dagger|}$ to obtain the noise floor of the interferometer in units of LO path length noise.

To determine the longitudinal motion requirement for the optics that combine the LO and signal beam the number of optics and, since there are many uncorrelated noise sources in LIGO, a factor of ten safety margin needs to be included. The requirement assumes, based on the current design, that there will be two mirrors at 45° which are used to steer the AS and LO beams. As the optics reflect at 45° , the path length change due to the optics motion is multiplied by $\sqrt{2}$. The beams are combined on a beam splitter, and the motion of this beam splitter needs to be counted twice. Figure 5.5 illustrates this.

For a 98 mW LO beam and a 5 mW signal beam, the noise floor of the interferometer in units of LO path length is shown in Figure 5.6. The requirement for the longitudinal motion of optics in the LO or signal path at 100 Hz is $5 \times 10^{-17} \text{m}/\sqrt{\text{Hz}}$.

To determine the amount of isolation required, the typical motion of the optical table needs to be compared to the longitudinal motion requirement for the optics. A typical motion of the optical benches at 10 Hz is $10 \times 10^{-1} \text{m}/\sqrt{\text{Hz}}$ [109]. To achieve this with passive isolation, triple suspensions will be used. These suspensions are

known as HAM Relay Triple Suspensions (HRTS) [110].

5.3.4 Summary of LO path length Stability

Details of the LO path were discussed and analysed. The LO will be derived from a beam picked off from the POP port, and a sketch of how the signal and LO will be combined is shown in Figure 5.2. Starting with the linear response function between a differential strain and a signal at the AS port and the strain sensitivity curve for A+, we were able to derive a relative path length stability requirement for the LO and signal beams. At 100 Hz this requirement is $5 \times 10^{-17} \text{m}/\sqrt{\text{Hz}}$. By comparing this motion to a typical motion of the optical table, we were able to derive an isolation requirement, shown in Figure 5.6. This longitudinal phase noise requirement led to triple suspensions being chosen to steer the LO and signal beams onto the combining beam splitter.

5.4 Conclusion

One of the major changes to LIGO for A+ will be the implementation of BHD. Combined with frequency dependent squeezing, BHD offers the possibility to optimise the detector for different gravitational wave sources. Along with this, BHD offers a way by which the coupling between the SRC length noise and DARM can be reduced.

BHD presents some technical challenges over DC readout. The LO phase noise must be kept low enough as to not contaminate the signal at the AS port with LO path noise. The requirement for the phase noise is shown in Figure 5.6. This

requirement can be met by using triple suspensions to deliver to LO and signal beams to the beam splitter they are combined on.

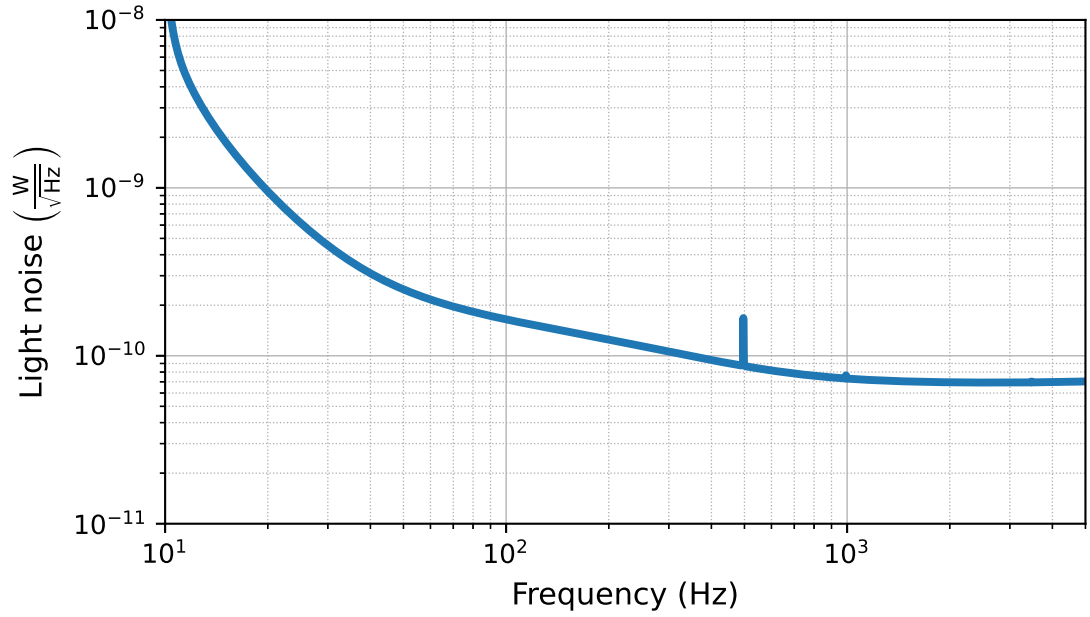


Figure 5.4: The intensity noise of the light at the AS port for the A+ detector. This was obtained by deconvolving the detectors sensitivity curve with the function shown in Figure 5.3. At frequencies above 1 kHz the noise is flat as it is dominated by shot noise. At lower frequencies the noise is increased relative to shot noise due to radiation pressure, coating thermal noise and other noise sources. The spike in the spectrum at 500Hz represents the presence of suspension modes (actually find out what this is).

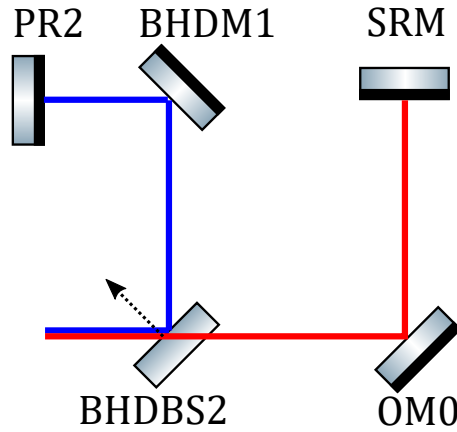


Figure 5.5: A sketch of the LO and signal beam paths. The LO beam path is shown in blue, and the signal beam is shown in red. A dashed arrow indicates the direction that the recombining beam splitter, BHDBS2, may move in. If BHDBS2 moves in this way, the phase of the signal beam has increased while the phase of the LO beam has decreased at the point where the signal and LO beams combine. This means that the recombination beam splitter motion needs to be counted twice in calculations of the phase noise requirement.

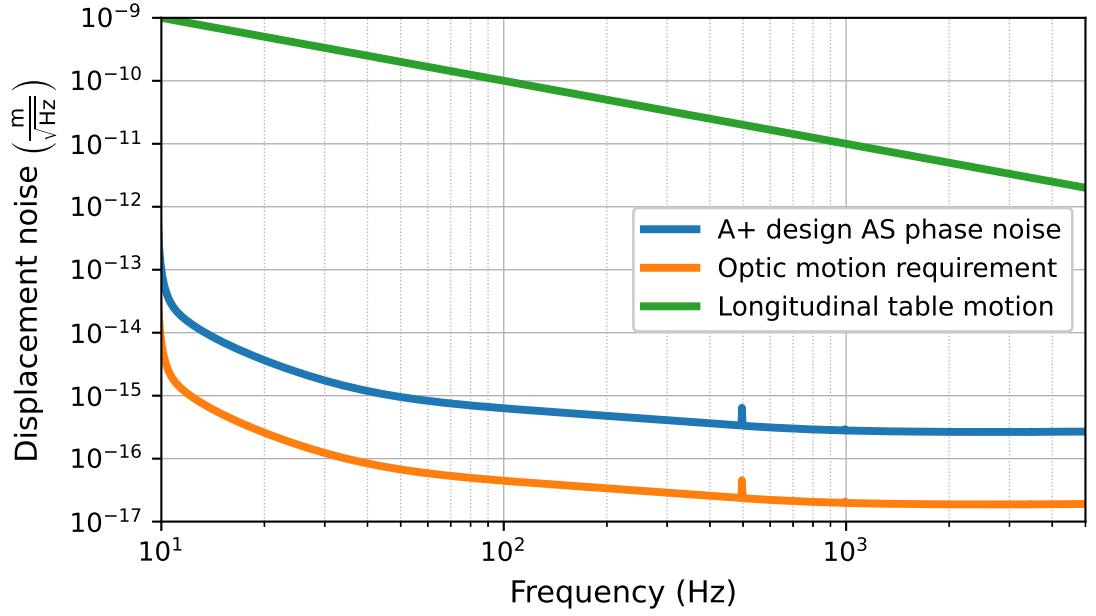


Figure 5.6: The phase noise between the LO and the signal beams causes noise. The sensitivity curve for A+ sets the limit on how much path length noise is acceptable between the AS and LO beams. The blue line shows sensitivity curve is shown in units of path length noise between the AS and LO beams, and the orange line shows the requirement of HRTS mirror motion. The table motion was modelled as $1/f$ noise.

Chapter 6

Design of the LIGO A+ HAM6

Layout

Fundamentally, the optical layout in HAM6 must combine the signal beam and the LO beam and sense the combined beams to form a balanced homodyne detector (BHD). The implementation of BHD for LIGO A+ requires a complete redesign of the HAM6 layout. This chapter is comprised of three sections. Section 6.1 is a description of the functionality and design constraints that all candidate HAM6 layouts must satisfy. In Section 6.2, the optical properties of the proposed HAM6 layout are examined. Section 6.3 is an analysis of how mode mismatches between the interferometer and the OMC can be corrected by the Active Wavefront Control (AWC). Section 6.4 is the conclusion for this chapter. This work was undertaken by a team made up of myself, Ken Strain, Russell Jones, Stephen Webster, Mark Barton and Eric Oelker.

6.1 The Constraints and Requirements for the Optical Layout Within HAM6

This section will discuss the functionality required from the optical layout within HAM6 and describe the practical constraints imposed by the existing hardware and infrastructure. HAM6 must contain an optical layout that combines the LO and signal beams as well as providing steering and mode-matching between the interferometer and the OMC. There are a few properties of the HAM6 layout that cannot be changed such as the entry point of beams and size of the optical table. The main design challenge was achieving close to 45° of Gouy phase between the optics for mode-matching, as this requires a lot of space relative to the size of the optical table. .

6.1.1 Constraints

As the constraints give a starting point for all designs, they will be discussed first¹. These include: where beams enter the HAM6 chamber, the size of the nominal mode entering HAM6 and the size of the OMC waist. The size of the HAM optical table does restrict the design, however table extensions can be made depending on where they would be needed.

¹It could be argued that requirements are a constraint in a design. Here, the constraints are geometric properties which we have no freedom in, such as the input beam shape, the OMC geometry and the layout of the rest of the interferometer. A requirement, such as the desired minimum mode-matching, is a property which does not have a set ‘physical form’. For instance, while Gouy phase does correspond to a physical distance, not all 45° Gouy phase spaces are the same physical size.

6.1.1.1 Beam Entry Points

The entry point of the signal and LO beams in principle can be anywhere in HAM6; however, in reality they are fixed due to the location of HAM6 with respect to the interferometer; the location of the septum windows between HAM5 and HAM6; and due to the many other components of the interferometer that the signal and LO beams need to avoid. The LO beam is steered with BHD1 onto BHDBS2, and the signal beam is steered with OM0 onto BHDBS2. These components are shown in Figure 5.2. This is required to provide the degrees of freedom to co-align the signal and LO beams. There is a freedom in choice for the locations of OM0 and BHDBS2, but there is no choice in the entry points into HAM6 for the signal and LO beams.

6.1.1.2 Input and Target Beam Parameters

The nominal beam parameter at the SRM is $q_{\text{SRM}} = -3.55 + 1.09i \text{ m}$ [111], so the design of the HAM6 mode-matching optics is based on this number. As discussed in Chapter 4, the SRM beam parameter could differ from its nominal value. Chapter 4 describes how geometry errors in the SRC cause the arm cavity and the SRC to be mismatched. This means that the radius of curvature and beam size at the SRM will not correspond to the eigenmode of the SRM, and the beam parameter at the SRM that should be considered corresponds to the arm cavity mode. This set of likely modes, shown in Figure 4.11, was taken into consideration when choosing a layout for HAM6 since different candidate layouts may have been more or less robust to beam parameter errors at the SRM.

The current OMC design [112], with trivial modifications, will be used to remove junk light from the signal beam, so there is no choice in target waist size. The

nominal OMC waist size is $485\,\mu\text{m}$. The beam enters the OMC through a flat mirror, and the waist is located 140 mm from this mirror. There is a negligible amount of astigmatism inherent in the design of the OMC. Mode-matching optics therefore have to match the nominal SRM and OMC modes.

6.1.1.3 Equal Path Lengths after BHDBS2

It was decided that the two beams after BHDBS2 would have the same path length to reduce the cost of the HAM6 optics. This may also improve the common-mode noise rejection ratio as asymmetries may enhance the effects of noise couplings. To achieve high levels of common mode noise rejection, both paths should be as similar as possible. Only one path needs to have imperfections for the common mode rejection ratio to be degraded. For instance, it may be a bad design choice to have a path with a mirror whose only purpose is to fold the layout so that a slightly better readout platform can be used. However, it should be noted that the amount of common mode rejection required is related to the classical noise of the laser and its shot noise limit. Since the LO in LIGO A+ will be derived from the PRC it will already be stable (shot noise?), so little common mode noise rejection will be needed. In addition to this, the layout will be in a clean environment with a low amount of residual gas; thus reducing mechanisms whereby imbalances due to path length difference can generate noise (may remove?). Therefore, the trade-off between mirror number and path length symmetry versus readout platform preference is not necessarily easy to evaluate.

6.1.2 Requirements

The HAM6 layout is required to provide steering and mode-matching for the signal-LO beam into the OMC. The allowable mode-mismatch is 2%. The losses due to components in HAM6 is summarised in Table 6.1 [113]. After BHDBS2, the layout must have a pair of mirrors separated by 90° and another pair separated by 45° of Gouy phase to provide orthogonal control over the beam pointing and mode-matching.

6.1.2.1 Mode Mathcing Losses

The goal for mode-matching between the arm mode and the OMC mode is $> 98\%$. Active mirrors, called Suspended Active Matching Stages (SAMS), will be used to correct for mode-mismatch; they will provide ± 100 mD of optical power tuning range [114]. Astigmatism degrades the mode-matching, and non-zero angles of incidence (AOI) of beams onto curved optics cause the reflected beam to become astigmatic (for a comprehensive description of astigmatism see [115]). This using a large AOI to increase the separation between optics in HAM6.

The role of AWC in HAM6 is to correct for thermal effects in other parts of the interferometer [116] such that the mode-matching between the arm mode and the OMC is maximised. Additionally, the AWC in HAM6 will correct for geometry errors in the layout after the SRM. Thermal changes in the interferometers mode may be corrected for by the Thermal Compensation System (TCS) [117]. In practice, operators and commissioners use each tunable aspect of the interferometer to improve its sensitivity to gravitational waves. For instance, at LLO it has been found that TCS can be used to decrease laser intensity noise coupling to DARM [118]. There-

fore, TCS may not be able to correct thermally-driven mode errors. The AWC in HAM6 may also be used for correcting geometry errors in the SRC. The optical layout and AWC functionality in HAM6 were designed with this flexibility in mind.

6.1.2.2 Gouy Phase Considerations

While in principle only two curved mirrors in each path to the OMC would be required to mode-match the beam from the interferometer to the OMC, even a single mirror may be able to do that, a lens in the OFI will be used to provide further Gouy phase separation between the two active mode-matching mirrors in each path after BHDBS2. An active lens was considered here, but concerns of OFI assembly heating [119] and the unavailability of flint glass meant that this idea was discarded. As well as this, the range of optical powers for the SAMS is large enough that it would be an over-design to implement an active lens in the OFI.

To provide orthogonal mode-matching control, the two SAMS should be separated by 45° . This Gouy phase separation requirement gives us a method of generating candidate designs, as this can be interpreted as needing the mode-matching mirrors to be as far apart as possible from each other on the HAM table. As the Rayleigh range of the beams is on the order of one meter, and most Gouy phase is accumulated within one Rayleigh range either side of a beam waist, a beam will have to travel a significant distance across the table before it acquires sufficient Gouy phase.

Additionally, to provide orthogonal alignment control for each beam path after BHDBS2, two of the mirrors in each of the said beam paths are required to be separated by 90° of Gouy phase. For all designs, the two mirrors for alignment were

always immediately after BHDBS2 and immediately before the OMC. It was found that placing the first mirror close to BHDBS2 often provided good layouts.

6.1.2.3 Readout Platform Suspension and Auxillary Components

The OMCs are expected to be the largest contributor of back-scatter from HAM6 to the interferometer [112,116]. By having the OMCs after the signal and LO beams are combined and by having the OFI protect the interferometer, the requirement on the motion of the OMCs is reduced compared to if they were before BHDBS2. Nevertheless, a single stage suspended platform is required. This is known as the readout platform suspension (RPS). Both OMCs will need to be on this platform, meaning all designs will need to have the OMCs fairly close to each other. Too large an RPS is mechanically infeasible as it will reduce space for other auxiliary components.

Additional sensors in HAM6 are required for controlling various degrees of freedom within the layout. Some of these degrees of freedom will be monitored using beams which are picked off from the main beam path, so additional mirrors will be required to steer these beams around the chamber. A set of smaller optics will be installed to provide diagnostic beams which can be imaged with CCDs. A shutter will be required to protect the hardware in HAM6 from being damaged by the large amounts of light that can potentially be transmitted through the SRM. Appendix ?? describes how the longitudinal and pointing degrees of freedom associated with BHD may be sensed and controlled May remove the appendix and refer to stephens document, although it is nice to have all this information in the same book.

6.1.3 Summary

In summary, the main constraints for the design of HAM6 are where the beams enter the chamber; the size of the optical bench; the input mode and the OMC design. There must be low amounts of loss; therefore, layouts with large AOI onto curved mirrors should be avoided. The optical layout within HAM6 constitutes a nontrivial design task as it is desirable to have orthogonal control over the mode-matching and beam direction. This requires the optical axis to be of a similar length as the maximum spacing permitted by a HAM table. In addition to this, the HAM6 layout needs to leave space for all the other sensors and beams used for control. This means that all designs will use nearly all of the table space.

HAM6 loss Source	Target for O5
OMC throughput loss	< 2%
OM3 transmission (or equivalent) for sensing	1–2%
PD quantum efficiency	1%
Mode mismatch to the OMC	< 2% (as low as possible)
Any other optical loss	< 1%

Table 6.1: Target for losses in HAM6. Reproduction of [113]

6.2 The Chosen HAM6 Optical Layout

The design that is being taken forward at the time of writing is shown in Figure 6.1. Optically, there are many layouts which could have satisfied the requirements and constraints laid out in Section 6.1 . The choice of layout for HAM6 was ultimately determined by mechanical factors such as table space, number of suspensions and preference of RPS design. The optical properties of this layout are summarised in Table 6.2. The mirrors OMA1, OMB1, OMA2, OMB2 etc. will be referred to as

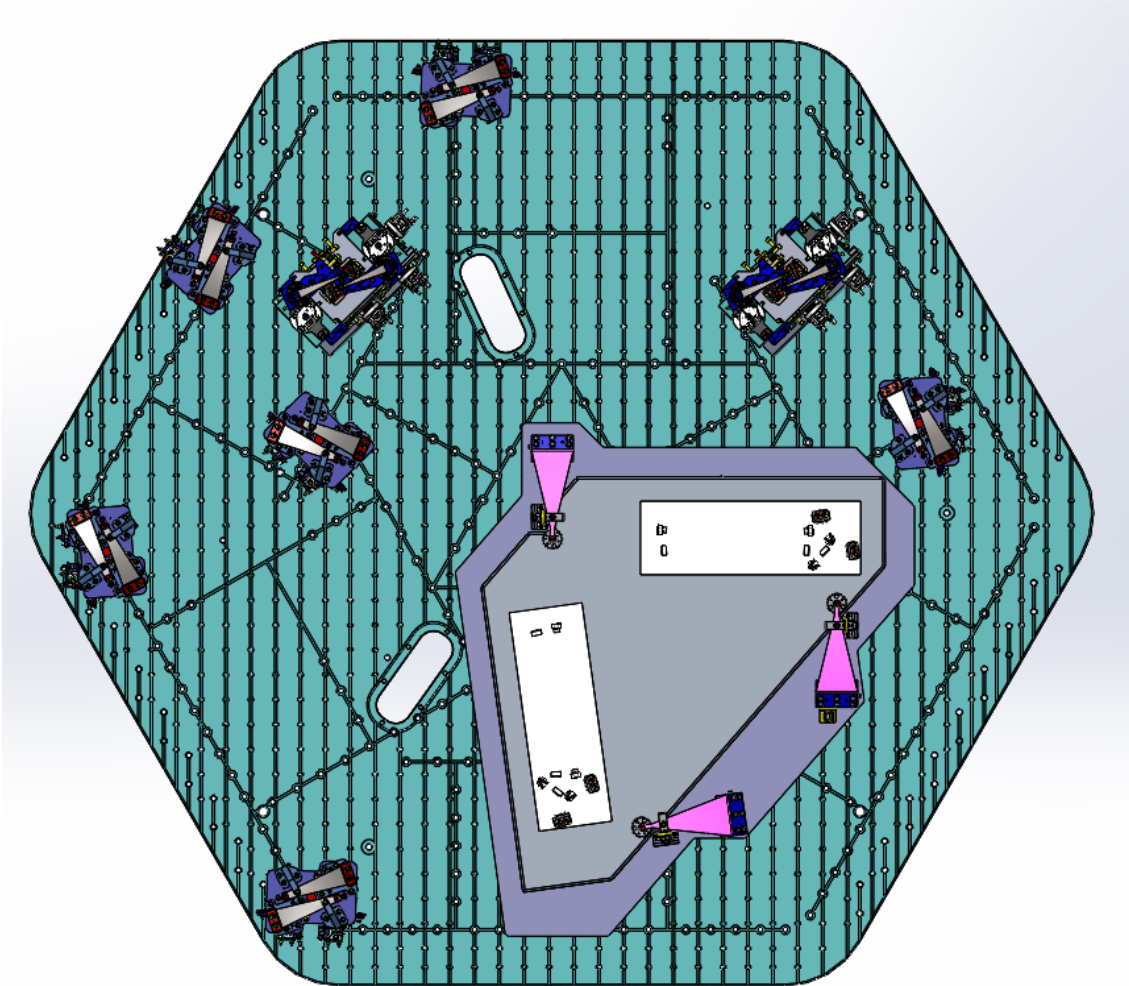


Figure 6.1: Place holder graphic. The pentagram layout. CAD with beam path will be inserted later. Caption will note the main beam path, the bits of hardware mentioned in the text, and will point out that there are lots of other doodads and thingamabobs all over the place which make it tricky to fit everything in HAM6.

OMx1 and OMx2 as, usually, it is not important to make a distinction between the A and B paths.

The beam size and Gouy phase as a function of position along the optical axis is shown in Figure 6.2. Significant points along the optical axis are also shown in this figure. The size of the beam at the final mode-matching optic matches the size of the target mode, as the size of the beam cannot be changed at a mirrors surface. The beam is always less than 1.2 mm within HAM6, an important fact to remember when considering how close structures can be to the beam path.

The astigmatism loss was calculated to be $< 0.05\%$; for detail in this type of calculation, see Appendix C. This astigmatism loss is plotted as a function of angle in Figure 6.4; above 10° the loss becomes significant. The compromise that is made when choosing a larger AOI is between the potential for larger Gouy phase separation between OMx1 and OMx2 at the expense of loss due to astigmatism. In reality the IFO and OMC will also have a small amount astigmatism so this calculation will not be entirely correct, however this effect is negligible compared to other losses. Losses of $< 0.05\%$ are considered unimportant, especially when compared to the other expected losses, some of which are expected to be at the many percent level, in the readout of the interferometer.

The focal lengths of the OFI lens, OMx1 and OMx2 where optimised so that there was as close to 45° Gouy phase separation between OMx1 and OMx2 as possible. This is shown in Figure 6.3. With the layout shown in Figure 6.1, it is possible to achieve 43° Gouy phase separation between OMx1 and OMx2. Other layouts could provide exactly 45° Gouy phase separation; however, this layout offers many

Optic	Power (1/m)	Focal length (m)	RoC (m)
OFI lens	-0.0583	-17.1552	-
OMx1	-0.3539	-2.8290	5.6580 (concave)
OMx2	-0.8737	-1.1445	2.2891 (concave)

Space Name	Optical Path Length (mm)
SRM to OFI lens	1120
OFI lens to HAM6 table edge	720
HAM6 table edge to OM0	432
OM0 to BHDBS2	905
BHDBS2 to OMxS	225
OMxS to OMx1	1240
OMx1 to OMx2	1590
OMx2 to OMCx waist	1220

Parameter	Value
SRM Complex Beam Parameter	$q_{\text{SRM}} = -3.55 + 1.09i \text{ m}$
OMC waist	485 μm
OMx1 to OMx2 Gouy phase	43°
OMxS to OMx2 Gouy phase	88°
OMx1 and OMx2 AOI	A: 10°, B: 6°

Table 6.2: Mechanical and optical properties of the LIGO A+ HAM6 layout.

practical benefits, mainly in relation to the RPS, over the other considered designs that outweigh an additional 2° of Gouy phase separation between OMx1 and OMx2. This layout also provides good Gouy phase separation for alignment control, with there being 88° between OMxS and OMx2.

In conclusion, the proposed HAM6 layout fulfils the criteria set in Section 6.1. It is shown in Figure 6.1. Near optimal mode-matching and alignment control can be achieved with this layout. A summary of the properties of the layout can be found in Table 6.2. The next stage of development for this layout involves the design and prototyping of parts, and the consideration of fine details such as cable routing.

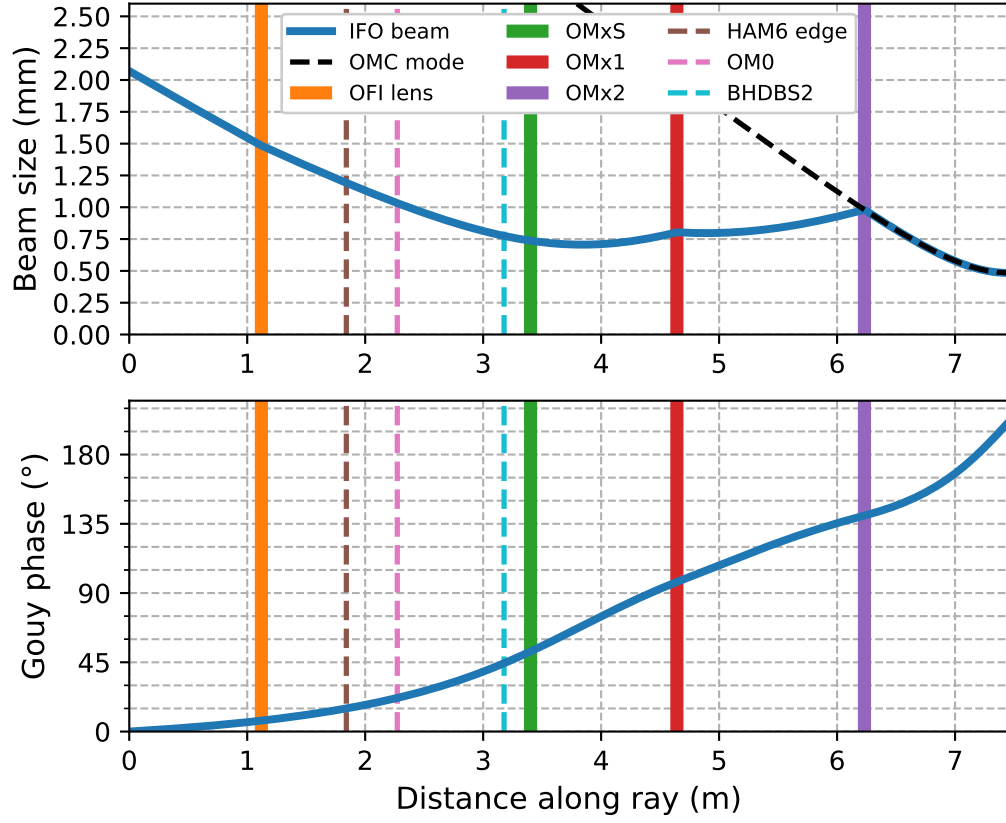


Figure 6.2: Top panel: the radius of the beam as it travels from the SRM ($z = 0$) to the waist of the OMC. Significant points along the axis are marked with vertical lines. Less significant, but useful, reference points are marked by dashed vertical lines. Bottom panel: The Gouy phase shift accumulated by the beam as it propagates along the optical axis.

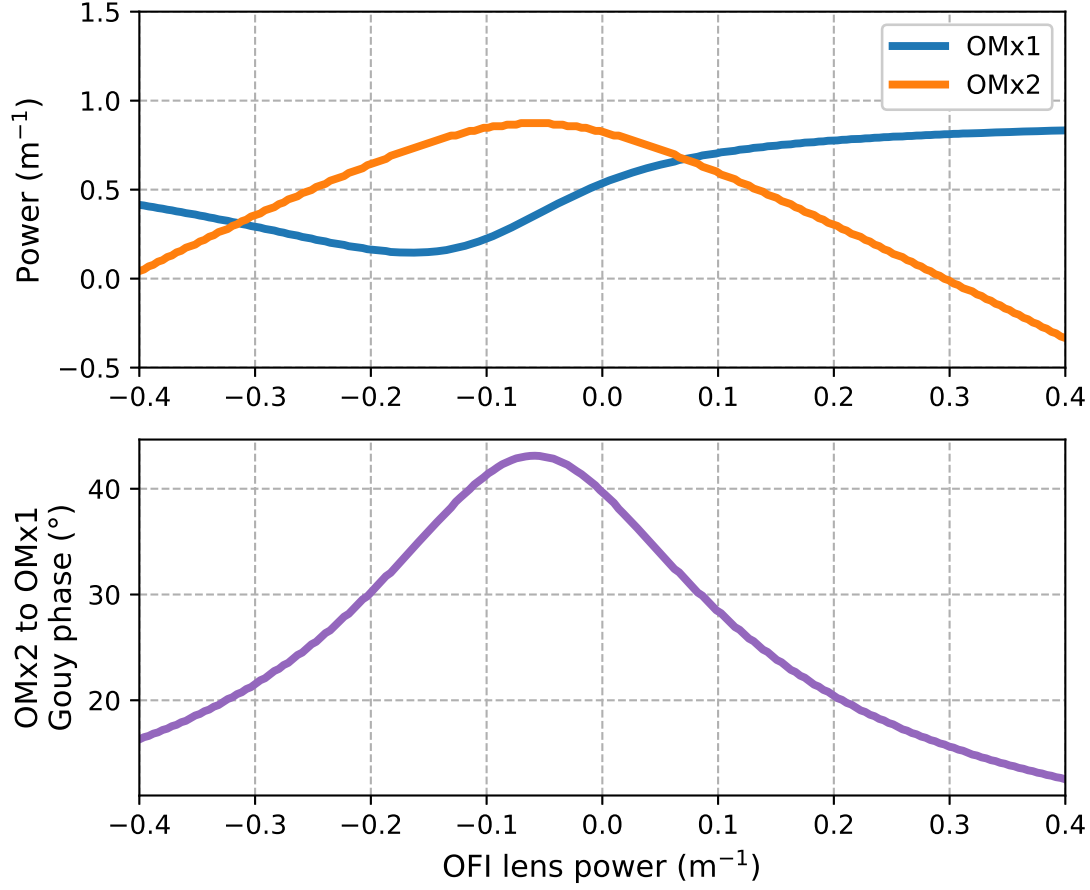


Figure 6.3: Top panel: The choice of OFI lens power affects the combination of OMx1 and OMx2 power that is required for 100% mode-matching. Bottom panel: The best choice of OFI lens corresponds to where the Gouy phase separation between the pair of OMx1 and OMx2 is closest to 45° . The most Gouy phase separation between OMx1 and OMx2 achievable by this layout is 43° .

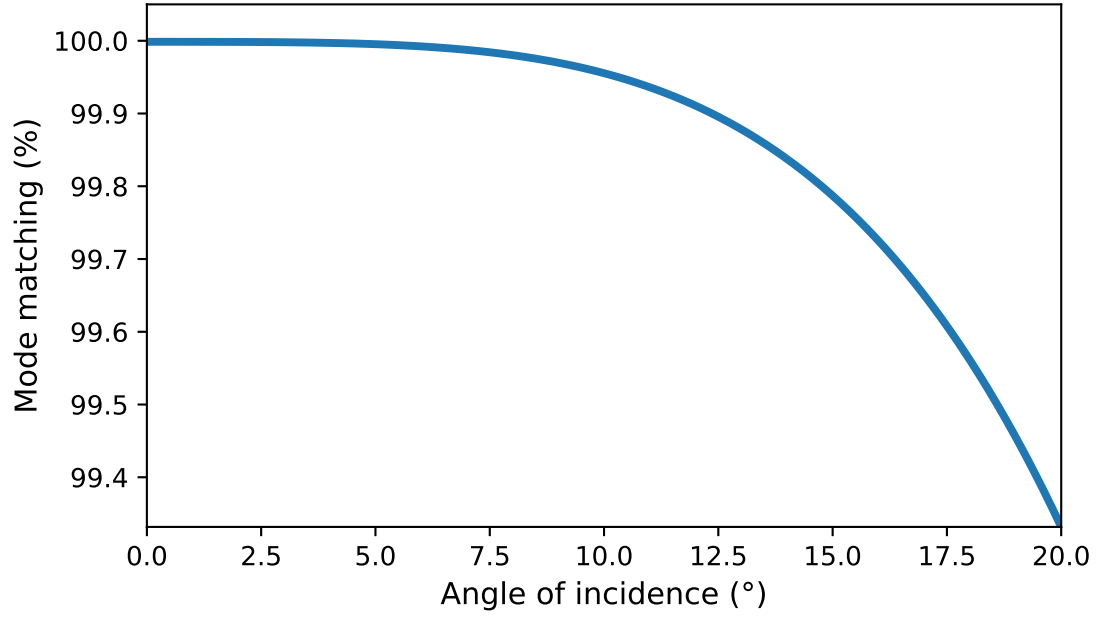


Figure 6.4: Astigmatism is a source of loss. When designing this layout, to explore how sensitive it was to astigmatism, the AOI for the curved mirrors was altered. Using equation (6.2), the loss was computed. This effect is negligible up to 10° after which it starts to become significant compared to the losses summarised in Table 6.1.

6.3 Visitations of the Robustness of the A+ HAM6 Layout to Mode Mismatch

It is expected that there will be some mismatch between the beam from the interferometer and the OMC due to geometry errors within the interferometer. Additionally, the thermal state of the interferometer may affect the arm mode beam parameter at the SRM. To visualise the effectiveness of the mode-matching control provided by the layout's active optics, it is useful to consider a 2D space representing Gaussian beams. The region of this space that can be mode-matched by the layout's mode-matching optics and a distribution of modes that those optics may have to correct for can be compared to gain insight into the effectiveness of the layout. It was found that the layout's active mode-matching optics will cover the vast majority of expected input beams.

The mode-matching between two laser beam modes, or a laser mode and a cavity mode, describes how similar those two modes are. In general, a combination of misalignment and differing beam shape can be considered to be a mode-mismatch. However, mode-matching typically refers to the overlap of the two modes if they were co-aligned, thus it is a measure of how similar in shape the two modes are.

6.3.1 A Description of Mode-Matching in Terms of HG Modes

Mode-matching is usually, to first approximation, thought of as a difference in the beam's waist size and the beam's waist location. First order mode-mismatch can be modelled as light scattering from the beam's, or cavity's, fundamental mode into its $\text{HG}_{20} + \text{HG}_{02}$ mode. If a cavity is designed such that the fundamental mode

will be resonant while the $\text{HG}_{20} + \text{HG}_{02}$ mode will be reflected then the effect of mode-mismatch between the cavity and a signal carrying beam is to decrease the signal light that is transmitted by the cavity; since, some of the signal carrying light will be scattered into the reflected $\text{HG}_{20} + \text{HG}_{02}$ mode. Sensing the $\text{HG}_{20} + \text{HG}_{02}$ mode therefore gives a measure of the mode-matching between two lasers.

A change in waist size is equivalent to an in-phase coupling between the fundamental mode and the $\text{HG}_{20} + \text{HG}_{02}$, whereas a change in waist location is a coupling to the $\text{HG}_{20} + \text{HG}_{02}$ which is 90° out of phase with the fundamental mode [67]. As the $\text{HG}_{20} + \text{HG}_{02}$ acquires 2ϕ compared to the fundamental mode, where ϕ is the Gouy phase acquired by the fundamental mode, spacing beam curvature actuators at 45° will give an orthogonal basis for control over mode-matching. This means that the beam size and beam waist, or two combinations of them that are orthogonal, can be independently controlled.

6.3.2 Calculating Mode-Matching Using The Gaussian Beam Parameter

Two perfectly mode-matched beams will have identical beam parameters at the same point along the optical axis. The overlap integral between two fields, E_1 and E_2 ,

$$\mathcal{M} = \frac{|\int_A E_1^* E_2 dA|}{\int_A |E_1|^2 dA \int_A |E_2|^2 dA}, \quad (6.1)$$

can be used to determine the amount of mode-matching, $\mathcal{M}$, between two astigmatic modes. Solving this integral gives

$$\mathcal{M} = \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}, \quad (6.2)$$

where $q_{1,2}$ is the beam parameter of the electric field $E_{1,2}$. This is derived in Appendix C. The solution to equation (6.2) is always $0 < \mathcal{M} \leq 1$. Alternatively, mode-matching may be computed in terms of the amplitude of the higher order modes relative to the fundamental mode [120]. The mode-matching can be calculated at any point along the optical axis of the layout as mode-matching must be preserved throughout it; however, it makes sense to calculate the mode-matching in a physically significant plane such as the input plane, a plane containing a curved optic or the plane containing the waist of the target beam.

6.3.3 WS Space Visualisations

A Gaussian beam can be described by its beam size, w , and defocus, S [121]. A WS space plots a Gaussian beam in terms of these two parameters. The effect of focussing elements on Gaussian beams can be clearly seen in this representation of them. Because a curved mirror can only change the curvature of a beam, at the mirrors surface the beams before and after reflection must have the same size. The process of a Gaussian beam, described by $q_1(S_1, w_1)$, being reflected from a mirror with power S_m to produce a new Gaussian beam, q_2 , can be written as

$$\frac{1}{q_2} = S_m + \frac{1}{q_1} = (S_m + S_1) - i \frac{\lambda}{\pi w_1^2}. \quad (6.3)$$

From equation (6.3) it can be seen that a mode moves in the S direction when it reflects from a curved mirror. This representation of Gaussian beams has the advantage of showing the beam in terms of intuitive quantities. The downside to this representation is as the beam propagates through the layout, the path that is traced through the WS space by the mode is fairly complex.

There are two other ways that a beam parameter could be graphically represented. Since q is a complex number, it could be represented in terms of its real and imaginary components. This makes seeing the effect of propagation straight forward, but the effect of a focussing element is harder to visualise. Alternatively, plotting $1/q$ in terms of $\lambda/\pi w^2$ and S produces plots where the effect of focussing elements and propagation are easy to visualise [122, p.680]. Curved mirrors and lenses result in a change in defocus, and propagated modes follow circular paths. As these figures are being used to visualise mode-matching and the effect of actuators on the beam size (rather than $\lambda/\pi w^2$) and defocus, the WS representation was chosen.

To illustrate how actuators move a mode in WS space, consider two lenses with variable power separated by 45° of Gouy phase. As the two lenses are orthogonal, the waist size and waist location of the mode exiting this layout will be orthogonally controllable. In a WS space the orthogonality of two actuators can be seen by the angle between the two vectors that represent each actuation. If the two lenses are orthogonal then the actuation vectors will be orthogonal. Figure 6.5 is a sketch of the WS space representation for a simple optical layout consisting of two active lenses separated by 45° Gouy phase.

The mode-matching requirement between the interferometer and the OMC for LIGO A+ is $> 98\%$. It is useful to plot contours of the mode overlap in the WS space between a mode and a target mode. Note that these contours can only tell you about the overlap between a mode and the target mode; the mode-matching between two arbitrary modes can not be deduced from these contours [121]. For example two modes may be mismatched by the same amount from the target in orthogonal directions in WS space. They cannot be 100% mode-matched as these two modes are different from each other.

6.3.4 Visualising how Active Mode-Matching Optics Expand the Area of WS Space Representing Acceptable Input Modes

To visualise the effectiveness of the actuation provided by the active elements of the HAM6 layout, the region of the WS space at the SRM from which the target mode-matching space can be achieved was computed. To do this, first the ring of modes in WS space that is 98% mode-matched to the OMC after reflection from the second active mirror, OMx2, was considered. If OMx2 had a static radius of curvature then the effect of OMx2 would be to move the ring vertically in the WS plane. However, the ring can also be stretched in the S direction as OMx2's radius of curvature can be changed. This can be seen, crudely, by applying $S_{\text{OMx2}} \pm \Delta S$ to the original $> 98\%$ mode-matching ring to produce two new rings separated by $2\Delta S$.

To generate the region from which the target region can be reached by OMx2, the target region should be cut by a line which intersects the ring at the extrema of the beam size. To maximise the amount of WS space covered by the actuator, the upper target region should have $+\Delta S$ applied to it, whereas for the lower region the actuation should be $-\Delta S$. This results in a region of WS space which is larger than the initial target region. All modes in the new region can be at least 98% mode-matched to the OMC. The new region was then propagated to the first mirror, OMx1, where the same procedure can be applied to further expand the region. This process is shown in Figure 6.6.

The modes from the SRM that can be matched to the target is shown in Figure 6.7. To visualise how successful the layout is at handling the likely modes from the interferometer (this set of modes was investigated in Chapter 4), the interferometer modes were compared to the region of WS space containing modes which can be corrected such that they meet the $> 98\%$ mode-matching requirement. It was found that almost all likely modes will be covered by the layout, as the modes near the edge of the interferometer mode continuum are more unlikely to occur than the ones near the centre. Its worth noting that the in Figure 6.7, modes at the yellow end of the spectrum may not be physical as they correspond to an unstable SRC.

For all candidate layouts analysed there was no major difference in the amount of modes that could be mode-matched. In principle, the requirement for having 45° of Gouy Phase separation between OMx1 and OMx2 does not maximise the amount of *likely* interferometer modes that can be mode-matched. It is possible to shape the region of WS space covered by the layout to maximise the probability

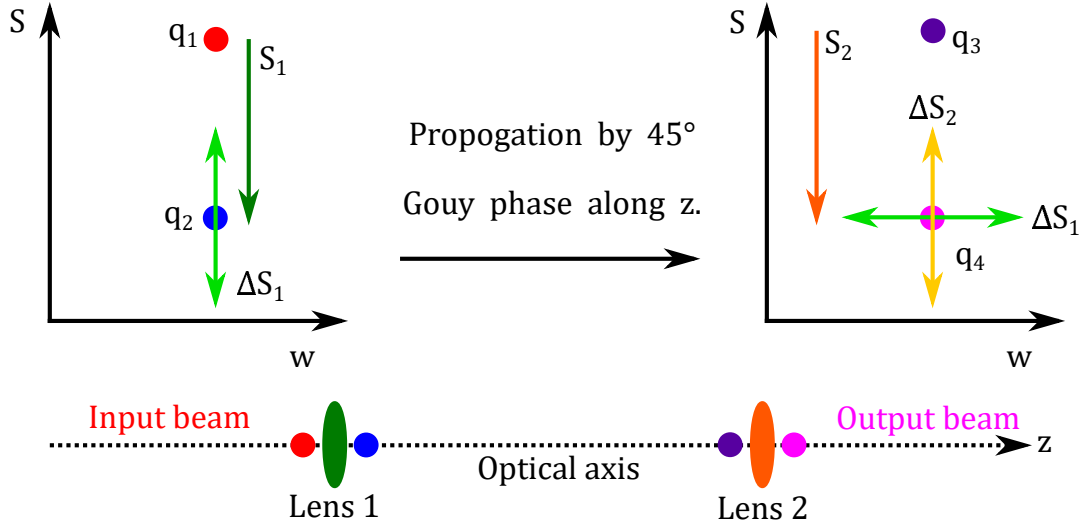


Figure 6.5: Sketches of the WS representation for a Gaussian beam interacting with different parts of a simple optical layout consisting of two active lenses with powers $S_{1,2} \pm \Delta S_{1,2}$. The left hand axis shows the WS plane at Lens 1. The input beam q_1 is indicated by a red dot, and the effect of the lens on q_1 is shown by the dark green arrow S_1 . This results in a new mode q_2 which is shown as a blue dot. A light green arrow indicates the actuation of Lens 1. The right hand axis shows the WS plane at Lens 2. The input mode at Lens 2, q_3 , is shown as a purple dot, and the output mode, q_4 , is shown as a pink dot. The static and dynamic power of Lens 2 is indicated by the orange and yellow lines. As Lens 1 and Lens 2 are separated by 45° of Gouy Phase, the actuation provided by Lens 1 results in a change in the beam parameter which is orthogonal to the actuation provided by Lens 2, provided that the actuations are small enough. Beneath the two WS space diagrams is a sketch of the optical axis and where the modes are along it.

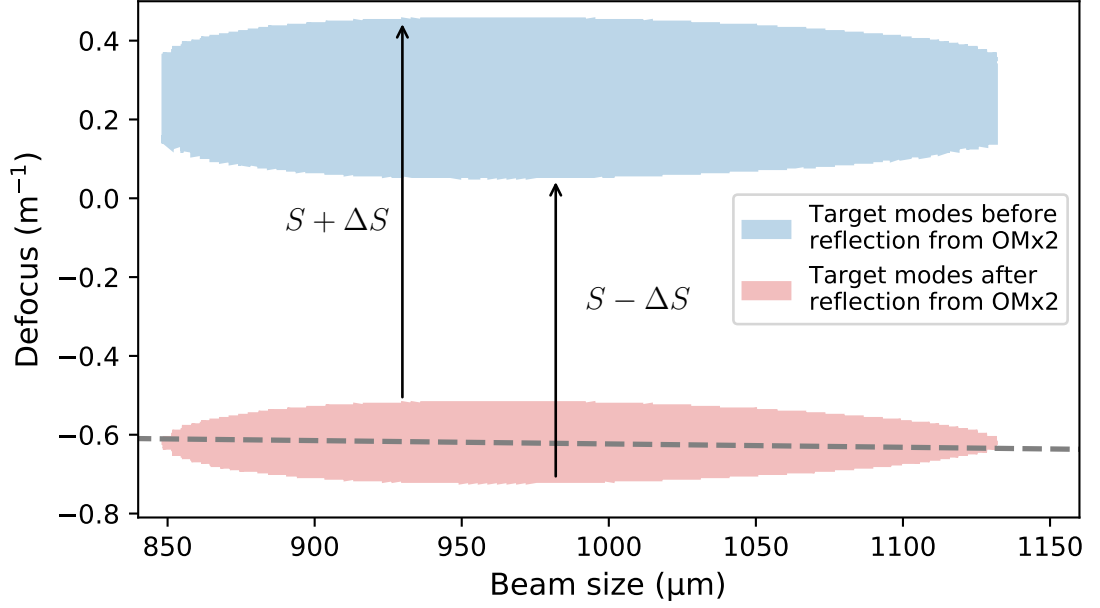


Figure 6.6: This is the WS space at the OMx2 plane. The effect of OMx2 in the plane of OMx2 is to move a mode vertically in WS space. After reflection from OMx2, a mode must be at least 98% mode-matched to the OMC as there are no further active mode-matching elements in the layout of HAM6. Modes with at least 98% are shown by the red region. The blue region represents which modes can be reached from the red region by either $S_{\text{OMx2}} \pm \Delta S$, and was constructed by applying transformations to the modes in the red region. The red region was split into two parts, ‘upper’ and ‘lower’, indicated by the dashed line that connects the two extremes in width of the region. To create the lower edge of the blue region, modes in the lower edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} - \Delta S$ applied to them. To construct the upper edge of the blue region, the upper edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} + \Delta S$ applied to them.

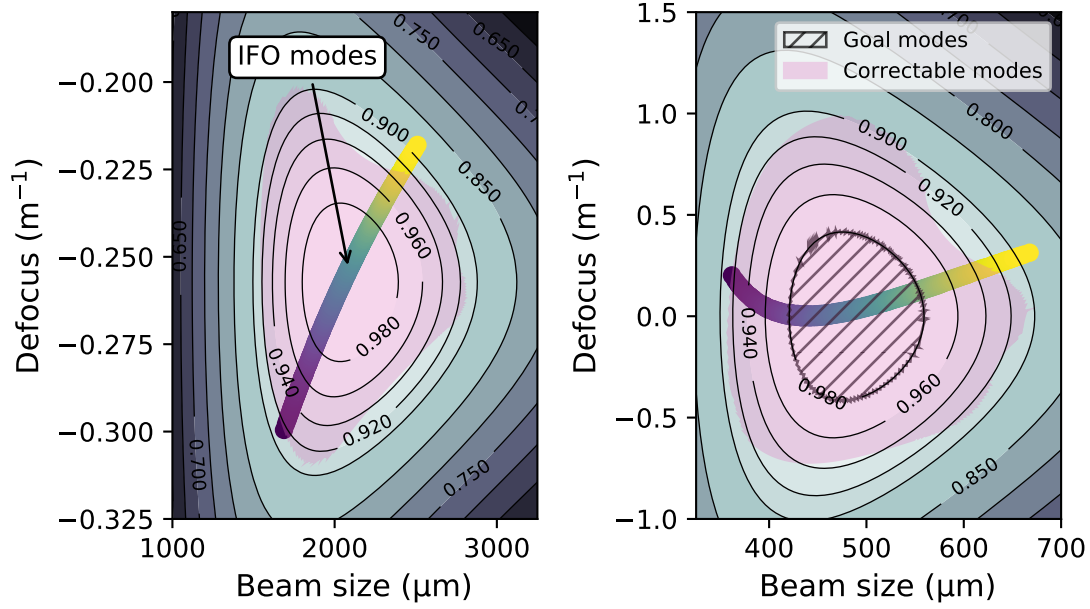


Figure 6.7: Left panel: the set of modes that could be present at the SRM, as described in Chapter 4, is shown by the purple-blue-green-yellow curve. The pink region shows which modes can reach 98% mode-matching by using the AWC in HAM6. The contours indicate the mode-matching between a mode (W, S) and the OMC mode. Right panel: this shows the same modes as the left panel, except they have been transformed by the optical layout to the plane of the OMC waist. Modes within the pink region can be actuated on so they move to the goal region, which is indicated with a hatched area.

that the arm mode will be at least 98% mode-matched to the OMC by the layout. For example, if it was found that arm modes were most likely to be characterised by an error in beam waist size at the OMC, the active optics should be arranged so that they can actuate predominantly on beam size at the OMC. However, choosing a layout which favours flexibility in one degree of freedom over the other may lead to unforeseen beam parameter errors being uncorrectable.

The shape of the pink region in Figure 6.7 is nominal; when the HAM6 layout is constructed the pink regions shape will be altered by geometric errors in OMx1 and OMx2. Nevertheless, Figure 6.7 is encouraging since it shows that the actuation provided by OMx1 and OMx2 is highly tolerant to mode-matching errors generated by errors in the SRC.

6.4 Conclusion

The functionality required from the BHD layout was discussed, and a proposed layout that satisfies these requirements is shown in Figure 6.1. The optical properties of this layout were investigated. By selection of the right combination of mode-matching optics, near orthogonal control of the signal-LO mode-matching and beam alignment to the OMC was achieved.

The effect that the AWC had on the region of WS space that contains modes which satisfy the $> 98\%$ mode-matching requirement space was investigated. WS diagrams were used to represent how robust the layout is to initial mode-mismatch. Figure 6.7 shows that the AWC provides a huge amount of mode-matching range. This means that the layout can tolerate large changes in the beam parameter at the

SRM.

Chapter 7

In Air BHD experiments

This chapter is pedagogical in style, where the MZ is used to demonstrate shot noise being related to the wavelength of light. The in air MZ had unequal arm powers to create a signal LO interference style signal. From this chapter, will show shot noise limited measurement in terms of metres, and discuss various noise sources in the lab. By introducing a peak . The experiment was modelled, and there is a calculation based analysis as well.

Chapter 8

In-vacuum experiments

8.1 Introduction

The suspended one looked at misalignments effect on signal, as well as being a more interesting control problem.

- made a suspended MZ in vacuum to get shot noise at lower frequencies and have control over angle of mirrors (angular dither)
- control theory important here
- investigated amplitude modulation
- investigated frequency noise with this as well (with frequency modulated laser)

Chapter 9

Characterisation of InGaAs

Photodiodes for detecting 2 μm Light

Chapter 10

Characterisation of InSb Photodiode

Chapter 11

Conclusion

Appendix A

FINESSE Simulations

The following simulations were used in chapter 4. Section A.1 shows how the input beam parameter was determined so that single bounce simulations and simulations with arm cavities had the same beam parameters. Section A.2 contains tables of the spacings and properties of the optics for the LLO interferometer.

A.1 Mode Matching the Laser to the Arm Cavity

To find suitable numbers to use for the `gauss` command in the simulation of single bounce and locked arm beam measurements (see section A.3 and figure 4.2), the following simulation was run. It uses the same Y arm and beam splitter as the main simulation from section A.3, however it does not have an X arm or the SRC. Without the `cav` command, finesse will default to using a 2mm beam with its waist at $z = 0$ with respect to the node at the laser. The following beam parameters were found with a `bp` detector placed at the laser's node:

- $w_0 = 0.00864064287177005$ m

- $z = 1339.83530090614$ m

To figure out if the sign of z is correct, consider what happens if the beam parameter detector is moved 1m towards the cavity to **bs_a**:

- $w_0 = 0.00864064287177005$
- $z = 1338.83530090614$ m

z increasing suggests moving from **n1** to **bs_a** causes one to be closer to the waist. Intuitively, we know this is the case since the geometry of the cavity means the waist is within the cavity, and moving towards the cavity means you are moving towards the cavity waist. In other words, we know the beam should be getting smaller as we move from the laser to the cavity, which matches the behaviour seen here.

However, setting the beam parameter at the laser with the first set of numbers gives *terrible* mode matching. The simulation below investigates this. A photodiode which is masked from the TM00 mode

```
pd HOM bs_d
mask HOM 0 0 0
```

was placed at **bs_d** , which has the beam reflected from the arm cavity in it. The value of z for the **gauss** was varied and the power incident on **HOM** was measured. To check that enough HOM were being used, the simulation was repeated with several **maxtem** commands. The results had a quadratic fitted to them, and the minimum was taken as the optimal value. This is shown in figure A.1. The values:

- $w_0 = 0.00864064287177005$ m

- $z = -1339.83683935$ m

give more than 99.99999% mode matching, which for this work is plenty. This is close to the original values that we obtained, except for a minus sign. This is due to the way FINESSE computes the beam parameter.

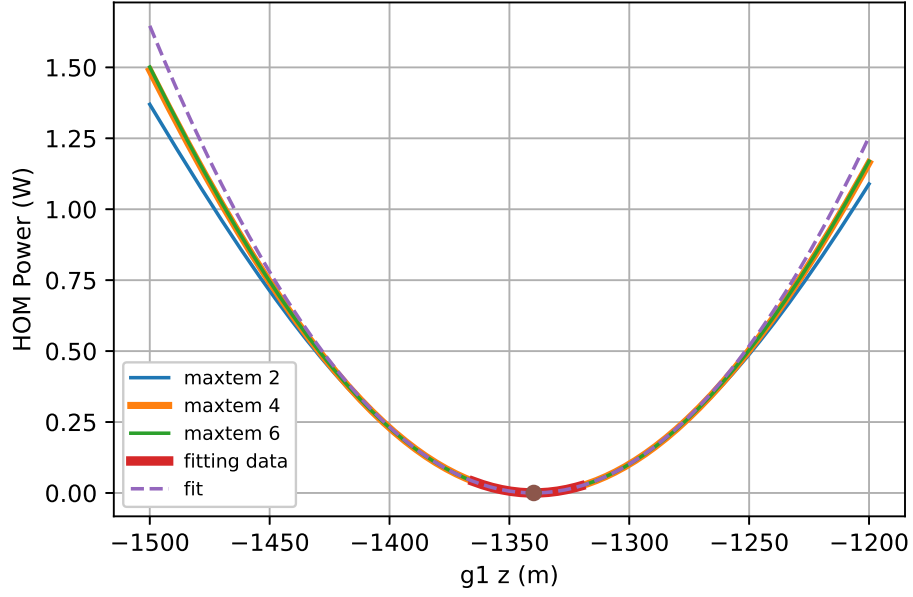


Figure A.1: Optimisation of parameters used in the `gauss` command to mode match the laser to the arms in the simulations described in chapter 4 and section A.3. The blue, green and orange lines show how the mode matching changes as the beam waist location is changed. As the mode matching becomes larger, more HOM are required, as the blue and green/orange lines diverge. A quadratic was fitted to the data highlighted in red, and the minimum value was used as the waist location parameter for the `gauss` command.

A.2 Interferometer Parameters

The simulated interferometer uses ‘design’ arms and the LLO SRC. The simulated arm cavity geometry is based on the design values to avoid contrast defects. The following subsections contain tables specifying the design and measured values for each component of the simulation.

A.2.1 Radii of Curvature of optical components

The optics in the SRC are simulated with the values from table A.2. The arm cavity optics have been simulated using the values from table A.1.

Optic Name	RoC (m)	Focal length (m)	Reference
SRM	−5.6938	−2.8469	[123]
SR2	−6.427	−3.2135	[123]
SR3	36	18	[123]
ITM	1934	967.0	[123]
ETM	2245	1122.5	[123]

Table A.1: The aLIGO design values of the RoC of the optics inside the SRC and the arm.

Optic Name	RoC (m)	Focal length (m)	Reference
SRM-02	-5.673 ± 0.002	-2.8365 ± 0.001	[124]
SR2-04	-6.406 ± 0.006	-3.203 ± 0.003	[125], [124]
SR3-01	35.97 ± 0.012	17.985 ± 0.006	[126], [124]
ITM-08	1938.44	969.22	[124]
ETM-15	2245.4	1122.7	[124]

Table A.2: The measured values of the RoC of the optics at LLO that are inside the SRC and the arm.

A.2.2 Spacings

The nominal values with a diagram can be found in [127] for the H1/L1 spacings. This information is also in [123]. The L1 Zemax parameters are found in [128]. Table A.3 contains the distances derived from the Zemax coordinates. Most spacings do not require sub mm accuracy, however SR3 to SR2 does since the Rayleigh range of this beam is very small (about 4 cm). The design value spacings are shown in table A.4. The values used in an earlier FINESSE model of LLO [108] are shown in table A.5. The thickness of each optic is found in table A.6.

Space Name	Length (m)
EMTY to ITMY	3994.4850
ITMY to CP	0.02
CP to BS HR	4.8471
BS AR to SR3	19.3659
SR3 to SR2	15.4435
SR2 to SRM	15.7562

Table A.3: Measured spacings between optics at LLO [128]

Space Name	Length (m)
EMTY to ITMY	3994.5
ITMY to CP	0.05
CP to BS HR	4.8497
BS AR to SR3	19.368
SR3 to SR2	15.4607
SR2 to SRM	15.726

Table A.4: Design value for spacings [127]

Space Name	Length (m)
EMTY to ITMY	3994.515
ITMY to CP	0.02
CP to BS HR	4.847
BS AR to SR3	19.3661
SR3 to SR2	15.4435
SR2 to SRM	15.7566

Table A.5: Values used in an earlier FINESSE model of LLO [108]

Optic	Thickness (m)
ITM	0.200
CP	0.100
BS	0.0685
Angled BS	0.1202

Table A.6: Thickness of optics between the ITM and the SRM.

A.2.3 Nominal Simulation

The nominal simulation values is shown in table A.7.

Parameter	Value	Description
SRM RoC	−5.673m	Measured RoC of SRM-02 installed at LLO
SR2 focal length	−3.203 m	Measured focal length of SR2-04 installed at LLO
SR3 focal length	17.985 m	Measured focal length of SR3-01 installed at LLO
ITM RoC	1934m	Design Value
ETM RoC	2245m	Design Value
ITM thermal lens	50 km	
ITM to ETM distance	3994.5m	Design Value
ITM to BS Y	4.8471m	As-built Value for Y arm
ITM to BS X	4.8471m	As-built Value for Y arm
BS to SR3	19.3659m	As-built Value
SR3 to SR2	15.4435m	As-built Value
SR2 to SRM	15.7562m	As-built Value
ITM T	0.0148	Power transmission coefficient
SRM T	0.324	Power transmission coefficient
ITM L	10 ppm	Scatter loss of ITM HR surface
ETM L	10 ppm	Scatter loss of ETM HR surface

Table A.7: Nominal values for the major components of the simulation. Design values were used for the arms to remove the contrast defect that would occur otherwise. The as-built values for the SRC were used to replicate the behaviour of the SRC at LLO more closely. The ITMs that are simulated have a thickness of 0.1 m and a refractive index of 1.45. The thermal lens is implemented as shown in figure 4.2 .

A.3 FINESSE Code

This section contains the FINESSE code used in chapter 4.

A.3.1 Michelson Interferometers with Thin Beamsplitter

This FINESSE code describes a Michelson interferometer where the beam splitter is implemented with a single `bs` command. This simulation will produce an incorrect result as the phase at the beamsplitter is not properly accounted for.

```
l l1 1 0 n1
s sin 1 n1 na

bs thin_bs 0.5 0.5 0 0 na nb nc nd

s Yarm 1 nb nY
m Ymirror 1 0 0 nY dump

s Xarm 1 nc nX
m Xmirror 1 0 0 nX dump

pd AS nd

xaxis Xmirror phi lin 0 180 100
```

A.3.2 Michelson Interferometers with Thick Beamsplitter

This FINESSE code describes a Michelson interferometer where the beam splitter is implemented with 3 `bs` and 2 `space` commands. This is the proper way to implement a beamsplitter in FINESSE.

```

l l1 1 0 n1
s sin 1 n1 na

bs thick_bs_1 0.5 0.5 0 0 na nb nsub1 nsub3
s bs_subX 1p nsub1 nsub2
s bs_subY 1p nsub3 nsub4
bs thick_bs_X 0 1 0 0 nsub2 dump nc dump
bs thick_bs_AS 0 1 0 0 dump nsub4 dumo nd

s Yarm 1 nb nY
m Ymirror 1 0 0 nY dump

s Xarm 1 nc nX
m Xmirror 1 0 0 nX dump

pd AS nd

xaxis Xmirror phi lin 0 180 100

```

A.3.3 Main Simulation

The following FINESSE code was used to simulate mode mismatches in the SRC.

```

l l1 5M 0 n1
gauss beamParaml1 l1 n1 0.00864064287177005 -1339.83683935

maxtem 12

s space1 1 n1 bs_a

```

```

#####
#bs beam_splitter 0.5 0.5 $phi_BS 0 bs_a bs_b bs_c bs_d

bs BSfront 0.5 0.5 0 0 bs_a bs_b nBSi1 nBSi3
s BSsubstrate1 1p 1.45 nBSi1 nBSi2
s BSsubstrate2 1p 1.45 nBSi3 nBSi4
bs BSback1 0 1 0 0 nBSi2 dump bs_c dump
bs BSback2 0 1 0 0 nBSi4 dump bs_d dump

#####
## Y ARM

s s_bs_to_ITMy 4.8471 bs_b nITMyARa

#ITMy
m1 ITMyAR 1 0 $phi_ITMy nITMyARa nITMyARb
s sITMy_thickness 0.05 1.45 nITMyARb nITMyTLa
lens ITMyTL 50k nITMyTLa nITMyTLb
s sITMy_thickness2 0.05 1.45 nITMyTLb nITMyHRa
m1 ITMy 0.0148 10u $phi_ITMy nITMyHRa nITMyHRb
attr ITMy Rc -1934

s space_arm_y 3994.5 nITMyHRb nETMy

m1 ETMy 0 10u $phi_ETMy nETMy dump
attr ETMy Rc 2245

cav army ETMy nETMy ITMy nITMyHRb

#####

```

X ARM

s s_bs_to_ITMx 4.8471 bs_c nITMxARa

#ITMx

m1 ITMxAR 1 0 \$phi_ITMx nITMxARa nITMxARb

s sITMx_thickness 0.05 1.45 nITMxARb nITMxTLa

lens ITMxTL 50k nITMxTLa nITMxTLb

s sITMx_thickness2 0.05 1.45 nITMxTLb nITMxHRa

m1 ITMx 0.0148 10u \$phi_ITMx nITMxHRa nITMxHRb

attr ITMx Rc -1934

s space_arm_x 3994.5 nITMxHRb nETMx

m1 ETMx 0 10u \$phi_ETMx nETMx dump

attr ETMx Rc 2245

cav armx ETMx nETMx ITMx nITMxHRb

#####

SRC

s bs_to_SR3 19.3659 bs_d nSR3a

lens SR3 17.985 nSR3a nSR3b

s SR3_to_SR2 15.4435 nSR3b nSR2a

```

lens SR2 -3.203 nSR2a nSR2b

s SR2_to_SRM 15.7562 nSR2b nSRMa

m1 SRM 0.324 0 0 nSRMa nSRMb
s SRM_substrate 0.07 1.45 nSRMb nSRMc
m SRM_AR 0 1 0 nSRMc nSRMd
attr SRM Rc -5.673

#cav src ITMx nITMxHRa SRM nSRMa

#####
#N.B these tunings don't represent how an IFO is kept on the dark
#fringe
const phi_ITMx 0
const phi_ITMy 0
const phi_ETMx 0
const phi_ETMy 0
const phi_BS 0
#####

beam SRM_trans nSRMd
#mask SRM_trans 0 0 0

bp SRM_trans_w0 x w0 nSRMd
bp SRM_trans_w x w nSRMd
bp SRM_trans_r x r nSRMd

bp laser_w x w0 n1
bp laser_z x z n1

```

```

beam ITMy_refl nITMyHRa
beam ITMy_refl nITMyHRa

bp ITMy_refl_w x w nITMyHRa
bp ITMx_refl_w x w nITMxHRa

pd p_in bs_a*
pd p_ref bs_a

pd p_out nSRMd

pd p_out_hOM nSRMd
mask p_out_hOM 0 0 0

ad ad_HOM_2_y 2 0 0 nSRMd
ad ad_HOM_2_x 0 2 0 nSRMd

gouy src_gouy y sITMx_thickness sITMx_thickness2 s_bs_to_ITMx
      bs_to_SR3 SR3_to_SR2 SR2_to_SRM

trace 64
phase 2

xaxis SRM_trans x lin -10 10 200

```

Appendix B

Algorithm For Finding a Three Optic Solution with 45° Gouy Phase

Separation between two of the Optics

To find a mode matching solutions between the SRM and the OMCs which have 45° phase separation for the AWC optics, an algorithm was developed. To visualise how this algorithm works see figures B.2 and B.1. Since a beam can be described by two parameters, its defocus S and width w , the modes can be shown in a 2-D space [121] (figure B.2). The action of a lens or curved mirror is to move the mode in the S direction. It is helpful to define a mode continuum along which w and/or S varies. The principle by which this algorithm works is: The continuum of modes after optic 1 and the continuum of modes just before optic 3 can be easily obtained since they must have a defined beam size, so they only vary in defocus. These two continuum are propagated to optic 2, and then its just a case of finding how to move from one continuum to an other.

The continua at optic 2 do not necessarily have to have the shape that they are sketched with in figure B.2, however if the blue-green and red-yellow continuum extend far enough they will always wrap around. It is also not guaranteed that every mode in the red-yellow can be reached from the blue-green continuum, and vice versa. Since the mapping between the blue-green and red-yellow continuum is not one to one, there is some choice in you can make. Both ‘branches’ of the continua, where the continuum turns back on itself in the w direction, do not need to be analysed at the same time, but a better solution may be found by looking at different combinations of branches. This is useful in setting a limit on the size of the continuum you search.

In our implementation of this algorithm, the continua are actually discrete, which means that when finding a mapping at optic two it is not guaranteed that a mode in the red-yellow continuum will have a corresponding mode of the same with in the blue-green continuum. Because of this granularity, if you were to plot optimal values of optic 2 and optic 3 power as a function of optic 1 power, you would see jumps in the optimal lens choices. This is especially bad around where the turning point in the continua is, since the modes defocus changes quickly while their size does not.

B.0.1 Method

This section delineates the method described in section B and can be visualised using figure B.2.

Define the target mode at the optic 3. In figure B.2, this is mode l at optic three.

Define a set of modes which share the same beam size as the target mode at optic 3 but have varying radii of curvature. In figure B.2, this is the red-yellow continuum at optic three.

Propagate the set of modes defined at optic 3 back to the optic 2. In figure B.2, this is the red-yellow continuum at optic 2.

Define the initial beam parameter and propagate this to optic 1. In figure B.2, this is mode i at optic one.

Define a set of modes that have the same beam size as the initial beam at optic 1 but differ in defocus. In figure B.2, this is the blue-green continuum at optic one.

Propagate the modes defined at the optic 1 to optic 2. In figure B.2, this is the blue-green continuum at optic two.

Calculate the difference in optical power between mode i and a mode in the blue-green continuum (e.g. mode j). In figure B.2, mode j represents just one possible mode in this continuum. Save this power.

Find a mode in the red-yellow continuum which is closest in size to mode j, call this mode k. In figure B.2, there is another possible choice on the red-yellow continua, however only one of them needs to be considered.

Calculate the difference in optical power between mode j and mode k at optic 2. Save this power.

Calculate the difference in optical power mode k and the target mode at optic 3. Save this power.

Calculate the Gouy phase separation of optic 2 and optic 3.

Repeat for every mode in the blue-green continuum, or until a gouy phase separation of 45degree has been found.

The number of modes that are considered and the range of modes in the continua should be tailored for the optical layout that is being considered. Our problem was with metre scale spacings between optics, and with metre scale Rayleigh lengths. It was found that continua with a range of the order 1-10 dioptres were suitable. This does not mean that an adaptive optic would need this actuation range, rather it allowed for the static radii of curvature for each optic that gives optimal Gouy phase separation between the active mode matching elements to be found. Using this algorithm, we were able to rapidly develop many possible HAM6 layouts. This allows for more time to consider various engineering factors (suspension size and the like) and how to adapt the mirror strengths accordingly to accommodate for small changes in a tentative design.

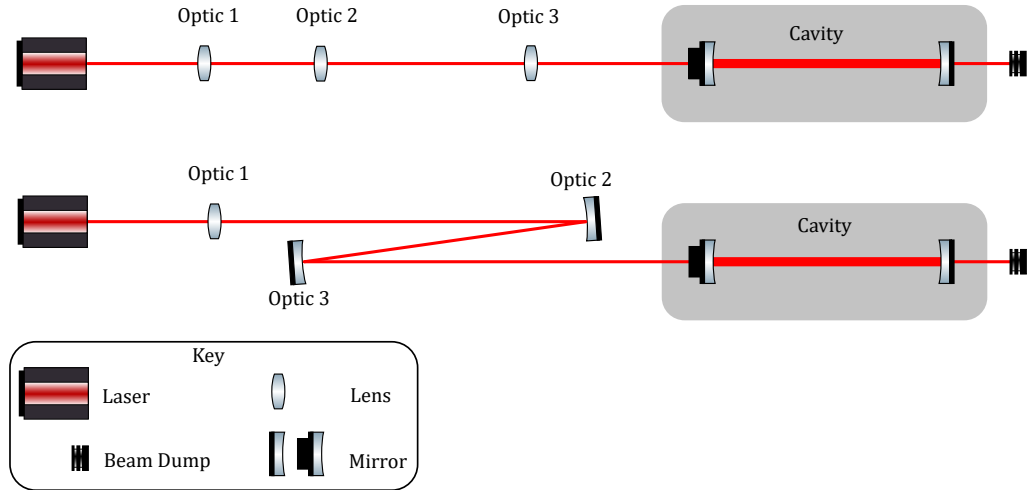


Figure B.1: Top layout: Three lenses used to achieve mode matching to the cavity. These are shown as positive lenses, however negative lenses can also be used when designing mode matching systems. Bottom Layout: A lens and two curved mirrors to achieve mode matching to a cavity. The optics 1,2 and 3 correspond to the ones shown in figure B.2. The initial and target modes are defined in these examples by the laser mode and the cavity mode.

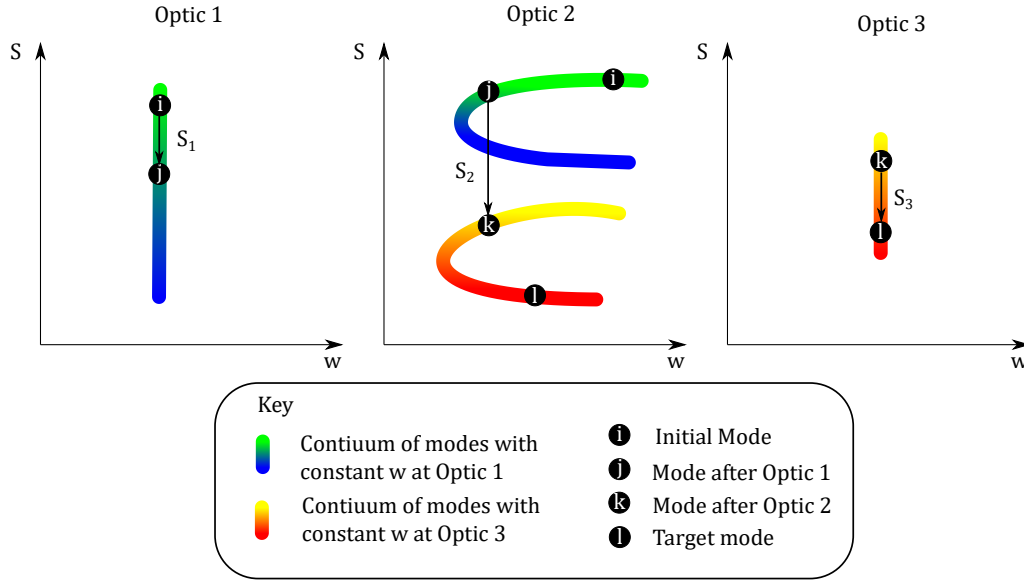


Figure B.2: Sketch of the WS phase space representations of a Gaussian beams at the three positions along the optical axis where a mirror or a lens could be. Examples of such layouts are shown in figure B.1. The optical power of the first optic is given by the difference in the defocus between the initial mode (i) and the second mode (j). The power of the second optic is found by finding the difference in defocus from the blue-green continuum to the red-yellow continuum. There are 2 optical powers that will move the mode j to the red-yellow continuum, and two possible modes in the blue-green continuum that can from which k can be reached, giving 4 possible similar solutions. The algorithm can be set up so that it is only one way of moving between the blue-green and red-yellow continua by limiting their extent. The power of the third optic is found from the difference between the mode k and mode l at optic 3.

Appendix C

Mode Matching Between 2 Astigmatic Beams

C.1 Derivation of Mode Matching TM00 Beams

The [check this naming](#) electric field, E , of a TM00 Gaussian beam can be written as

$$E(x, y, z) = E_x(x, z)E_y(y, z) \quad (\text{C.1})$$

where

$$E_x(x, z) = \frac{1}{\sqrt{q_x(z)}} \exp\left(-ik \frac{x^2}{2q_x(z)}\right), \quad (\text{C.2})$$

$$E_y(y, z) = \frac{1}{\sqrt{q_y(z)}} \exp\left(-ik \frac{y^2}{2q_y(z)}\right). \quad (\text{C.3})$$

For a beam without astigmatism, $x^2 + y^2 = r^2$ and $q_x = q_y = q$ so:

$$E(r, z) = \frac{1}{q(z)} \exp\left(-ik \frac{r^2}{2q(z)}\right). \quad (\text{C.4})$$

The mode matching, $\mathcal{M}$, of two beams can be expressed by overlap of the two fields

$$\mathcal{M} = \frac{|\int_A E_1^* E_2 dA|}{\int_A |E_1|^2 dA \int_A |E_2|^2 dA}. \quad (\text{C.5})$$

The numerator of equation C.5 for TM00 beams can be written as:

$$\left| \int_A E_1^* E_2 dA \right| = \int_A E_1^* E_2 dA \int_A E_1 E_2^* dA \quad (\text{C.6})$$

$$= \int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{y1}^* E_{y2} dy \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx \int_{-\infty}^{\infty} E_{y1} E_{y2}^* dy. \quad (\text{C.7})$$

Since the calculation is the same for the x and y coordinates, we only need to show the solution for the x . The x terms of equation C.7 can be written as:

$$\int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}^*}} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \frac{1}{\sqrt{q_{x2}}} \exp\left(\frac{-ikx^2}{2q_{x2}}\right) dx \quad (\text{C.8})$$

$$\times \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}}} \exp\left(\frac{-ikx^2}{2q_{x1}}\right) \frac{1}{\sqrt{q_{x2}^*}} \exp\left(\frac{ikx^2}{2q_{x2}^*}\right) dx \quad (\text{C.9})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \exp\left(\frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{C.10})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*} - \frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{C.11})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2 (q_{x2} - qx1^*)}{2q_{x1}^* q_{x2}}\right) dx \times c.c. \right). \quad (\text{C.12})$$

Using the identity

$$\mathcal{I} = \int_{-\infty}^{\infty} \exp(iax^2) dx = \frac{i\pi}{a} \quad (\text{C.13})$$

and labelling

$$a = \frac{k(q_{x2} - q_{x1}^*)}{2q_{x1}^* q_{x2}} \quad (\text{C.14})$$

allows equation C.12 to be simplified to

$$\frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2(q_{x2} - q_{x1}^*)}{2q_{x1}^* q_{x2}}\right) dx \times c.c. \right) = \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{i\pi(2q_{x1}^* q_{x2})}{k(q_{x2} - q_{x1}^*)}} \sqrt{\frac{-i\pi(2q_{x1} q_{x2}^*)}{k(q_{x2}^* - q_{x1})}} \quad (\text{C.15})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{4\pi |q_{x1}|^2 |q_{x2}|^2}{k^2 |q_{x2} - q_{x1}^*|^2}} \quad (\text{C.16})$$

$$= \frac{2\pi}{k} \frac{1}{|q_{x2} - q_{x1}^*|}. \quad (\text{C.17})$$

This means that the numerator of equation C.5 is

$$\frac{4\pi^2}{k^2} \frac{1}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{C.18})$$

Since E_x and E_y are orthogonal, $|E|^2 = |E_x|^2 |E_y|^2$, an expression for the denominator of equation C.5 can be found by:

$$\int_A |E|^2 dA = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |E_x|^2 |E_y|^2 dx dy \quad (\text{C.19})$$

$$= \int_{-\infty}^{\infty} \frac{1}{|q_x|} \exp\left(\frac{ikx^2(q_x - q_x^*)}{2q_x q_x^*}\right) \times y \text{ term.} \quad (\text{C.20})$$

Using a similar approach as in the calculation of the numerator terms:

$$\int_A |E|^2 dA = \frac{1}{|q_x|} \sqrt{\frac{i2\pi |q_x|^2}{k(q_x - q_x^*)}} \frac{1}{|q_y|} \sqrt{\frac{i2\pi |q_y|^2}{k(q_y - q_y^*)}} \quad (\text{C.21})$$

$$= -\frac{2\pi}{k} \frac{1}{\sqrt{(q_x - q_x^*)(q_y - q_y^*)}} \quad (\text{C.22})$$

we get the mode matching $\mathcal{M}$ to be

$$\mathcal{M} = \frac{\frac{4\pi}{k^2}}{-\frac{2\pi}{k} \times -\frac{2\pi}{k}} \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)} \sqrt{(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|} \quad (\text{C.23})$$

$$= \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{C.24})$$

C.2 Effect of Error in the Radius of Curvature of a Mirror on the defocus of a Beam

The error in the optical power of a mirror due to an error in the radius of curvature of is

$$S = \frac{2}{R_m} \quad (\text{C.25})$$

$$dS = \frac{-2}{R_m^2} dR. \quad (\text{C.26})$$

The error in the beams defocus is the same as the error in the radius of curvature of the the mirror. The effect this has on the beam parameter q can be found by

defining a contour of modes distance dq from q i.e. $q + |dq| e^{i\theta}$, where dq is

$$q = \frac{q_i}{Sq_i + 1} \quad (\text{C.27})$$

$$dq = \frac{q_i^2}{(-Sq_i + 1)^2} dS. \quad (\text{C.28})$$

This gives you a family of modes which will contain the modes due to an error in the radius of curvature of the mirror, but since q is a complex number the contour you need to define to encapsulate the concept of an error in q also contains modes which are have different beam sizes as well as different defocuses. To do the calculation this way is cumbersome, it is much more straight forward to compute the effect of having a mirror ROC error by evaluating at every mirror

$$\begin{pmatrix} 1 & 0 \\ \pm dS & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -S & 1 \end{pmatrix} \begin{pmatrix} q \\ 1 \end{pmatrix} = \begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix} \quad (\text{C.29})$$

and then divide the top and bottom entries of the $\begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix}$ vector.

Appendix D

Light and Gaussian Modes

This chapter will show the maths...

- check normalisation constants
- check the formula for the electric fields are actually right!
- change formulas to be in terms of a mode function and something that is to do with its propogational along z, and amplitude.

D.1 Plane Waves

Light is electromagnetic radiation. From Maxwell's equations it can be shown that the electric field, $\mathbf{E}$, component of electromagnetic radiation satisfies the wave equation

$$\nabla^2 \mathbf{E} = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{E}. \quad (\text{D.1})$$

A monochromatic, coherent, plane wave solution for $\mathbf{E}$ satisfies equation (D.1):

$$\mathbf{E} = \hat{\mathbf{n}}E \exp(i(\omega t - \mathbf{k} \cdot \mathbf{r} + \varphi)), \quad (\text{D.2})$$

where $\hat{\mathbf{u}}$ indicates the polarisation, E is the complex field amplitude, ω is the angular frequency, $\mathbf{k}$ is the wave vector and φ is the absolute phase. The physical amplitude of the electric field oscillation is obtained from the $\Re(E)$. For optical systems is useful to define an optical axis. When the light is travelling along the optical axis, the $\mathbf{k} \cdot \mathbf{r}$ becomes kz . k is known as the wave number and is related to the wavelength of the light, λ by

$$k = \frac{2\pi}{\lambda}. \quad (\text{D.3})$$

For plane waves there is no dependence of $\mathbf{E}$ on the x and y coordinates. The absolute phase φ becomes relevant in calculations involving interfering beams. By inserting equation (D.2) into equation (D.1), we obtain the dispersion relationship

$$\omega = ck. \quad (\text{D.4})$$

The plane wave solution can be used to analyse many optical phenomena, such as amplitude modulation, phase modulation and interference.

D.2 Gaussian Modes

Light in an optical set-up is normally generated by a laser. Lasers produce almost monochromatic, coherent, light so in this sense they are quite similar to a plane wave, however lasers also have geometric properties. A laser produces light with an

approximately circular beam profile which is brightest at its centre. The spread of light power in a laser beam follows a Gaussian distribution, and the width of the beam profile is characterised by the radius at which $\sim 86\%$ of the light power is contained.

As the beam propagates along its optical axis, the beam size changes. A Gaussian beam has a point where it is smallest, this is known as the waist of the beam. A Gaussian beam can be completely characterised by the size of its waist and location along the waist along the optical axis. Whenever a Gaussian mode interacts with a simple optical component such as a lens or a mirror, the current Gaussian mode is transformed into a new Gaussian mode.

The following sections describe Gaussian modes and how they interact with real optical layouts with imperfections. The lowest order Gaussian mode will be derived, and the properties of this mode will be explained. The concept of the complex Gaussian beam parameter will be introduced as a way of encapsulating all the information needed to determine the properties of a Gaussian mode. Astigmatic Gaussian beams will be defined in terms of two complex Gaussian beam parameters. The first order Hermite-Gaussian (HG) mode will be shown to model the effect of misalignment, and then it will be described how all laser beams can be made up of a set of HG modes. Finally, the way that a Gaussian mode traverses an optical set-up will be expressed in terms of the ABCD matrix formalism.

D.3 The Paraxial Wave Equation

Short intro on what this is

To derive the paraxial wave equation, we apply the separation of variables to equation (D.1):

$$\mathbf{E}(\mathbf{r}, t) = E(\mathbf{r})T(t). \quad (\text{D.5})$$

Next we reform equation D.1 in terms of E and T

$$\frac{\nabla^2 E}{E} = \frac{1}{c^2 T} \frac{\partial^2 T}{\partial T^2}. \quad (\text{D.6})$$

By noticing that either side of the equation (D.6) depends on $\mathbf{r}$ or t , the left hand side of this equation must equal a constant, and the Helmholtz equation can be obtained:

$$\frac{\nabla^2 E}{E} = -k^2. \quad (\text{D.7})$$

Rewriting $E(\mathbf{r})$ in terms of two functions, **some how remove units from $u(\mathbf{r})$**

$$E(\mathbf{r}) = u(\mathbf{r}) \exp(-ikz), \quad (\text{D.8})$$

and substituting this into equation (D.7) yields

$$\left(\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u(\mathbf{r}) - 2ik \frac{\partial u}{\partial z} + \frac{\partial^2 u}{\partial z^2} \right) \exp(ikz) = 0. \quad (\text{D.9})$$

The paraxial approximation means the $\frac{\partial^2 u}{\partial z^2}$ is neglected as it is much smaller than the other terms. Applying this approximation produces the paraxial Helmholtz equation

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u(\mathbf{r}) - 2ik \frac{\partial u}{\partial z} = 0. \quad (\text{D.10})$$

The physical interpretation of $u(\mathbf{r})$ is a function that describes the spatial distribution of the complex amplitude of the light field. **reword this so it actually makes sense.**

D.4 The fundamental Gaussian mode

Rewrite E in terms of u .

The simplest geometry of a laser beam can be described by the fundamental Gaussian mode. This is a mode which has a circular intensity profile that follows a Gaussian distribution, and the surface of constant phase is curved. Substituting the simplest solution to (D.10) into equation (D.8) produces the fundamental Gaussian mode **find out what the amplitude constant is**

$$E(\mathbf{r}) = E_0 \sqrt{\frac{2}{\pi}} \frac{1}{w(z)} \exp(i\phi(z)) \exp\left(-ik \frac{x^2 + y^2}{2R_C(z)} - \frac{x^2 + y^2}{w^2(z)}\right) \exp(-ikz). \quad (\text{D.11})$$

This is characterised by a function $w(z)$, which gives the size of the beam at position z the optical axis, by $\phi(z)$, which gives the Gouy Phase of the beam at z along the optical axis, and by $R_c(z)$, which gives radius of curvature of the beam at z . The minimum value of $w(z)$ is known as the beam waist, or w_0 .

It is useful to express $w(z), \phi(z)$ and $R_c(z)$ in terms of the Rayleigh range z_R . The Rayleigh range characterises how fast the beam will spread out. A beam with a short Rayleigh length will expand quicker than a beam with a longer Rayleigh length. For example, in order for the beam to not be significantly clipped by the aperture of the test masses, the Rayleigh length of the Gaussian beam in the arm cavities at LIGO

has to be on the order of kilometres. The size of the waist determines the Rayleigh length, a smaller beam waist corresponds to a shorter Rayleigh length. If the waist of a Gaussian beam at $z = 0$, the position along the z axis by which the radius has increased by a factor of $\sqrt{2}$ is known as the Rayleigh range z_R , and is given by

$$z_R = \frac{\pi w_0^2}{\lambda}. \quad (\text{D.12})$$

The radius of the beam as it propagates along the optical axis can be written as

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_R} \right)^2}. \quad (\text{D.13})$$

The Rayleigh range can be used to approximately split beam into two regions: the near field for $z < z_R$ and the far field for $z > z_R$. In the far field equation (D.13) becomes linear and the beam size can be approximated as

$$w(z) \approx w_0 \frac{z}{z_R}. \quad (\text{D.14})$$

As a Gaussian beam travels propagates along the optical axis, it acquires an additional phase shift compared to a plane wave. This is known as the Gouy phase shift, $\phi(z)$, and it given by

$$\phi(z) = \tan^{-1} \frac{z}{z_R}. \quad (\text{D.15})$$

At either side of it waist, the fundamental Gaussian mode will be shifted $\pm\pi/2$ relative to a plane wave. An alternative way of putting this is the fundamental Gaussian mode will gain a π change as it goes through from $-\infty$ to ∞ compared to a plane wave. Most of this phase change is gained within $\pm z_R$ of its waist. The

Gouy phase results in the the phase velocity at the waist being greater for a Gaussian beam than it is for plane waves.

The radius of curvature, $R_c(z)$, in terms of the Rayleigh range is

$$R = z + \frac{z_R^2}{z}. \quad (\text{D.16})$$

For $z \ll z_R$ the radius of curvature can be approximated as infinite. As the beam approaches its Rayleigh length the radius of curvature decreases until it reaches its minimum at $z = z_R$. After this the radius of curvature begin to flatten out again, and for $z \gg z_R$ the radius of curvature can be approximated as z . The defocus, $S(z)$, of a beam is a more practical way of describing a beam as it is easier to compute the effects of a beam interacting with a focussing element of an optical layout in terms of the defocus than it is with the radius of curvature. The defocus is defined as the inverse of its radius of curvature

$$S(z) = \frac{1}{R(z)} \quad (\text{D.17})$$

The fundamental Gaussian mode can be expressed in terms of a complex beam parameter, $q(z)$. This parameter is useful for calculating how a gaussian beam is transformed by an optical system. From this parameter, all the physical quantities related to a Gaussian beam can be computed. The complex beam parameter is defined as

$$q(z) = z + iz_R. \quad (\text{D.18})$$

The inverse of equation D.18 can be written in terms of the curvature and radius of the beam:

$$\frac{1}{q(z)} = \frac{1}{R(z)} - i \frac{\lambda}{\pi w^2(z)}. \quad (\text{D.19})$$

Equation (D.11) can be re written in terms of this the complex beam parameter,

$$E(\mathbf{r}) = E_0 \sqrt{\frac{2}{\pi}} \frac{q_0}{w_0 q(z)} \exp(i\phi(z)) \exp\left(-ik \frac{x^2 + y^2}{2q(z)}\right) \exp(-ikz). \quad (\text{D.20})$$

The radius of the beam, gouy phase, and the defocus at a point z along the optical axis can be calculated from $q(z)$ using

$$w(z) = \sqrt{\frac{\lambda}{-\pi \Im\left(\frac{1}{q(z)}\right)}}, \quad (\text{D.21})$$

$$\phi(z) = \tan^{-1}\left(\frac{\Re(q(z))}{\Im(q(z))}\right), \quad (\text{D.22})$$

$$S(z) = \Re\left(\frac{1}{q(z)}\right). \quad (\text{D.23})$$

Mode matching integral

D.4.1 Astigmatism

If a beam has an elliptical rather than circular profile, then the beam is astigmatic. For simple astigmatism, a beam can be described with two complex Gaussian beam parameters, one for the yz plane and one for the xz plane. These are referred to as the tangential (xz) and the saggital (xy) planes. A gaussian beam with simple astigmatism will have a different waist size and location in the tangential and sagittal planes. This type of astigmatism arises due to gaussian beams passing through a

cylindrical lens or having a non-zero angle of incidence to a mirror.

An simple astigmatic gaussian beam can be split up into two orthogonal fields
(rewrite in terms of $u(\mathbf{r})$)

$$\begin{aligned}
E(\mathbf{r}) &= E(x, z)E(y, z) \\
&= E_0 \sqrt{\frac{2}{\pi}} \sqrt{\frac{q_{x0}}{w_{x0}q_x(z)}} \sqrt{\frac{q_{y0}}{w_{y0}q_y(z)}} \exp(i\phi_{Av}(z)) \\
&\quad \times \exp\left(-ik\frac{x^2}{2q_x(z)} - ik\frac{y^2}{2q_y(z)}\right) \exp(-ikz). \tag{D.24}
\end{aligned}$$

where $\exp(i\phi_{Av}(z))$ is the average of the Gouy Phase in the saggital and tangential planes. The equations for waist size, equation (D.21), Gouy phase, equation (D.22) and defocus, equation (D.23), remain the same. A beam with simple astigmatism can be though of as two independent beams, and their interaction with the optical layout can be calculated independently.

D.5 Gaussian Beams Traversing optical Layouts

As a Gaussian beam propagates through an optical layout, the beam parameter that describes changes. The effect of simple optical components can be modelled with ABCD matrices, and an entire layout can be modelled by multiplying a series of ABCD matrices together. Components that introduce astigmatism to the beam can be modelled as having separate ABCD matrices for the tangential and saggital planes. This formalism can also be used to compute cavity eigenmodes.

When a beam interacts with an optical component it is transformed from a mode deccribed by q_1 to a new mode q_2 according to

$$q_2 = \frac{Aq_1 + B}{Cq_1 + D} \quad (\text{D.25})$$

where $n_{1,2}$ are the refractive indicies of the medium $q_{1,2}$ are in. The coefficients A, B, C, D are defined by a coefficient matrix $\mathbf{M}$,

$$\mathbf{M} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}. \quad (\text{D.26})$$

To compute the coefficient matrix for a series of elements numbered 0 to n, the matrices for each individual element should be multiplied together

$$\mathbf{M} = \mathbf{M}_n \mathbf{M}_{n-1} \dots \mathbf{M}_0. \quad (\text{D.27})$$

The ABCD matrices for various components is summarised in table D.1.

D.6 Higher Order Modes and optical imperfections

The fundemental gaussian mode describes the electric field of a beam in an ideal optical system. Imperfections in the optical layout, such as misalignment and mode-mismatch, results in a different mode shape. If two modes are misaligned slightly, one will see a mode profile that looks like two spots instead of one. A small mode-mismatch results in a ring shaped mode profile. These imperfections in the layout and the resulting beam shapes can be modelled as the optical layout scattering light

Optical Component	ABCD Matrix
space	$\begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}$
Mirror or lens (normal incidence)	$\begin{pmatrix} 1 & 0 \\ -S & 1 \end{pmatrix}$
Mirror (AOI = α , Sagittal Plane)	$\begin{pmatrix} 1 & 0 \\ -\frac{S}{\cos \alpha} & 1 \end{pmatrix}$
Mirror (AOI = α , Tangential Plane)	$\begin{pmatrix} 1 & 0 \\ -S \cos \alpha & 1 \end{pmatrix}$

Table D.1: Table of the ABCD matrices for simple optical components.

into higher-order optical modes.

There are many solutions to equation D.10. One complete, orthonormal basis set of functions that can be used to construct solutions to D.10 are the Hermite-Gaussian (HG) modes. These modes have xy symmetry, and appear as 'square' shaped. These are useful for modelling misalignments, as this type of imperfection can be naturally broken into xy coordinates. Mode-mismatches are best described by the $r\theta$ coordinates, and so its natural to express these in terms of another orthonormal basis set known as the Laguerre-Gauss (LG) modes. However, if the beam is astigmatic then the mode no longer has the symmetry required in order to be modelled by LG modes.

Equation (D.10) can be solved using HG modes, which have the mathematical form of

$$u_{nm}(x, y, z) = u_n(x, z)u_m(y, z). \quad (\text{D.28})$$

The functions $u_{n,m}$ have the form

$$u_n(x, z) = \left(\frac{2}{\pi}\right)^{1/4} \left(\frac{\exp(i(2n+1)\phi(z))}{2^n n! w(z)}\right)^{1/2} H_n\left(\frac{\sqrt{2}x}{w(z)}\right) \exp\left(-i\frac{kx^2}{2R_c(z)} - i\frac{x^2}{w^2(z)}\right). \quad (\text{D.29})$$

where $H_n(x)$ a Hermite polynomial of the order n . By writing $u_{nm}(x, y, z)$ out explicitly,

$$u_{nm}(x, y, z) = \left(2^{(n+m-1)} n! m! \pi\right)^{1/2} \frac{1}{w(z)} \exp(i(n+m+1)\phi(z)) \\ H_n\left(\frac{\sqrt{2}x}{w(z)}\right) H_m\left(\frac{\sqrt{2}y}{w(z)}\right) \exp\left(-ik\frac{x^2+y^2}{2R_c(z)} - i\frac{x^2+y^2}{w^2(z)}\right), \quad (\text{D.30})$$

we can see that the higher order mode solutions to equation D.10 have the same waist size and radius of curvature to the fundamental mode, but different intensity distributions and additional Gouy phase. If n and m are set to zero, then equation D.11 is recovered. These HG modes interact with optical elements in the same way as the fundamental mode.

The eigen mode of a cavity is determined by its length and the radius of curvature of its end mirrors. The Gaussian mode that is the eigen mode of the cavity will have the same radius of curvature at the end mirrors of the cavity as the mirrors themselves. As higher order modes share the same radius of curvature as the fundamental

mode, these modes can also be resonant within the cavity. What separates these modes out in a cavity is the round trip Gouy phase that is acquired. Most cavities are designed such that the round trip Gouy phase is such that only one mode can be resonant at once, and all the other modes destructively interfere.

The Gouy phase separation of higher order modes is also important when designing alignment and mode matching control systems, as different types of errors have different phases. For example, angular and translational misalignment are both first order effects. This means they both transfer light from the HG_{00} mode into the HG_{10} . However, the complex amplitude of these misalignment situations is separated by $\pi/2$ rad of phase. Having two sensors separated by $\pi/2$ of Gouy phase, one can derive alignment control signals.

In general, a mode can be written as a superposition of HG modes

$$E(x, y, z) = \sum_{n=0}^{n=\infty} \sum_{m=0}^{m=\infty} A_m u(x) A_n u(y), \quad (\text{D.31})$$

where A is the mode amplitude. This form is useful when calculating the coupling between modes.

The following example of the amount of power lost due to two arms of an interferometer being misaligned. The field at the AS port of the interferometer can be written as

$$E_{\text{AS}} = \frac{A_0}{\sqrt{2}} \left(\frac{\Delta A_0}{A_{00}} u_0 + \sum_{m=1}^{\infty} \frac{a_m u_m}{A_{00}} \right), \quad (\text{D.32})$$

where A_0 is the incident field amplitude and A_{00} is the amplitude of the field in the arms. The change in the fundamental power due to an optical imperfection in one

of the arms is ΔA_0 . E_{AS} is normalised to the incoming power at the beam splitter. As the modes are orthogonal by calculating the power in the higher order modes, the power lost from the fundamental mode can be calculated. The amplitude of a mode can be written as

$$A_m = \frac{1}{\Gamma(m/2 + 1)2^m\sqrt{\pi}} \int_{-\infty}^{\infty} E(X)u_m(X)e^{-X^2/2}dX \quad (D.33)$$

where Γ is the Gamma function, X is the normalised distance $\sqrt{2}x/w$ and $E(X)$ is an arbitrary field distribution. For a misaligned beam with a purely angular misalignment θ , a linear phase gradient $\varphi(x) = 2\pi\theta x/\lambda$ must be applied to the regular field distribution. The field distribution at the waist can be written as

$$E(X) = \exp\left(iX\frac{\theta}{\theta_c} - \frac{X^2}{2}\right) \quad (D.34)$$

where the critical angle is $\theta_c = \lambda\sqrt{2}/\pi w$. Inserting this $E(X)$ into equation (D.33) gives

$$A_m = \frac{e^{-(\theta/\theta_c)^2}}{\Gamma(m/2 + 1)2^m\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(-X+i\frac{\theta}{\theta_c})^2}u_m(X)dX, \quad (D.35)$$

which can be solved analytically by using the relationship

$$\int_{-\infty}^{\infty} e^{-(z^*t)^2}u_n(t)dt = 2^n\sqrt{\pi}z^n \quad (D.36)$$

to give

$$A_m = \frac{(i\frac{\theta}{\theta_c})^m e^{-(\theta/\theta_c)^2}}{\Gamma(m/2 + 1)}. \quad (D.37)$$

The fundamental order is reduced by

$$\frac{A_0}{A_{00}} = \exp \left((-\theta/\theta_c)^2 \right) \quad (\text{D.38})$$

D.7 Concluding remarks

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